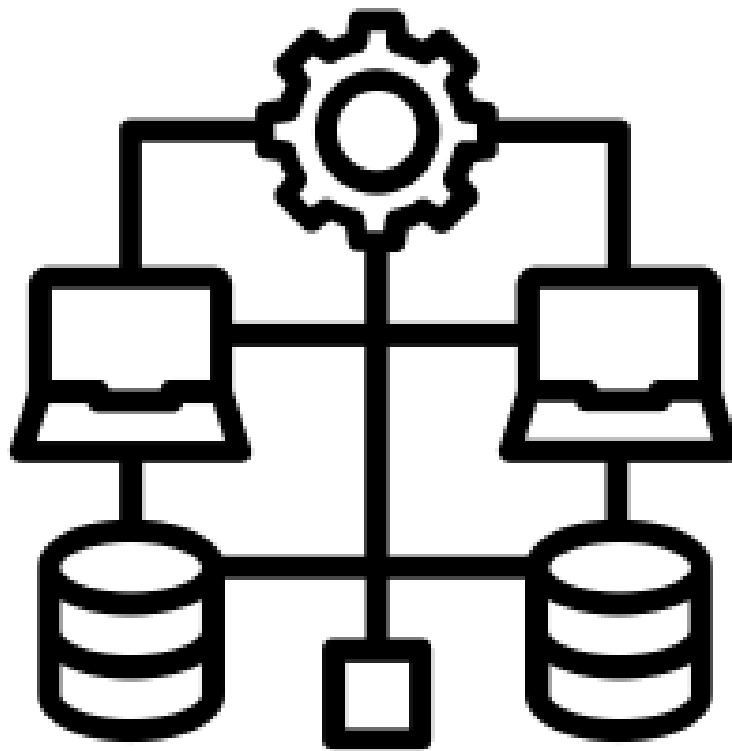


Lesson 2: Configure Network Services





Two Truths and a Lie

Objective:

In this activity, you will analyze and evaluate statements related to network services, specifically DHCP, DNS, and FTP. Two of the statements are true, and one is false. Your task is to identify the false statement and explain why it is incorrect.

Instructions:

1. Read the three statements carefully. Each set will include three statements about network services.
2. Identify the one statement that is false (the lie) in each set.
3. Explain why the false statement is incorrect, referring to your knowledge of network services.

Example Set:

1. **DHCP** automatically assigns IP addresses to devices on a network.
2. **DNS** stands for Dynamic Network Service.
3. **FTP** allows users to upload and download files from a server.

Answer:

The false statement is **Statement 2** because **DNS** actually stands for **Domain Name System**, not Dynamic Network Service.

Set 1:

1. **DHCP** automatically assigns IP addresses to devices on a network.
2. **DNS** translates domain names into IP addresses.
3. **FTP** is used to manage email on a server.

Answer:

Set 2:

1. **FTP** allows users to upload and download files from a server.
2. **DNS** translates domain names into IP addresses.
3. **DHCP** requires manual configuration of IP addresses for each device.

Answer:

Set 3:

1. **DHCP** assigns IP addresses dynamically, reducing the need for manual configuration.
2. **DNS** is responsible for securing network connections through encryption.
3. **FTP** uses ports 20 and 21 for data transfer and command control.

Answer:

Set 4:

1. **DHCP** can allocate IP addresses on a lease basis, meaning they may change over time.
2. **FTP** is mainly used for email communication between mail servers.
3. **DNS** servers contain a database of domain names and their corresponding IP addresses.

Answer:



What I Need to Know?

Objectives:

At the end of the lesson, the students should be able to:

1. **Analyze** the purpose and functions of the common network services (DHCP, DNS, FTP).
2. **Demonstrate** the configuration process for network services, including assigning IP addresses and setting up domain names.
3. **Reflect** on and evaluate their own experiences with network services and how the proper configuration will impact their daily technology use through an essay



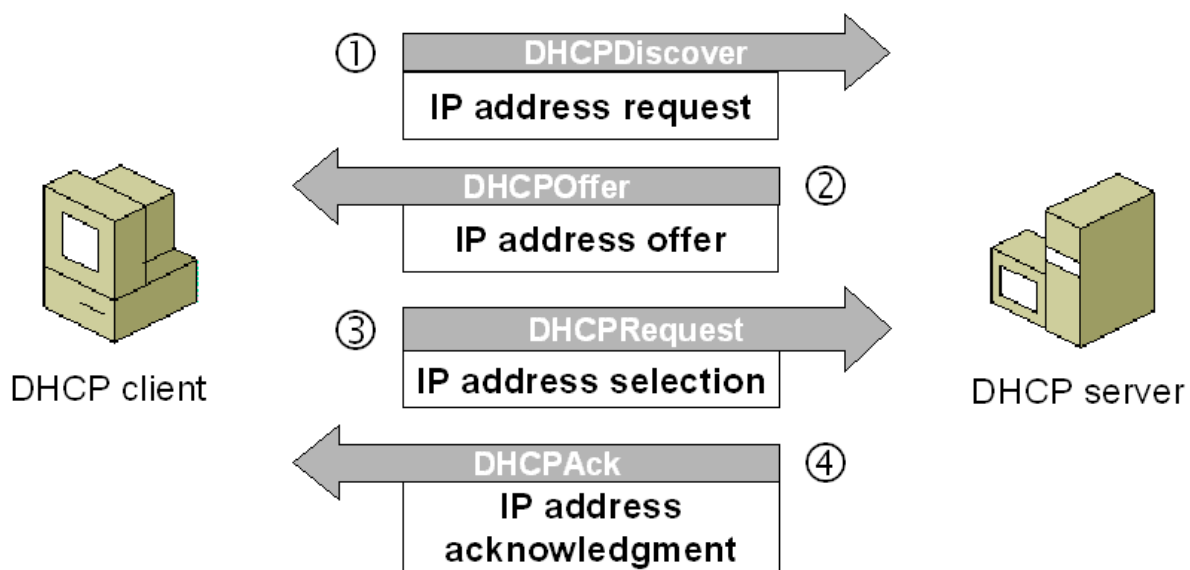
LET'S LEARN:

The Backbone of Network Communication

In the vast digital landscape of today, where millions of devices connect and communicate seamlessly, there exist invisible forces that ensure everything runs smoothly. Among them, three essential protocols play a crucial role: **DHCP, FTP, and DNS**. Each of these serves a unique purpose, working behind the scenes to keep networks functional and efficient.

Section 1: DHCP – The Network’s Silent Organizer

Imagine entering a vast library where every book is arranged without labels, and finding the right one is a tedious process. Now, imagine a librarian who, the moment you walk in, hands you a neatly written note with the exact location of the book you need. That librarian, in the world of networking, is **DHCP (Dynamic Host Configuration Protocol)**.



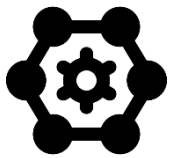
Purpose of DHCP

The fundamental role of DHCP is to automatically assign **IP addresses** and other network settings to devices. Without it, every computer, phone, and IoT device would require manual configuration, a time-consuming and error-prone task.

How DHCP Works

The process unfolds in four key steps:

1. **Discovery** – A new device joins the network and sends a request, asking for an IP address.
2. **Offer** – The DHCP server responds with an available IP address.
3. **Request** – The device confirms that it wants to use the offered IP address.
4. **Acknowledgment** – The DHCP server finalizes the assignment, allowing the device to communicate on the network.



Remember

Key Functions of DHCP

- **Automatic IP Assignment** – Eliminates the need for manual IP configuration.
- **Efficient Management** – Prevents address conflicts and recycles unused IPs.
- **Configuration Distribution** – Sends additional details like subnet masks, gateway addresses, and DNS settings.



Importance

Thanks to DHCP, users can connect to a network effortlessly, without ever realizing the complex orchestration happening behind the scenes.

Section 2: FTP – The Digital Bridge for File Transfers

In the digital world, sharing information is as vital as having access to it. Enter **FTP (File Transfer Protocol)**, a system designed to move files between computers with ease, long before cloud storage became a household term.

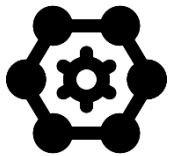
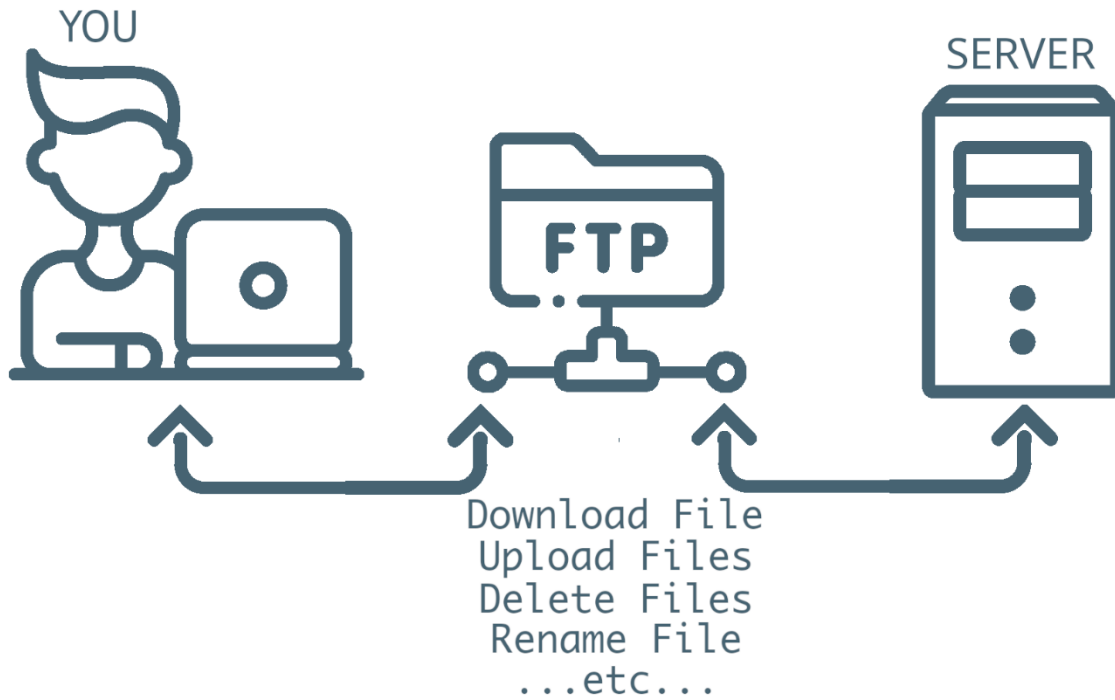
Purpose of FTP

FTP is a protocol used to **upload, download, and manage files** over a network. Whether it's a developer uploading a website to a server or a company backing up important documents, FTP ensures that files can be transferred efficiently.

How FTP Works

An FTP connection involves two parties:

- **The Client** – The user or system requesting to send or receive files.
- **The Server** – The destination system that stores or retrieves the files.



To transfer files securely, users log in with credentials (username and password). In some cases, **Anonymous FTP** allows access without login credentials, often for public file repositories.

Remember

Key Functions of FTP

- **File Upload & Download** – Enables users to send or retrieve files from a remote server.
- **Support for Multiple Modes** – Operates in **Active** (direct connection) or **Passive** (firewall-friendly) mode.
- **Resume Capability** – Allows interrupted transfers to continue instead of restarting.

- **Batch Transfers** – Sends multiple files simultaneously, increasing efficiency.



Importance

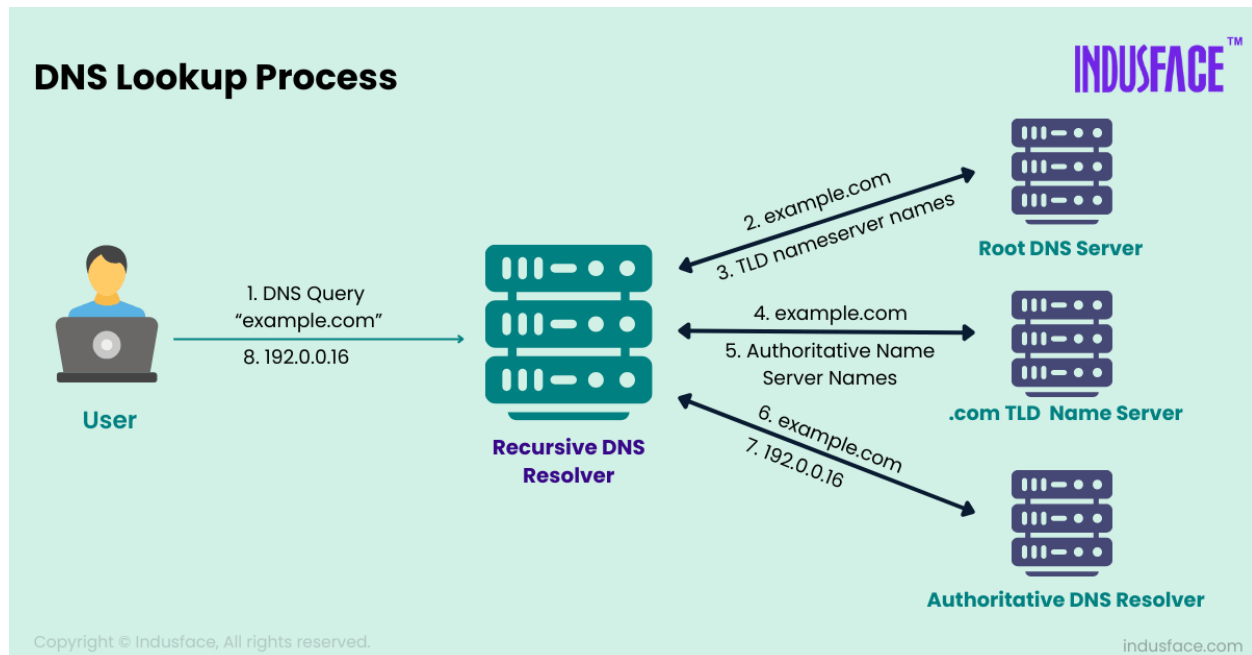
Though newer technologies like cloud storage have gained popularity, FTP remains a cornerstone for large-scale data transfers in businesses, web hosting, and digital archives.

Section 3: DNS – The Internet’s Address Book

If FTP is the bridge for transferring files, **DNS (Domain Name System)** is the map that guides users to their destinations. Without it, browsing the internet would be like trying to remember the exact coordinates of every house you want to visit.

Purpose of DNS

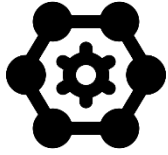
DNS is responsible for translating human-friendly domain names (like www.google.com) into numerical IP addresses (142.250.190.78). This makes the internet accessible without needing to memorize long strings of numbers.



How DNS Works

When you type a web address into your browser, the following happens:

- 1. Query Initiation** – Your device asks a DNS server, “What is the IP address of this website?”
- 2. Lookup & Resolution** – The DNS server checks its records and responds with the correct IP.
- 3. Connection Establishment** – Your browser connects to the website’s IP, allowing you to access the content.



Remember

Key Functions of DNS

- **Domain Name Resolution** – Converts website names into numerical addresses.
- **Caching** – Stores frequently accessed domain details to speed up browsing.
- **Load Balancing** – Distributes web traffic across multiple servers for better performance.
- **Security (DNSSEC)** – Protects against attacks like domain spoofing.



Importance

DNS is what makes the internet user-friendly. Instead of remembering long sequences of numbers, we can simply type a name and let DNS handle the rest.



Let's Try!

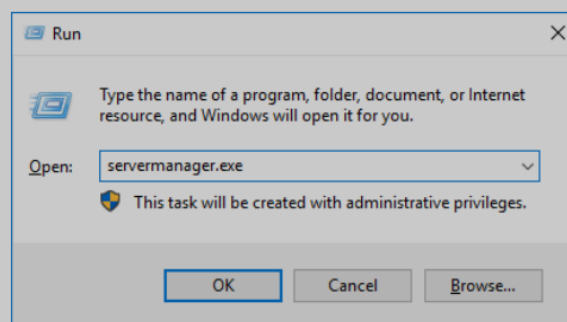
The following images would guide us in configuring our network services.

Before proceeding:

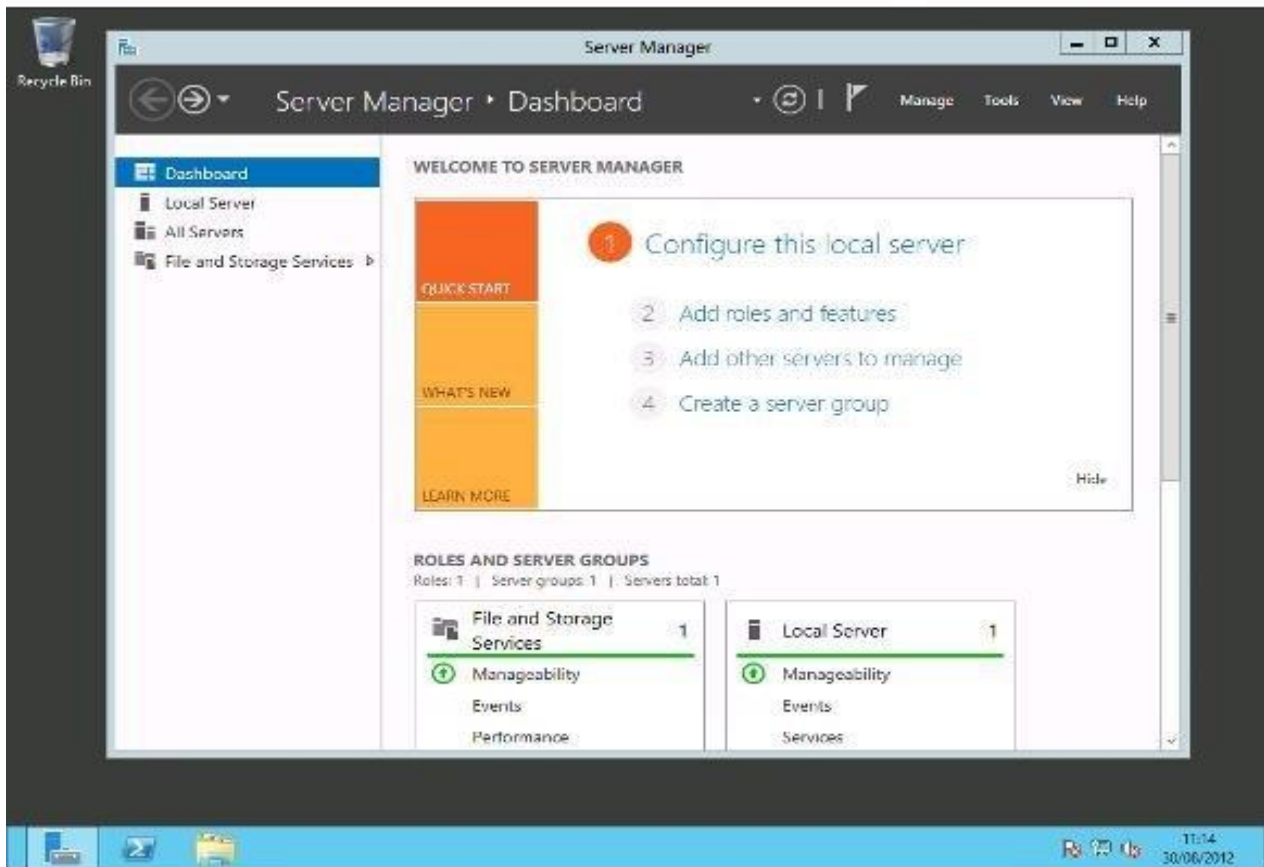
1. You must have administrative privileges on server.
2. Setup server with a static IP address.
3. Change the Windows Server name according to your company's naming standards.

After making sure of your administrative privilege, setting up a static IP address, and naming according to the standard. Make sure to check your cables if they are working properly.

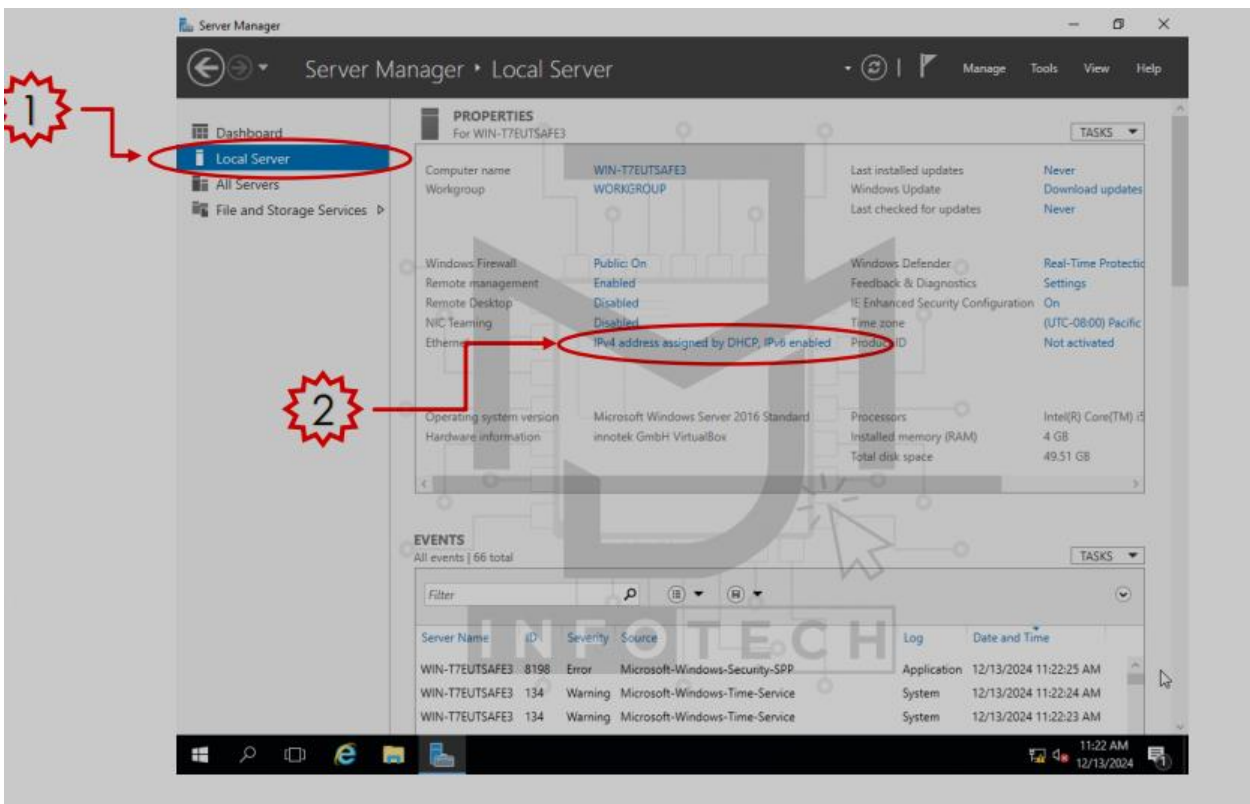
Open the Run box using Win+R, type servermanager.exe, and click OK



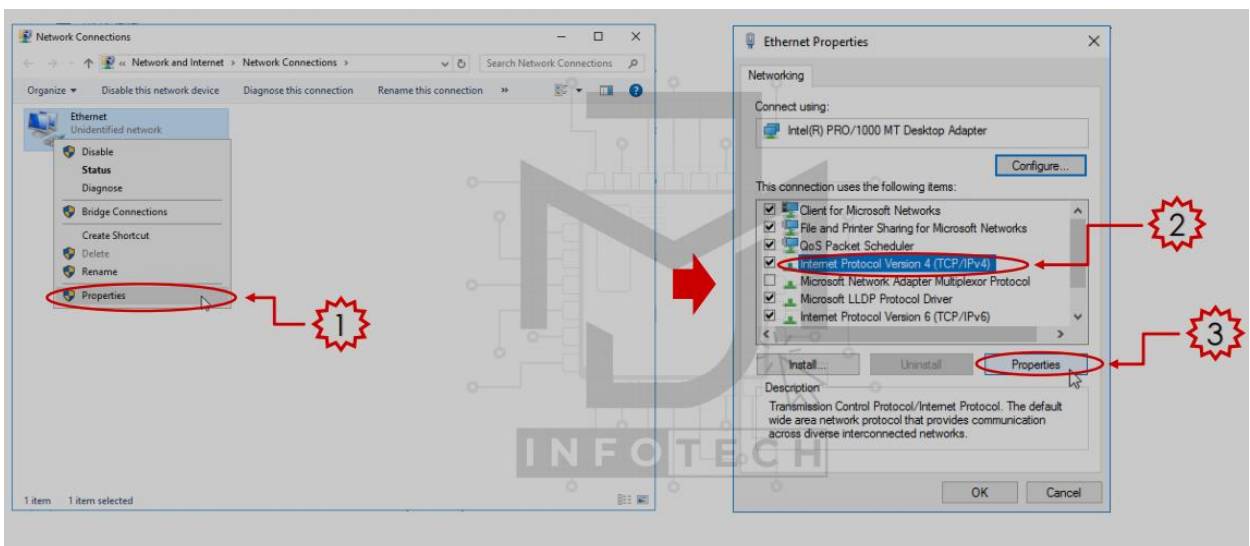
Or you can open the server manager by clicking the windows button in your keyboard and then look for the server manager



- On the Server Manager Dashboard click **Local Server**
- Tap the link beside the Ethernet
- Right click on the Ethernet and select Properties
- Select Internet Protocol Version 4 (TCP/IPv4) and select Properties
- Select on Use the following IP address:

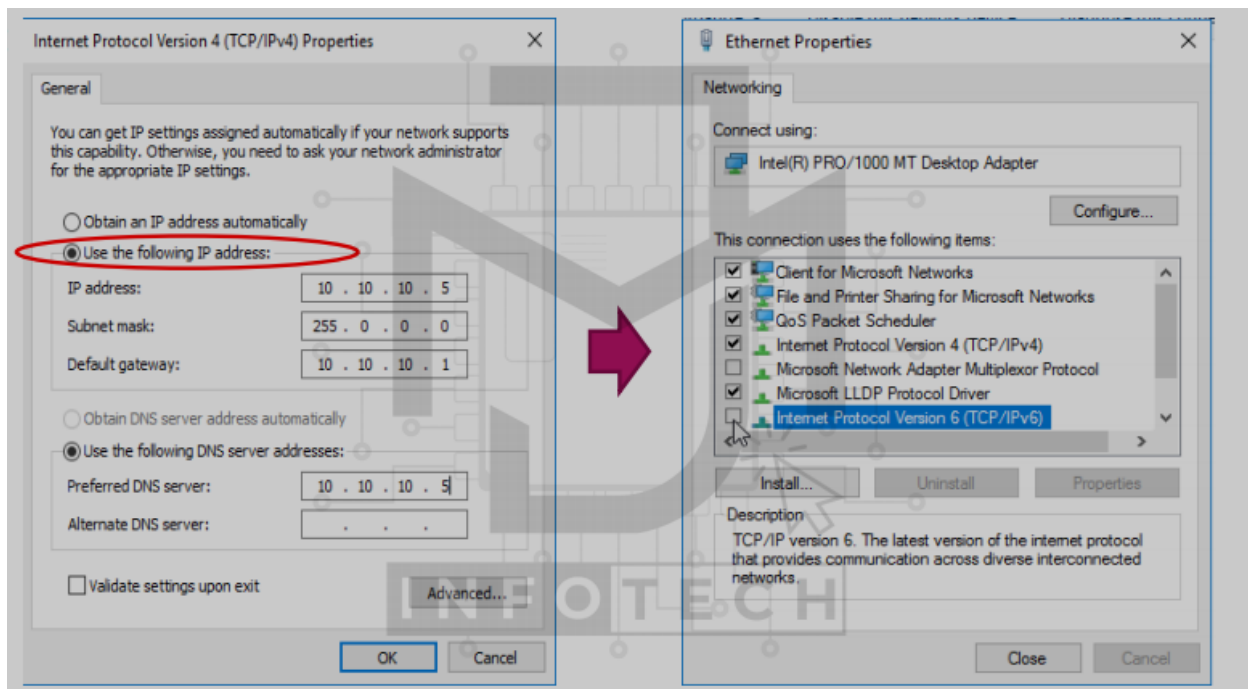


- Right click on the Ethernet and select Properties
- Select Internet Protocol Version 4 (TCP/IPv4) and select Properties

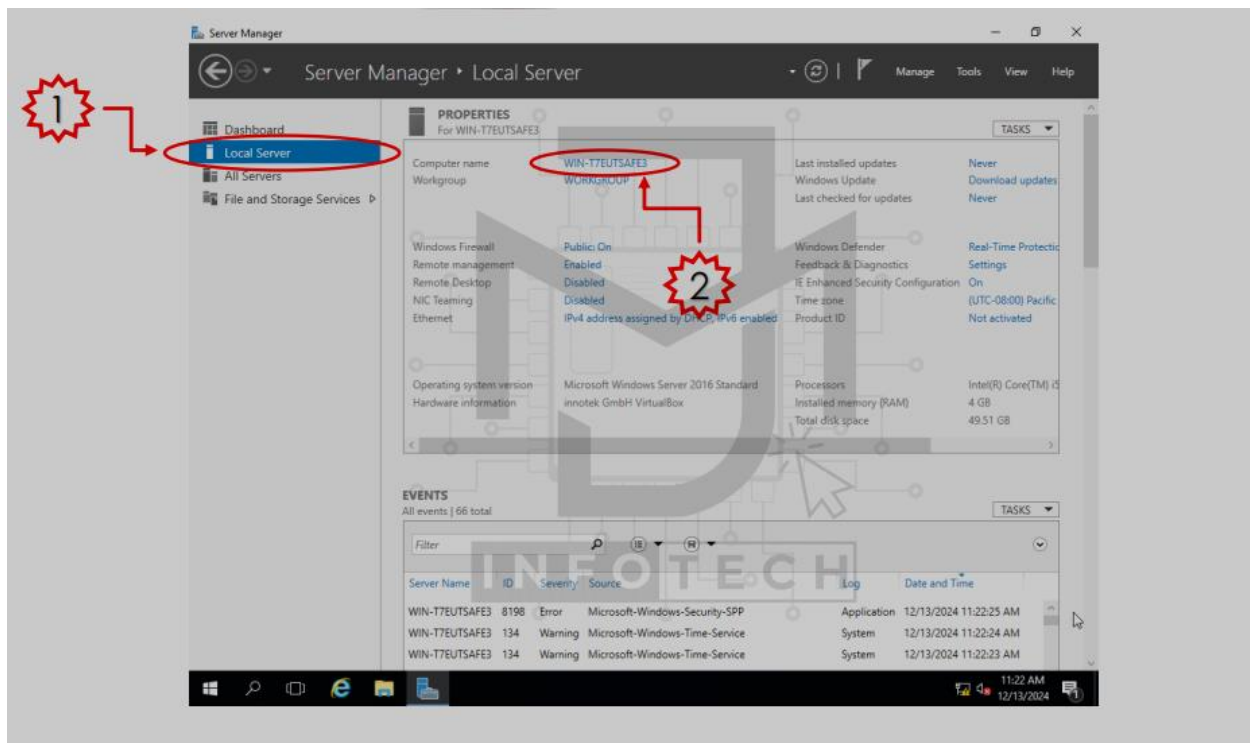


In this Demo we will be using this following IP address:

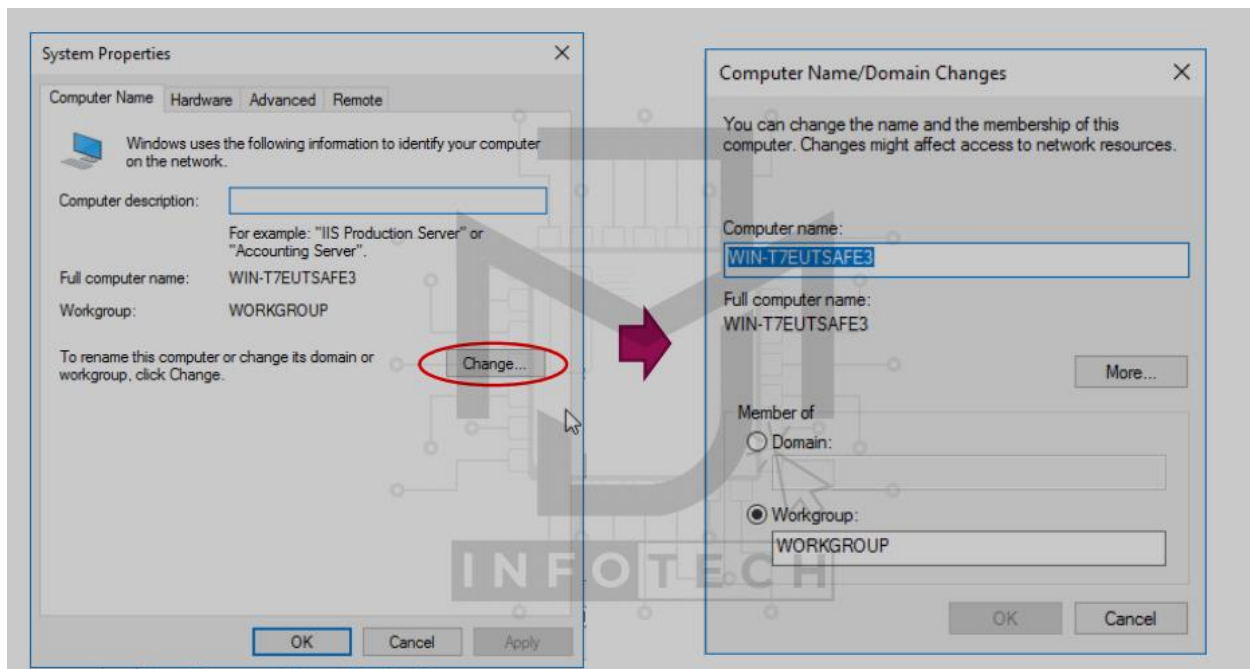
- Select on Use the following IP address:
- IP Address: – **10.10.10.5**
- Subnet Mask: – **255.0.0.0**
- Default gateway: **10.10.10.1**
- Preferred DNS Server: **10.10.10.5**
- Click OK to save
- Uncheck Internet Protocol Version 6
- Click Close to save



- On the Server Manager Dashboard click **Local Server**
- Tap the link beside the computer name to **Rename**

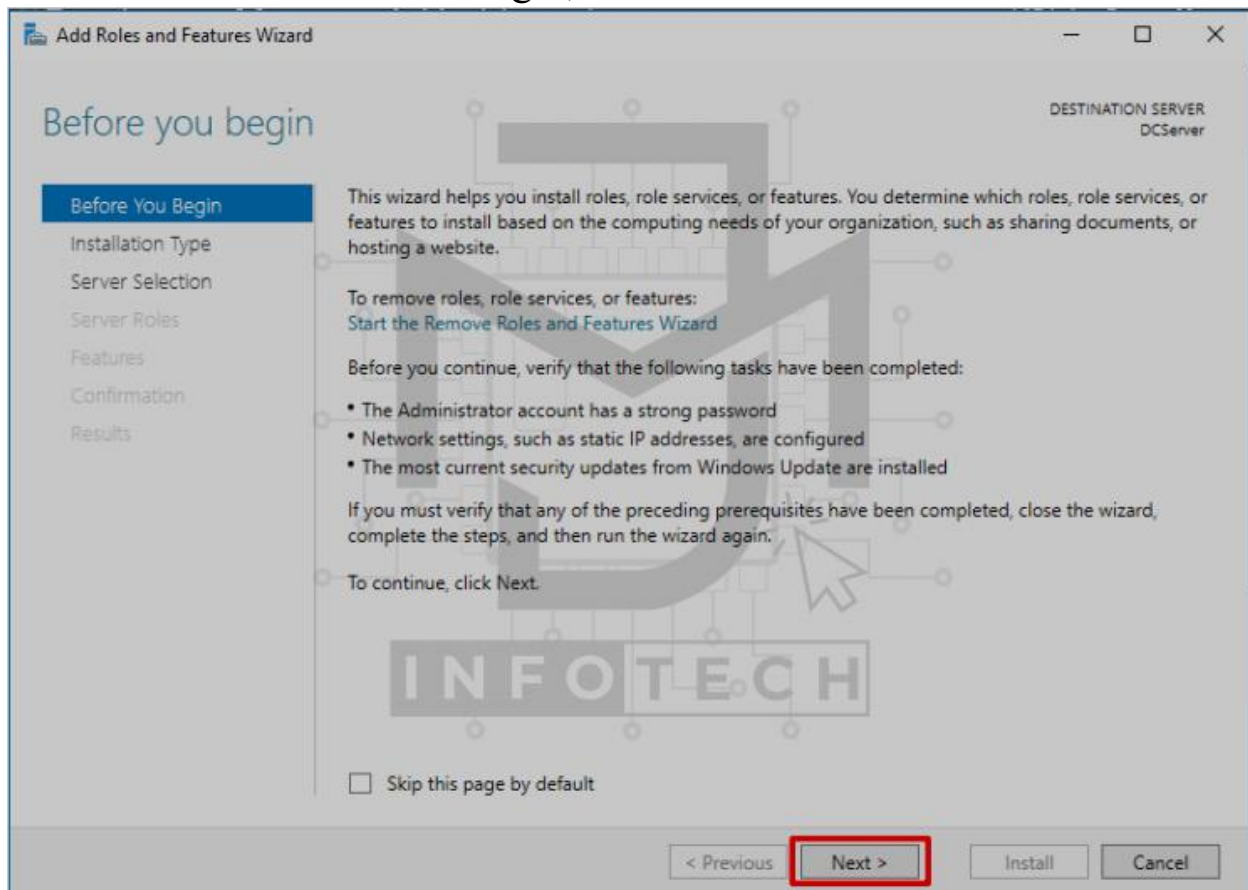


On the Computer name tab click change...

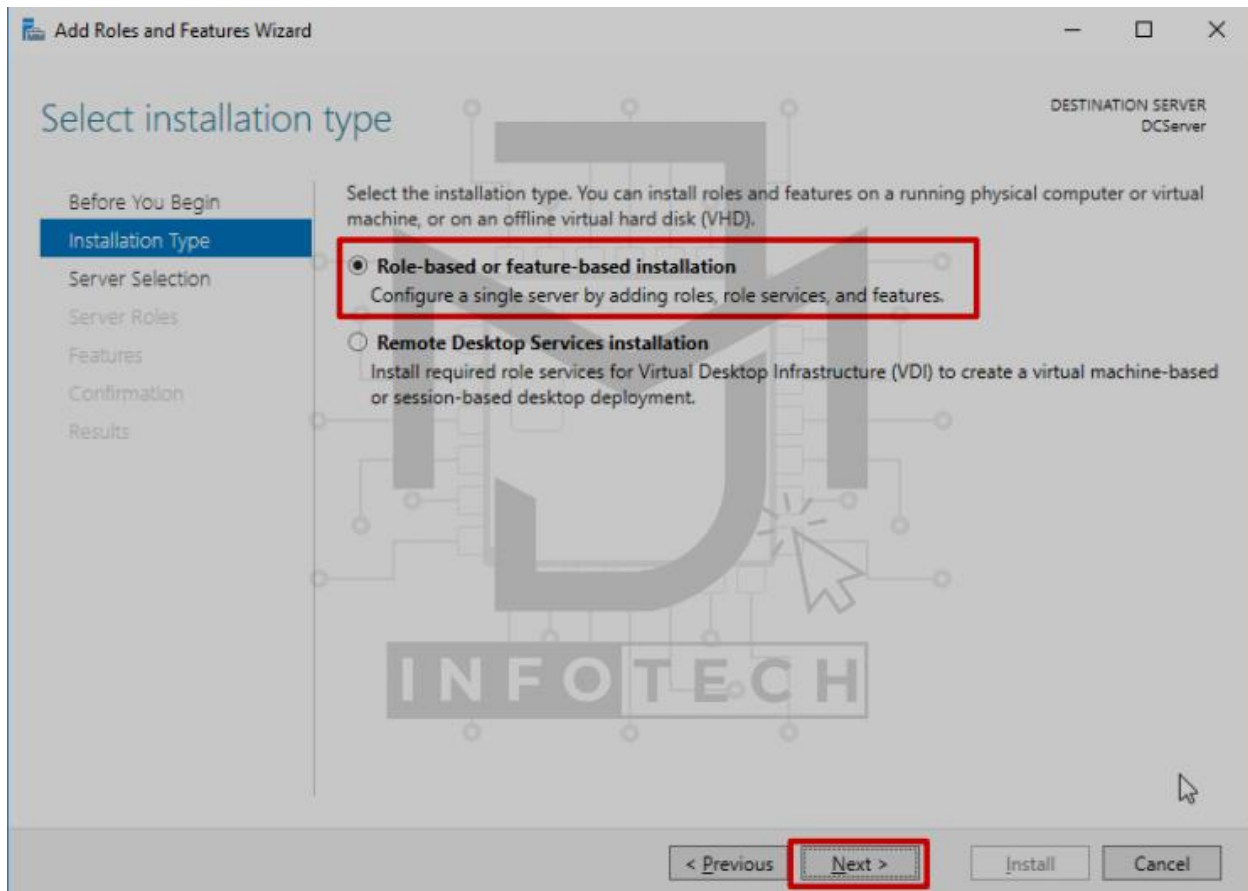


After renaming the PC to preferred name the PC will undergo a restart just proceed with it.

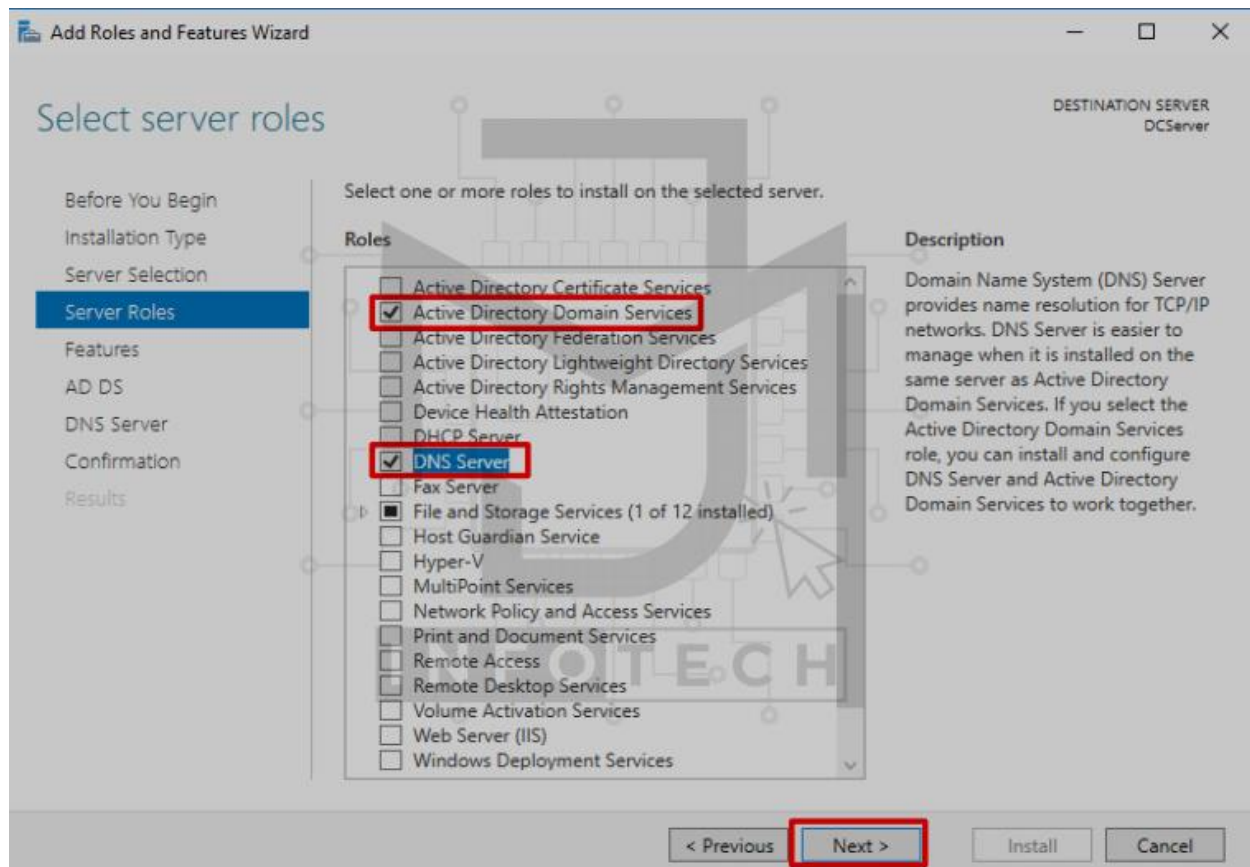
- In the Server Manager – Dashboard Click Add roles and features
- In the Before You Begin, click Next to continue



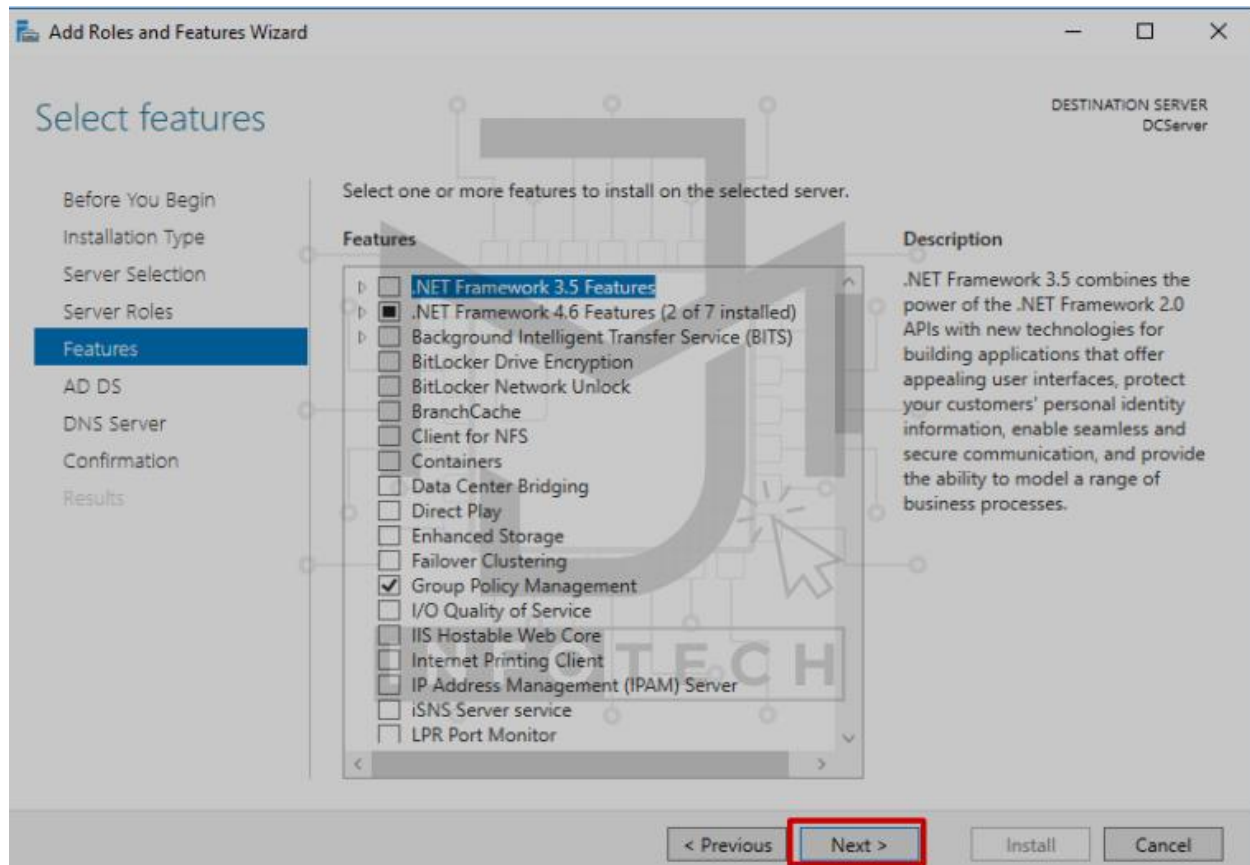
- In Installation Type select Role-base or feature-based installation
- Click Next to continue



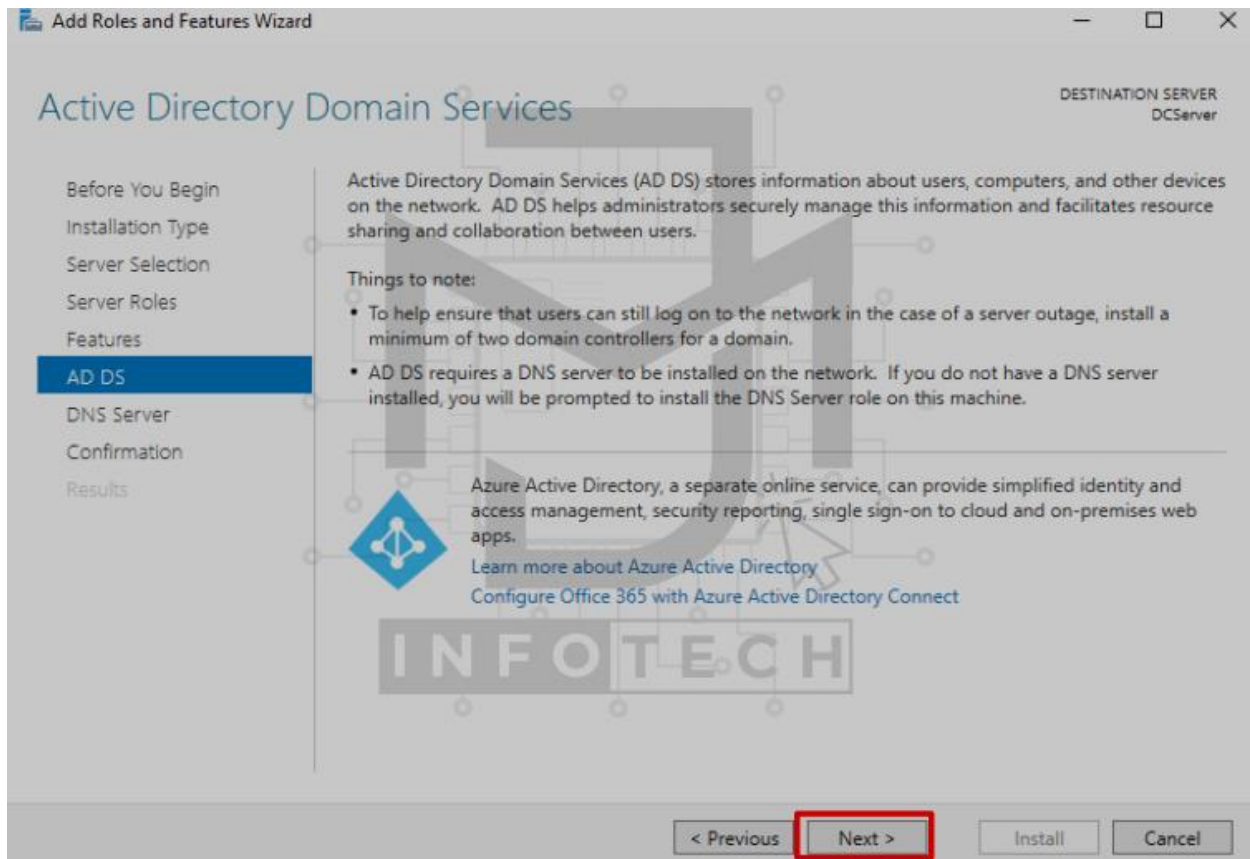
- On the Server Roles, Tick the following Roles
 - Active Directory Domain Services, then add features.
 - DNS Server, then add features.
- Click Next to continue



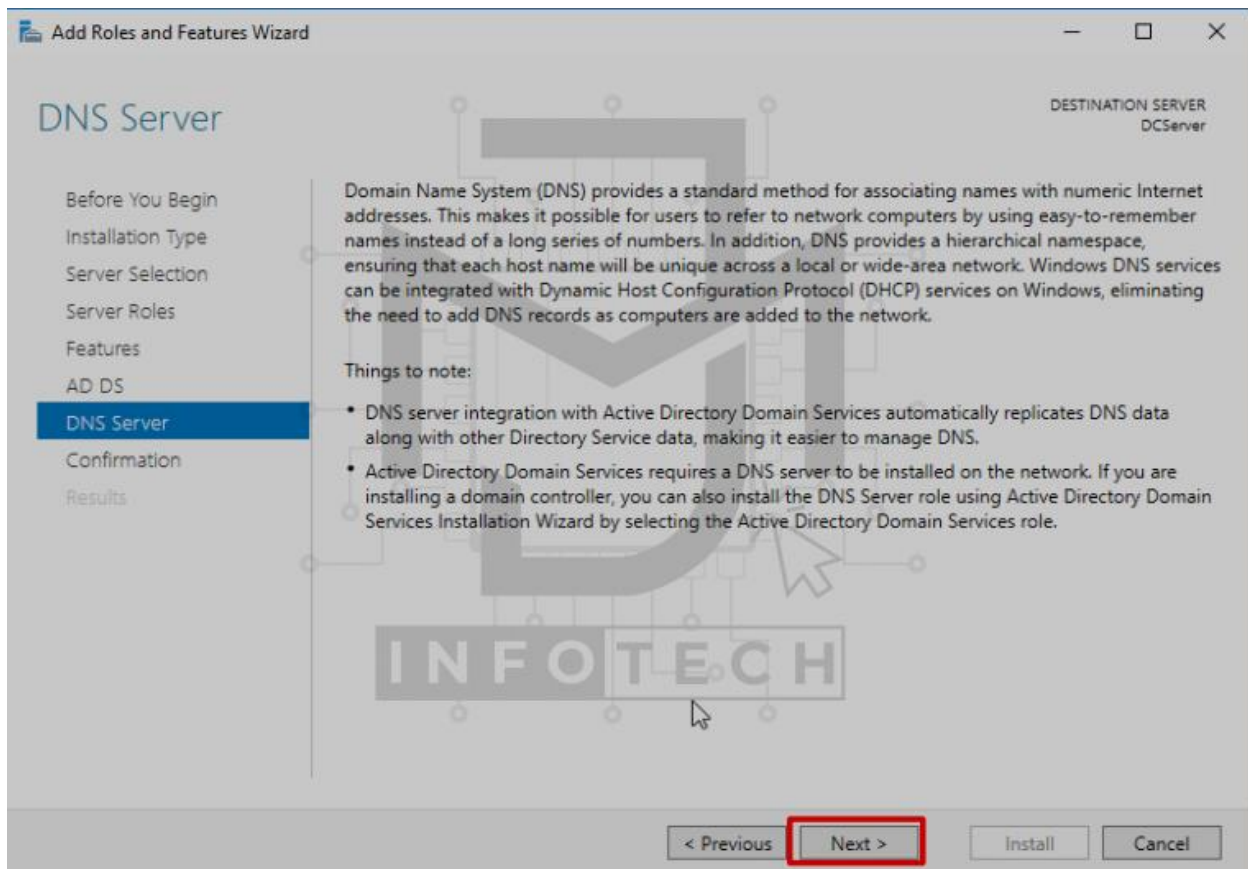
On the Features, click Next to continue..



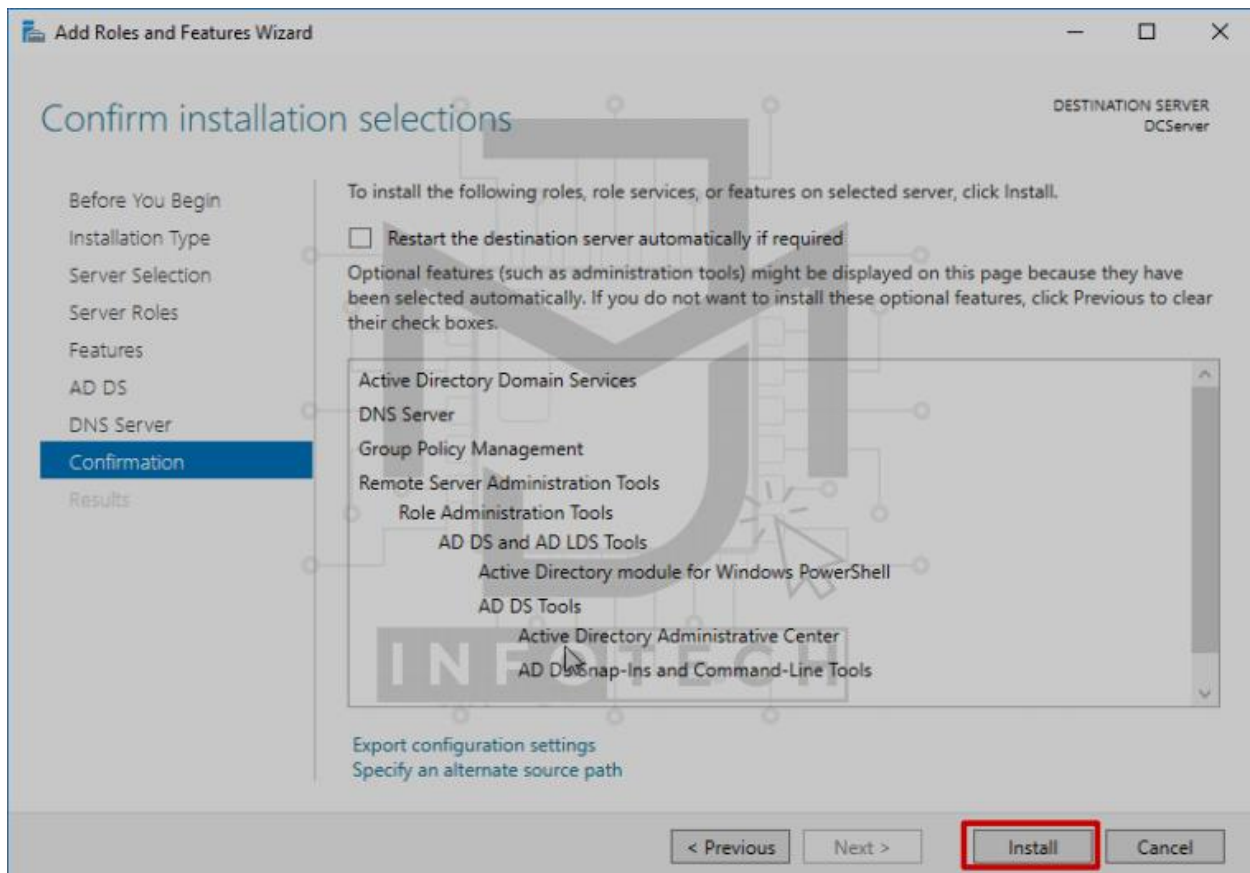
On the AD DS, click Next to continue..



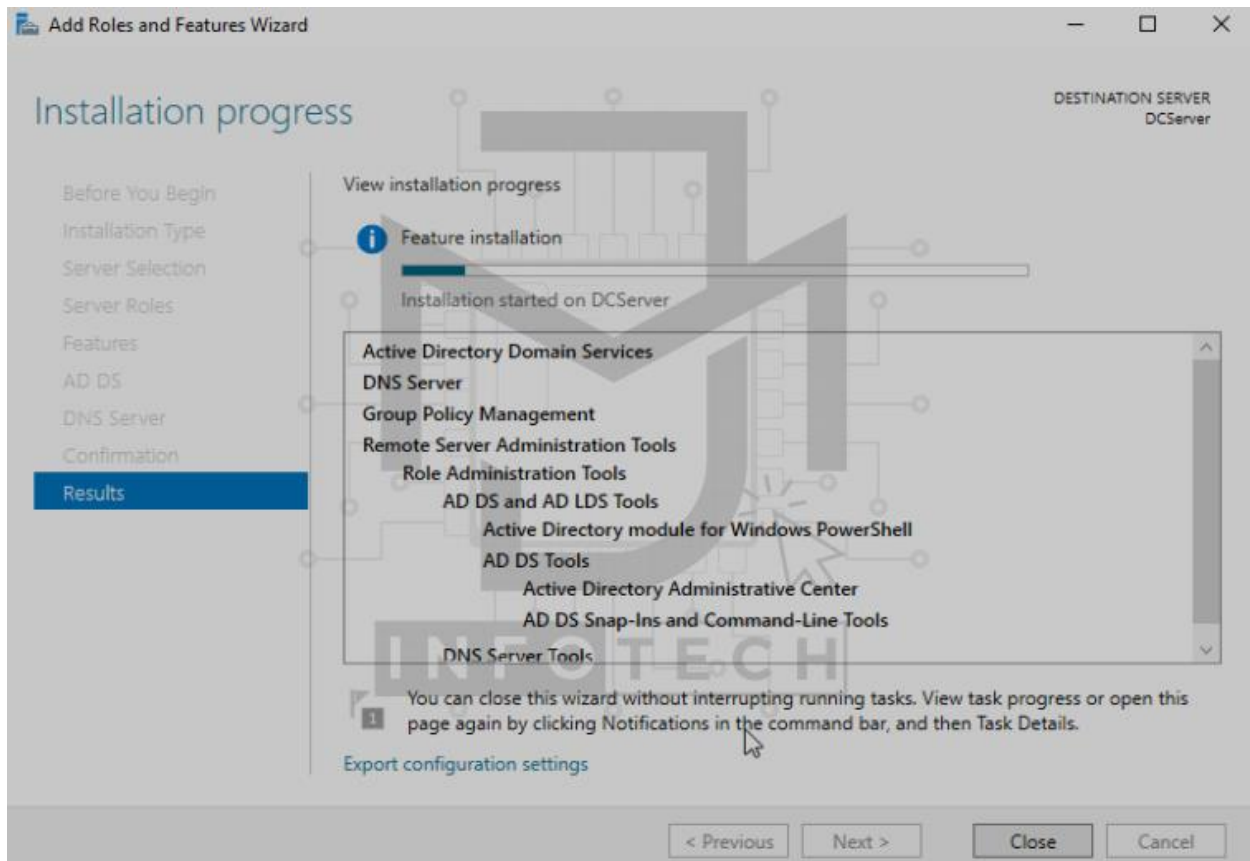
On the DNS Server, click Next to continue...



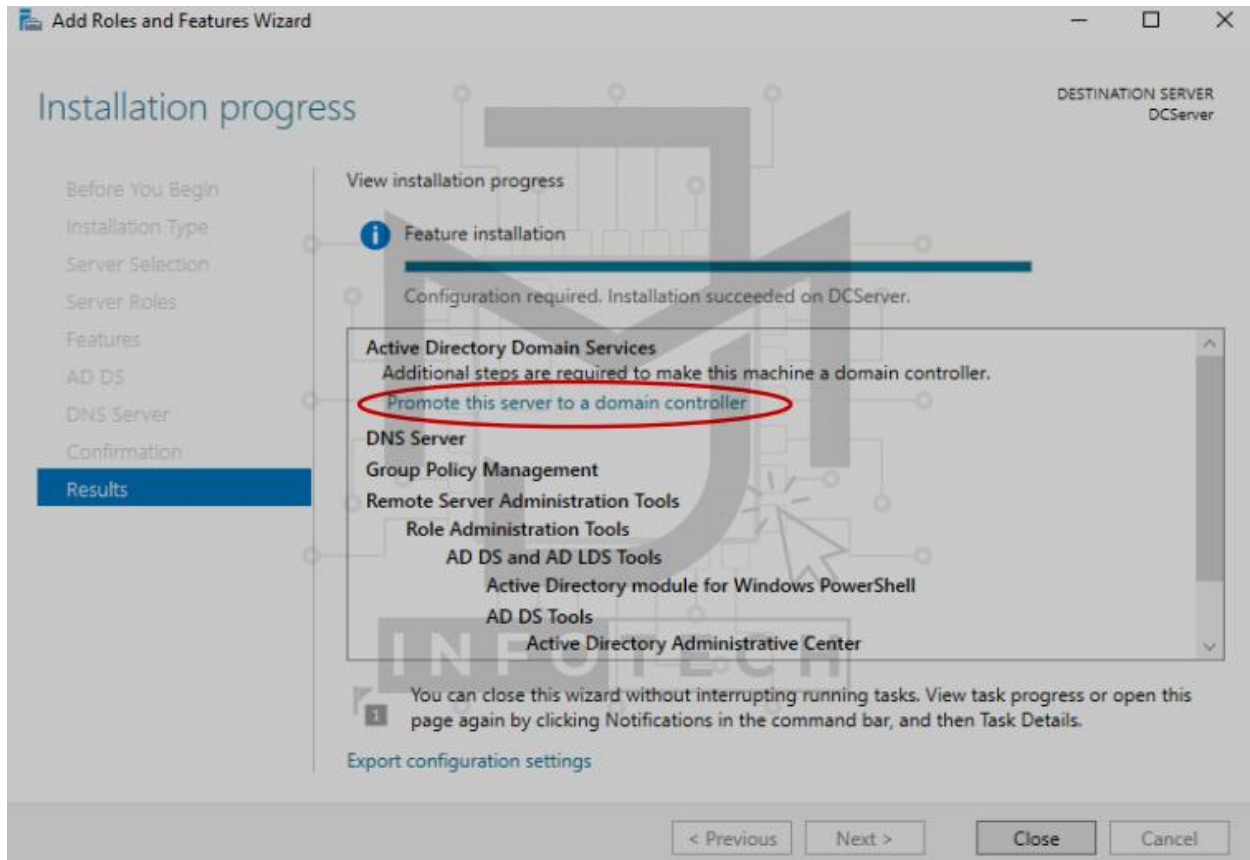
On the Confirmation, click Install to continue....



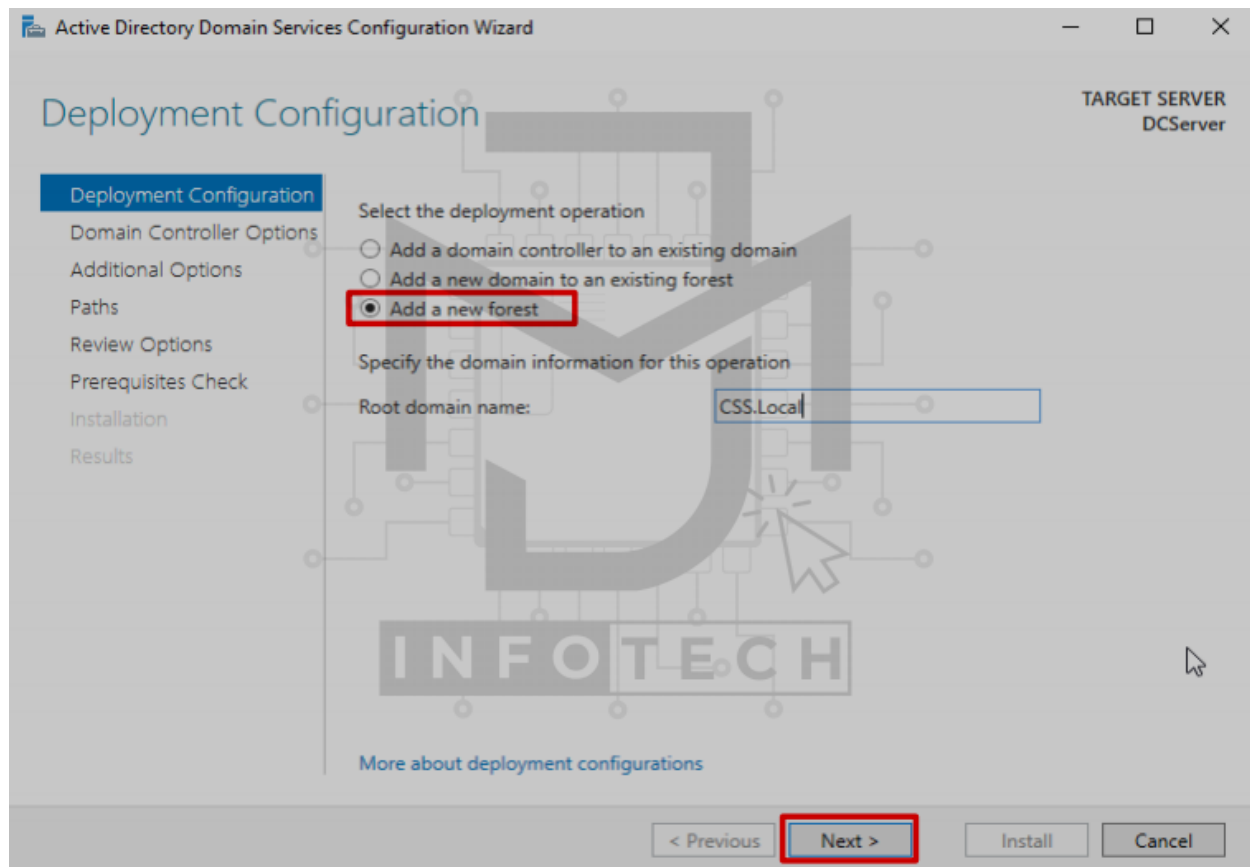
On the Results, just wait until the installation progress is completed, depending on your system, it takes 3 to 5 minutes to complete the installation.



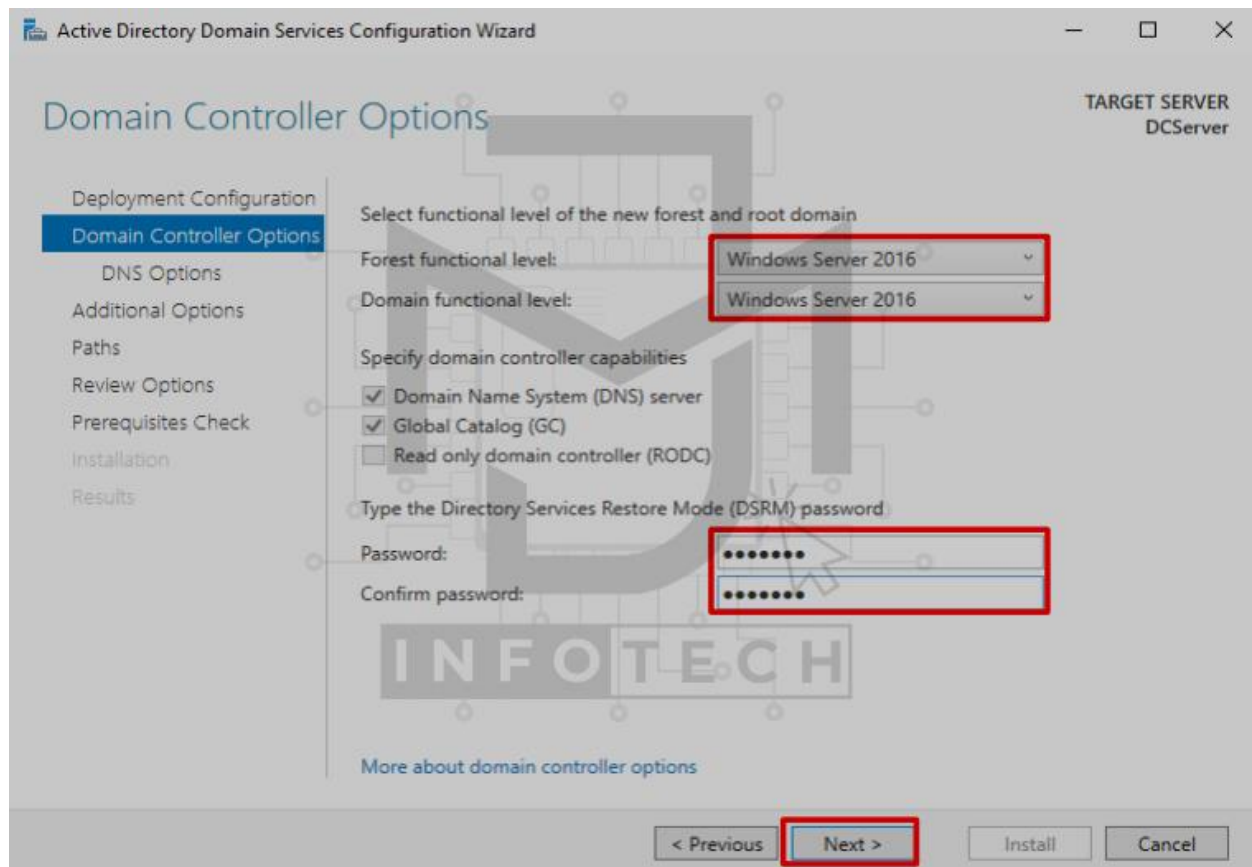
After the installation progress is completed, click to “Promote this server to a domain controller”



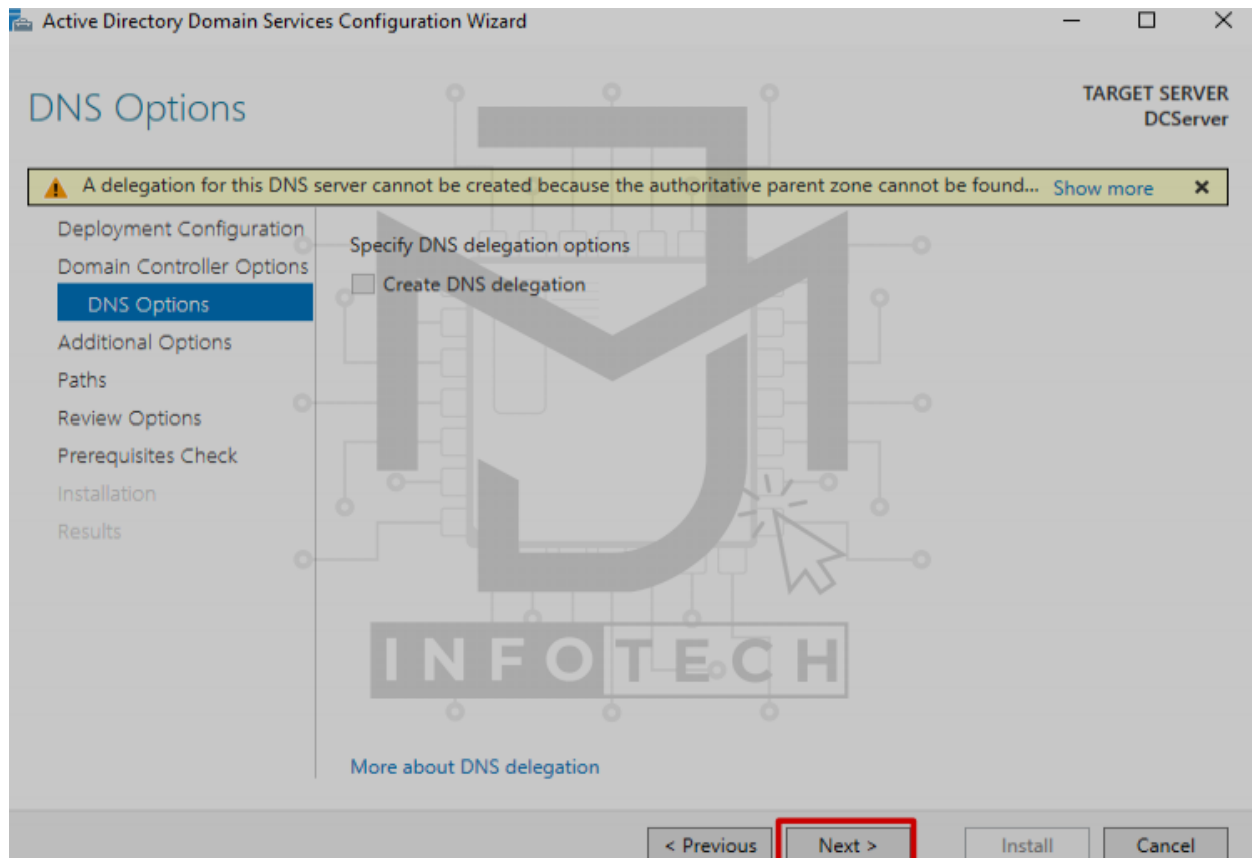
- On the Deployment Configuration, select “**Add a new forest**”
- On the Root domain name, in this demo we will be using “**CSS.Local**” without quote.
- Click Next to continue



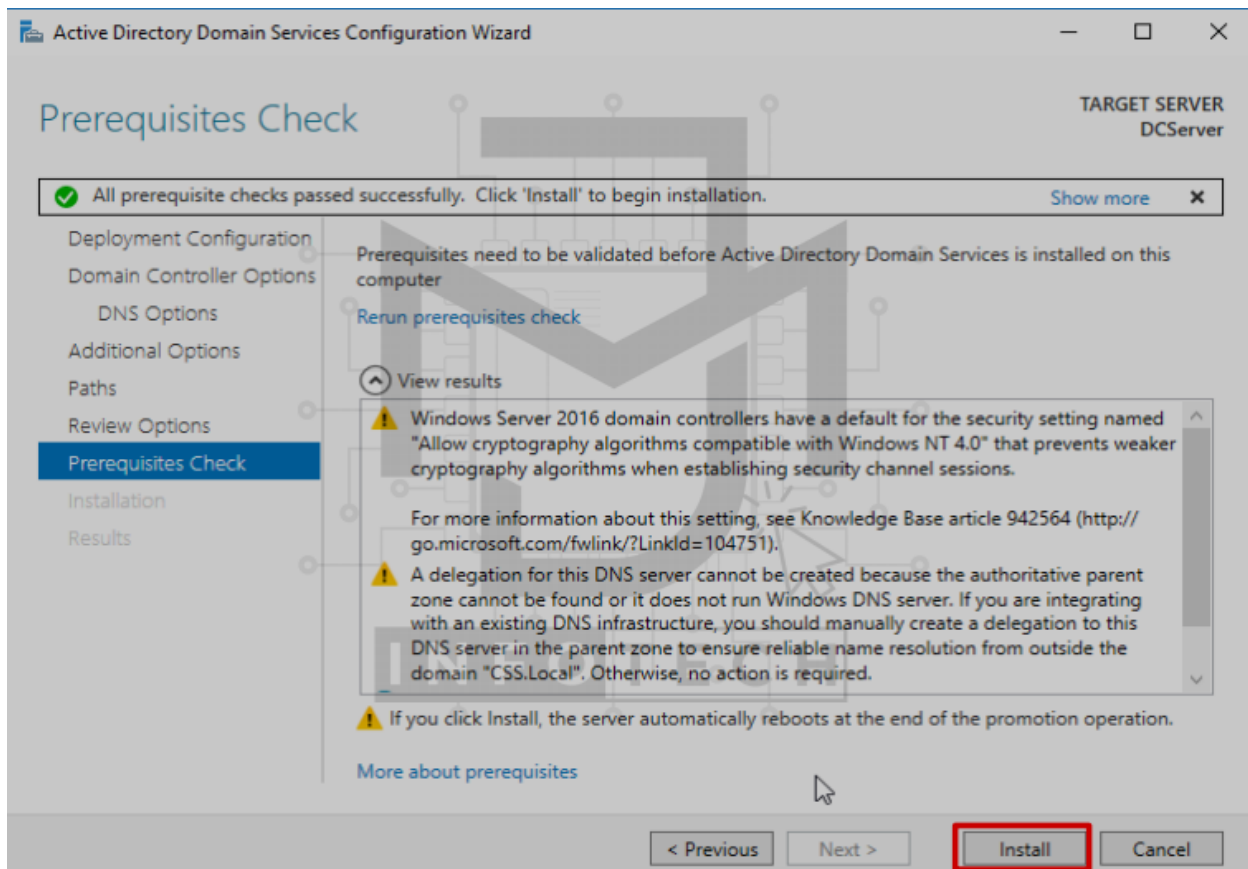
- On the Functional level of the new forest and root domain, select both **Windows Server 2016**
- On DSRM password, just type your **local server administrator password**
- Click Next to continue



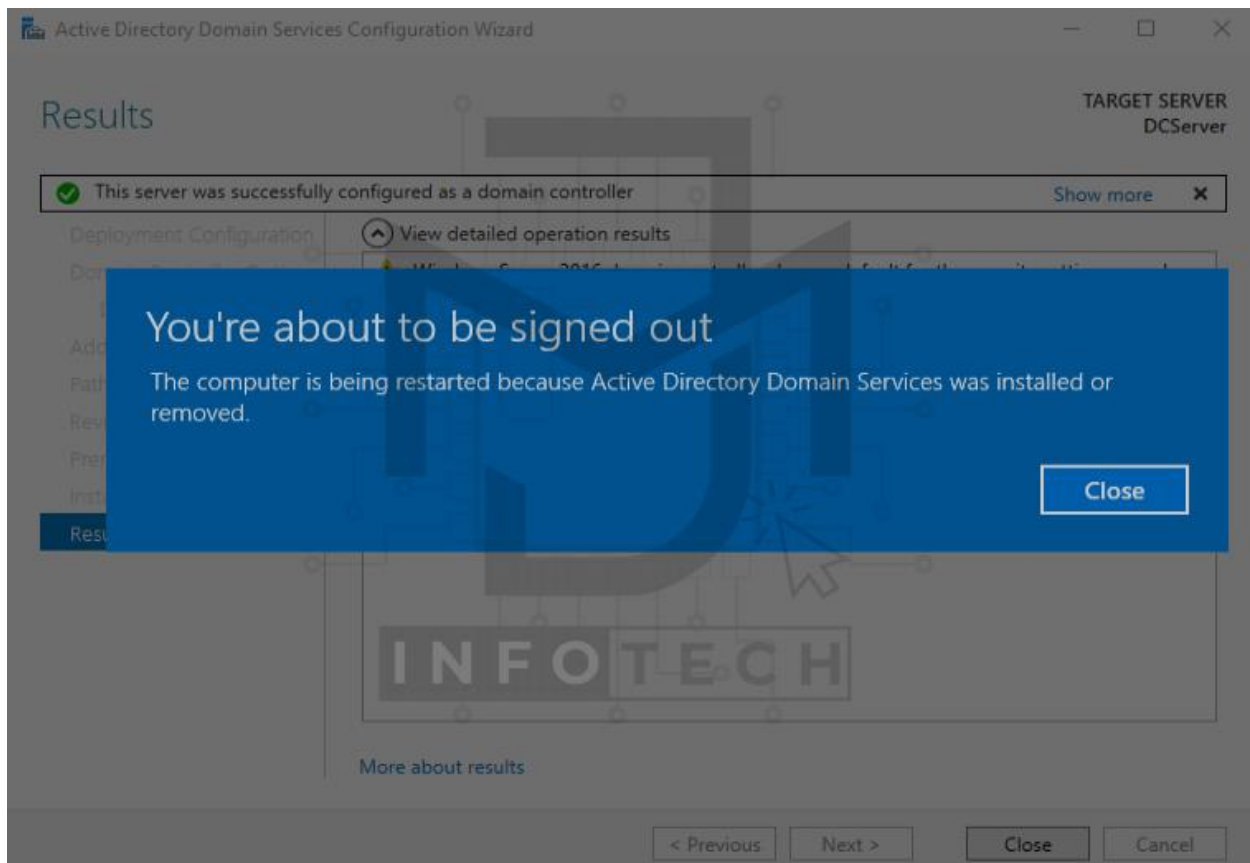
On the DNS options, click Next continue..



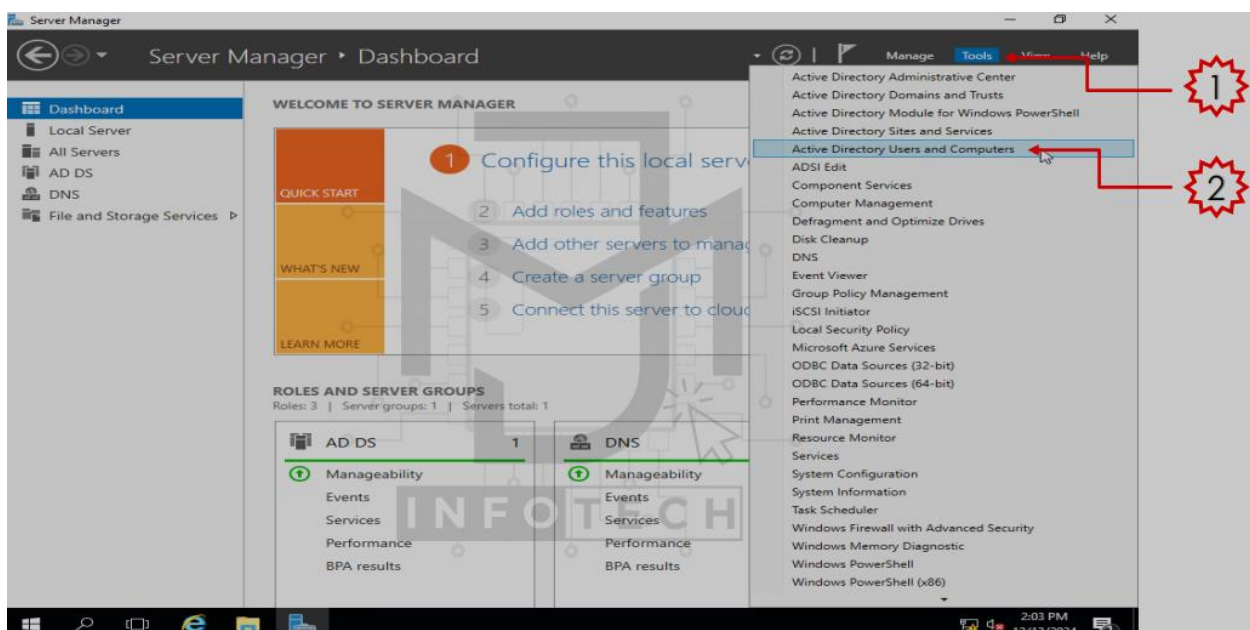
From DNS Options to Review Options just click next to continue.. And when you reach Prerequisites check, click Install.



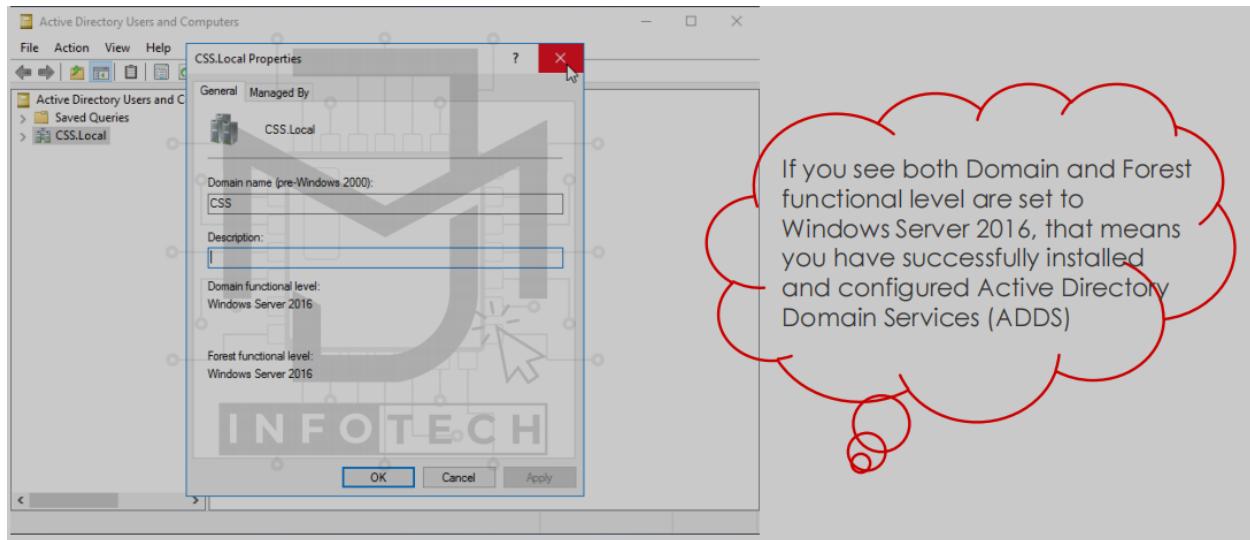
Wait until installation progression is completed. After the installation is complete the computer will prompt you for a restart.



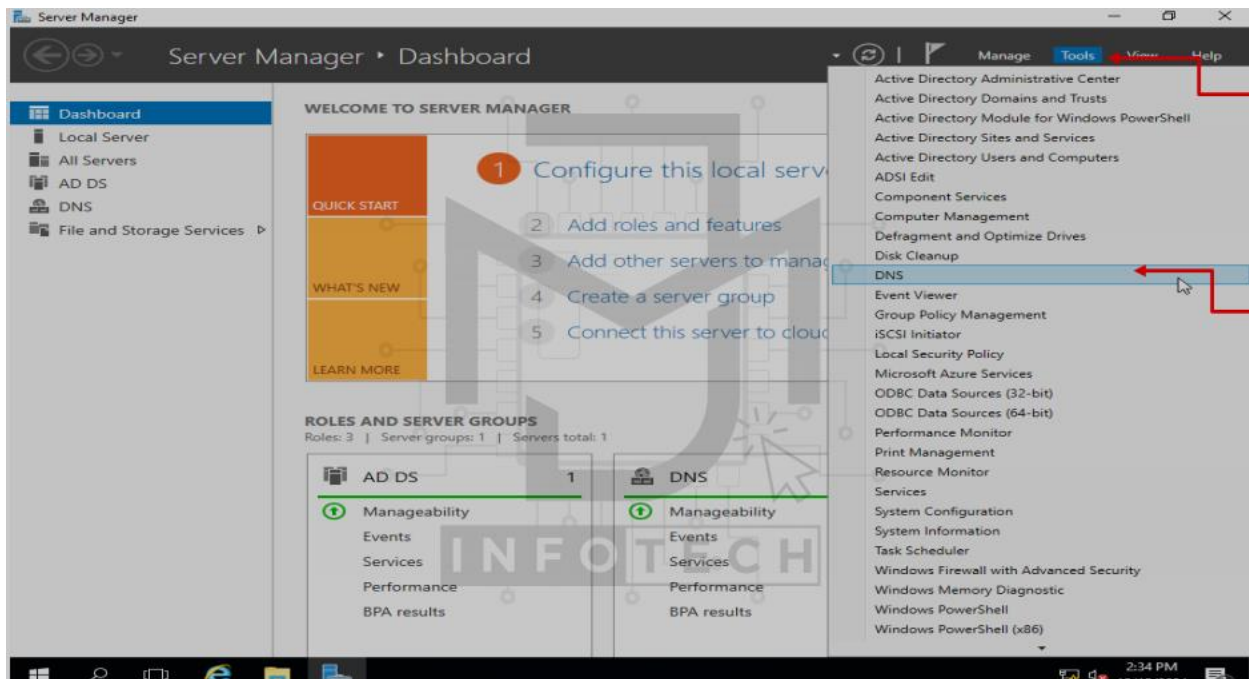
To verify if your AD DS is successfully configured or not. On Server Manager Dashboard click Tools, and select Active Directory Users and Computers.



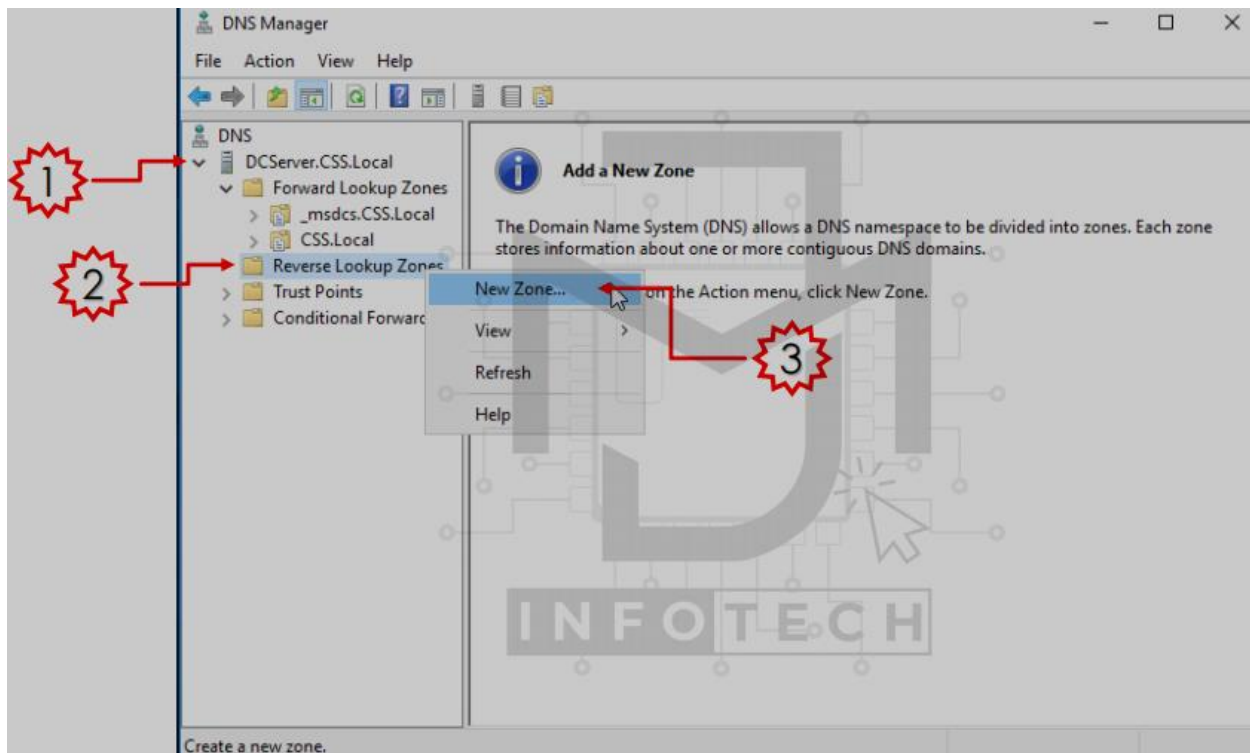
Right click on the CSS.Local (or whatever you named your locale is) then select properties.



On the server manager dashboard click tools and select DNS.

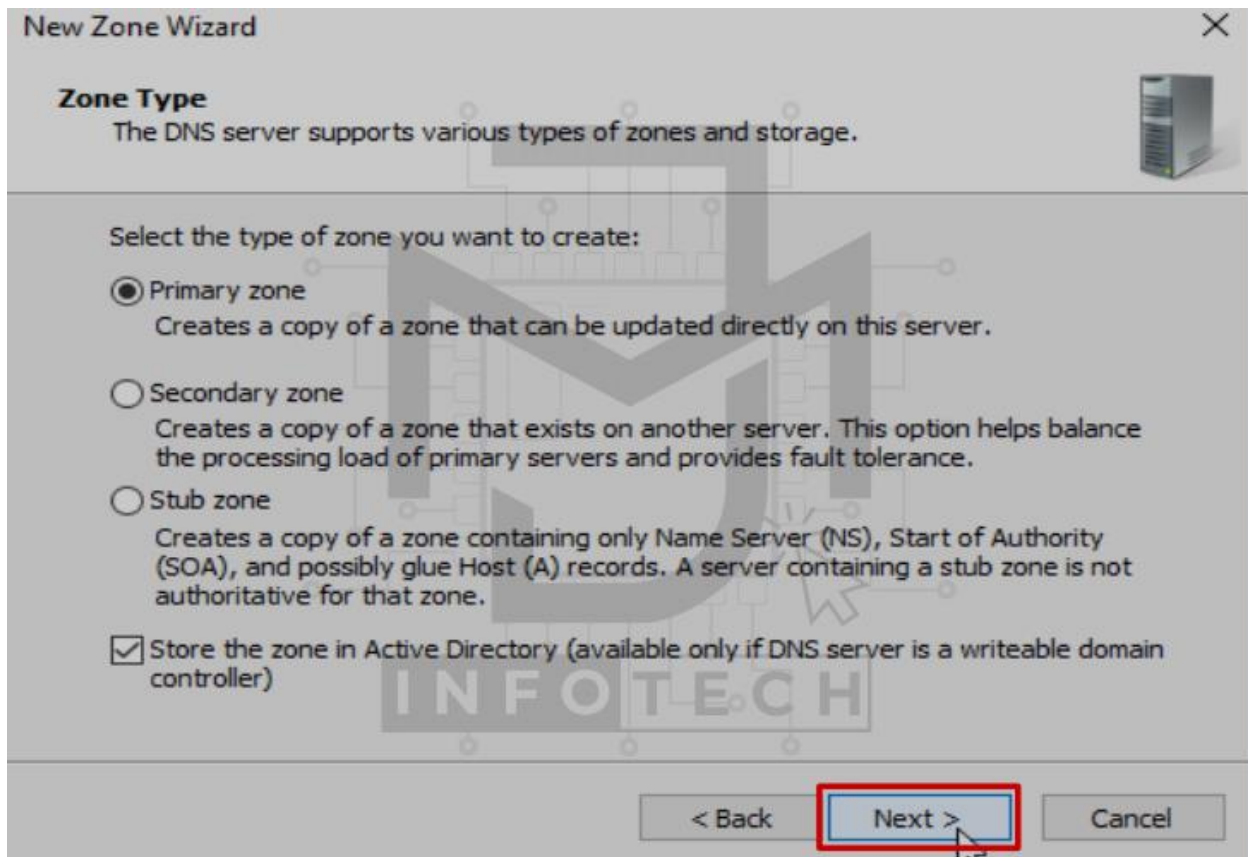


- On the DNS Manager, Expand your server "**DCServer.CSS.Local**" DCServer being the computer name and CSS.Local as the root domain name.
- Right-click on Reverse lookup zone and select New Zone

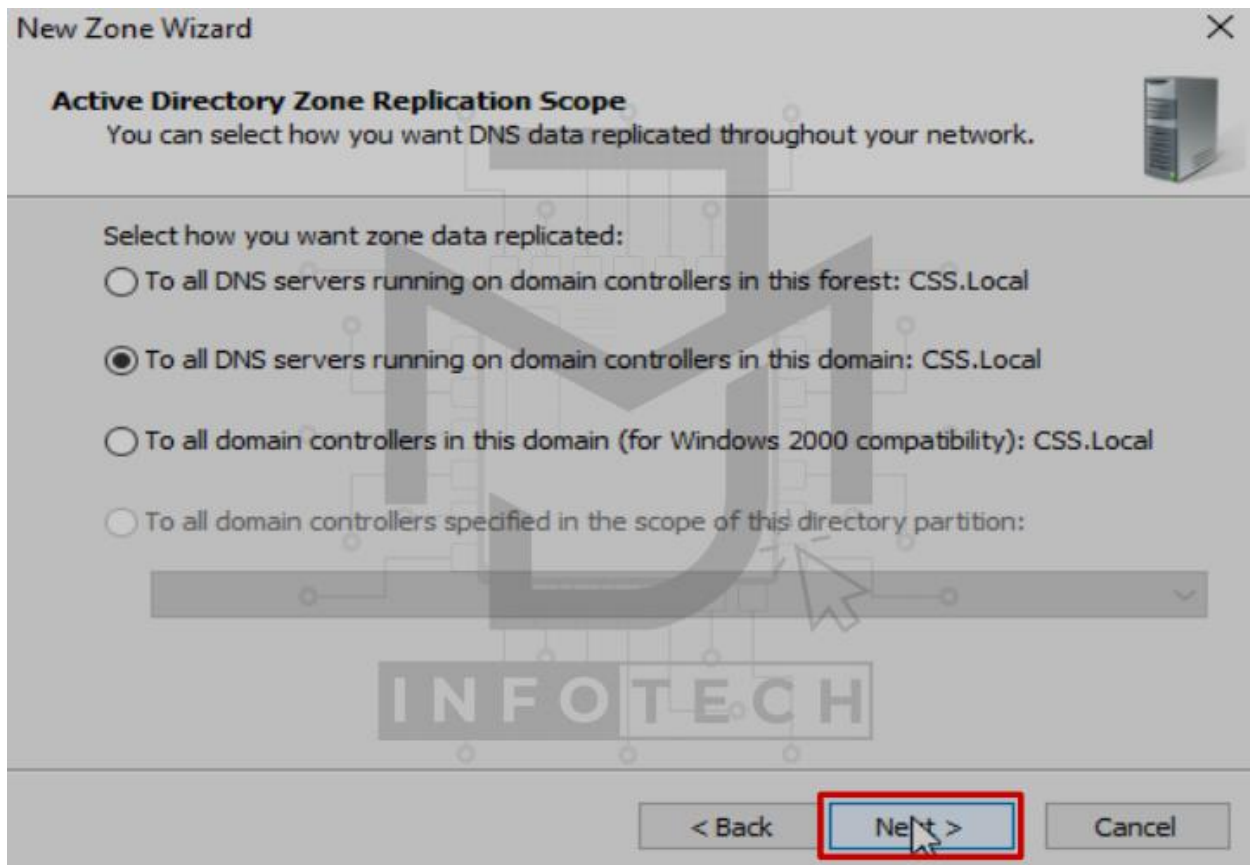


After New Zone Wizard opens up, click next to continue.

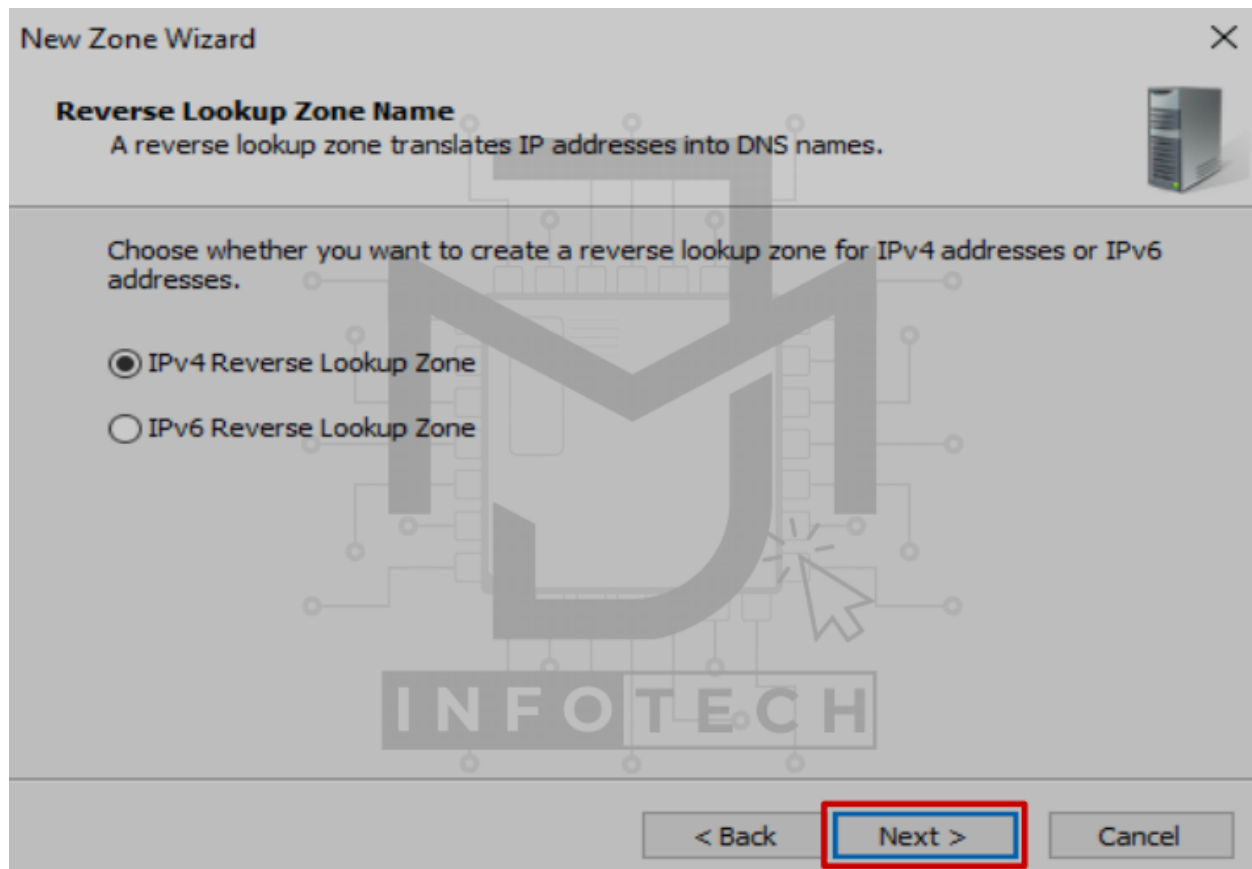
On the Zone Type, Select Primary Zone and click Next to continue.



**On the Active Directory Zone Replication Scope, select
To all DNS servers running on domain controllers in this domain:
CSS.Local, and click Next to continue.**

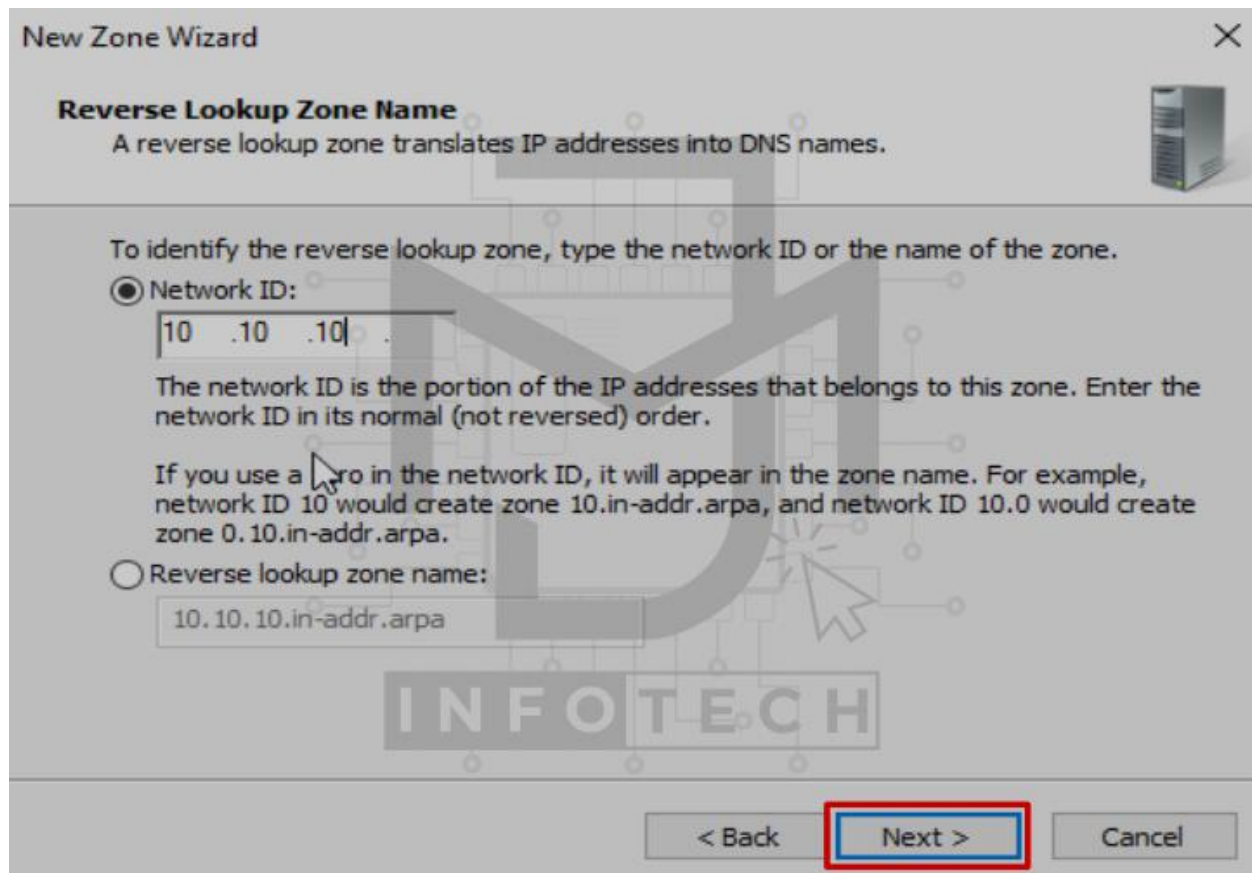


**On the Reverse Lookup Zone Name, select
IPv4 Reverse Lookup Zone,
and click Next to continue.**

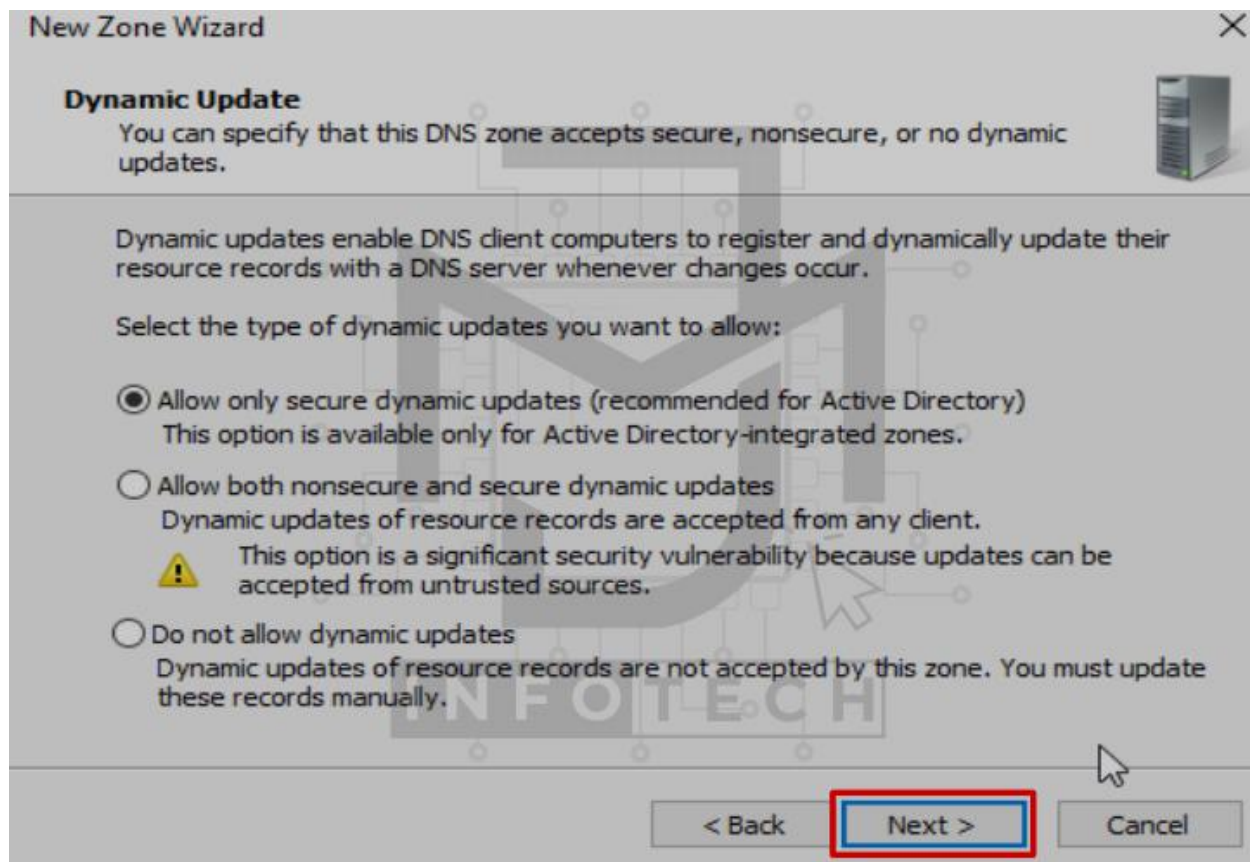


On the Network ID,

Type the first 3 octets of your IP address,
e.g. 10.10.10 and click Next to continue.



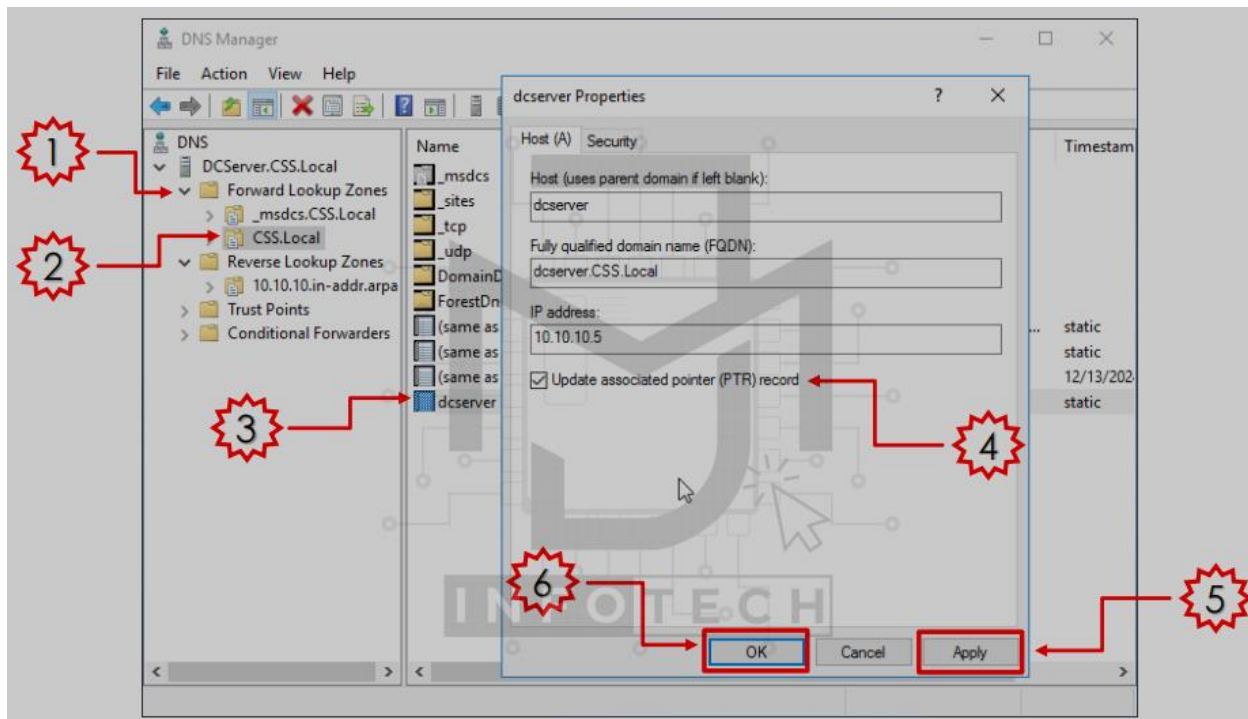
On the Dynamic Update select
Allow only secure dynamic updates
(recommended for Active Directory)
and click Next to continue.



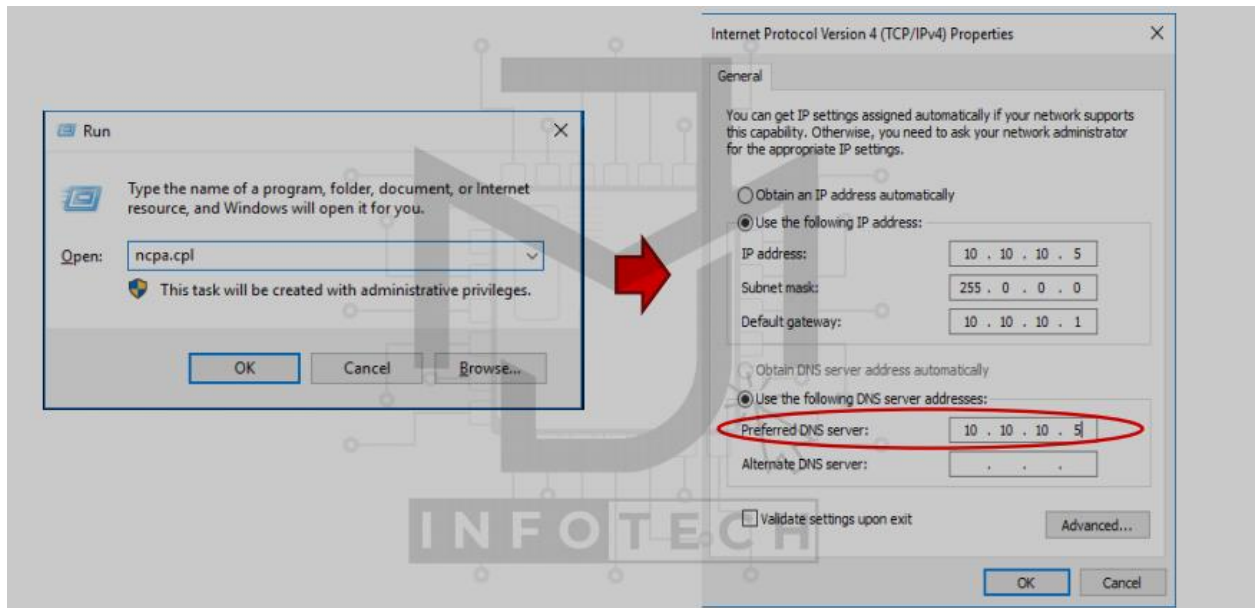
After process a window will pop-up saying Completing the New Zone Wizard, click Finish to continue.

- **On the DNS Manager, Expand Forward Lookup Zones**

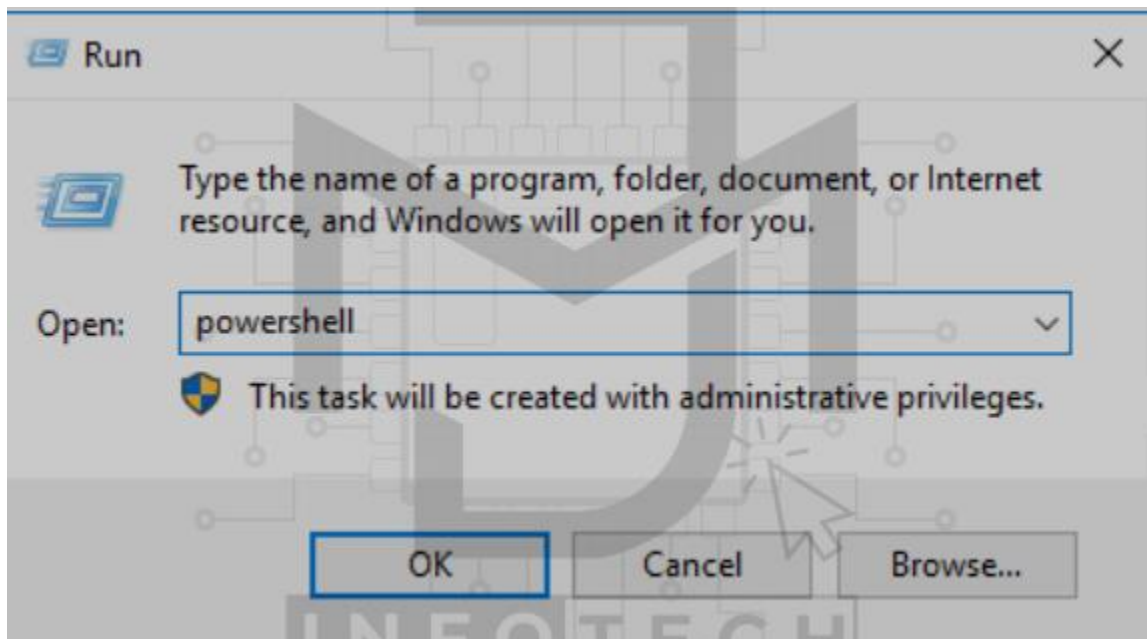
- **Expand CSS.Local**
- **Right click your server “dcserver” and select properties**
- **Tick on Update associated pointer (PTR) record**
- **Click Apply and OK to finish.**



- **Open the Run box using Win+R, type ncpa.cpl, and click OK**
- **Right click on the Ethernet and select Properties**
- **Select Internet Protocol Version 4 (TCP/IPv4) and select Properties**
- **On the Use the following DNS server addresses, from 127.0.0.1 change it to 10.10.10.5 (must be your server IP address)**
- **Click OK to finish.**



To test your **DNS Server** open **Powershell** by clicking windows button in your keyboard OR clicking windows + R buttons and then the Run windows will pop-up type powershell.



- 1st type **nslookup**, and press enter, and wait for reply
- 2nd type your domain name "**css.local**" without quotes, and wait for reply
- 3rd type your server IP address "**10.10.10.5**" without quote, and wait for reply
- The following reply should be as shown on the picture.

The screenshot shows a Windows PowerShell window with the following commands and outputs:

```

PS C:\Users\Administrator> nslookup
Default Server: dcserver.css.local
Address: 10.10.10.5

> css.local
Server: dcserver.css.local
Address: 10.10.10.5
Name: css.local
Address: 10.10.10.5

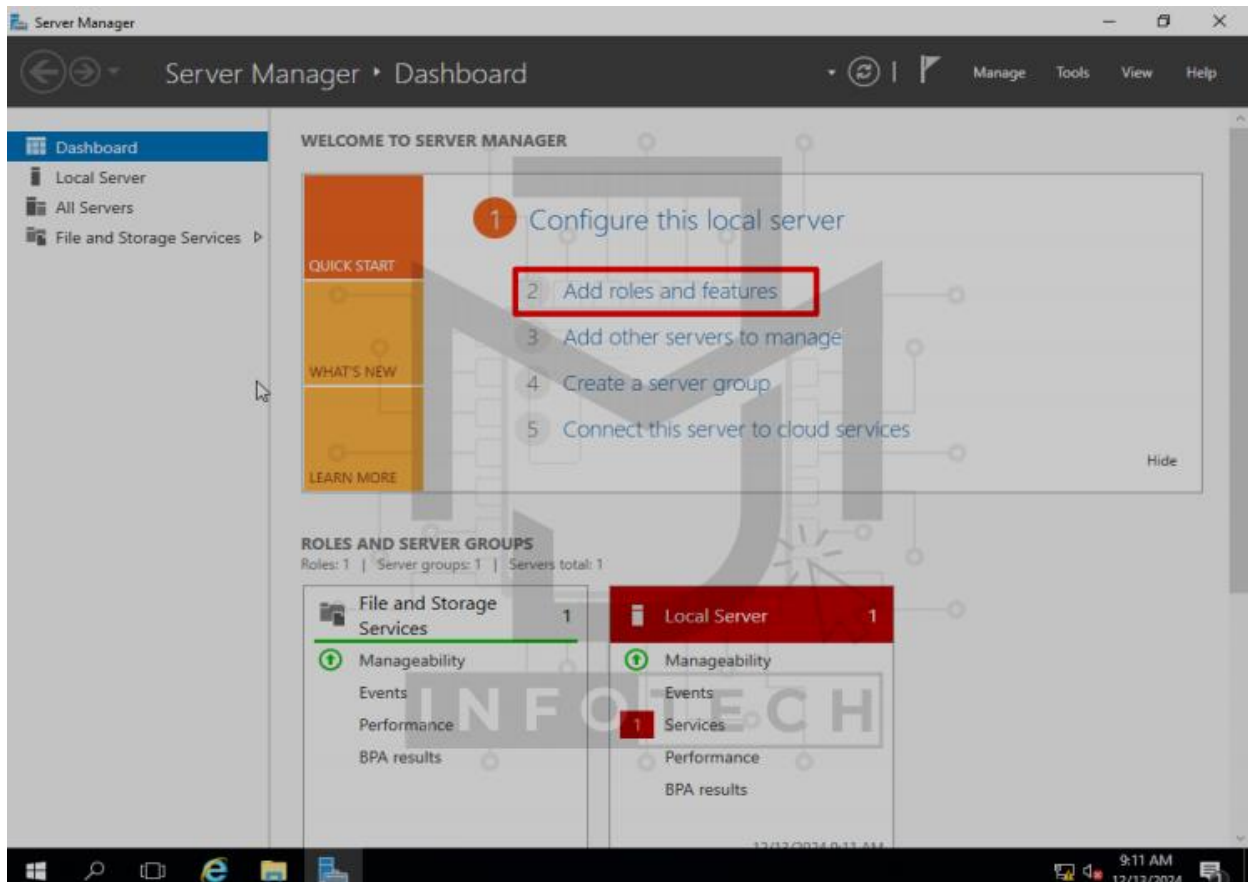
> 10.10.10.5
Server: dcserver.css.local
Address: 10.10.10.5
Name: dcserver.css.local
Address: 10.10.10.5

```

Handwritten annotations in red include:

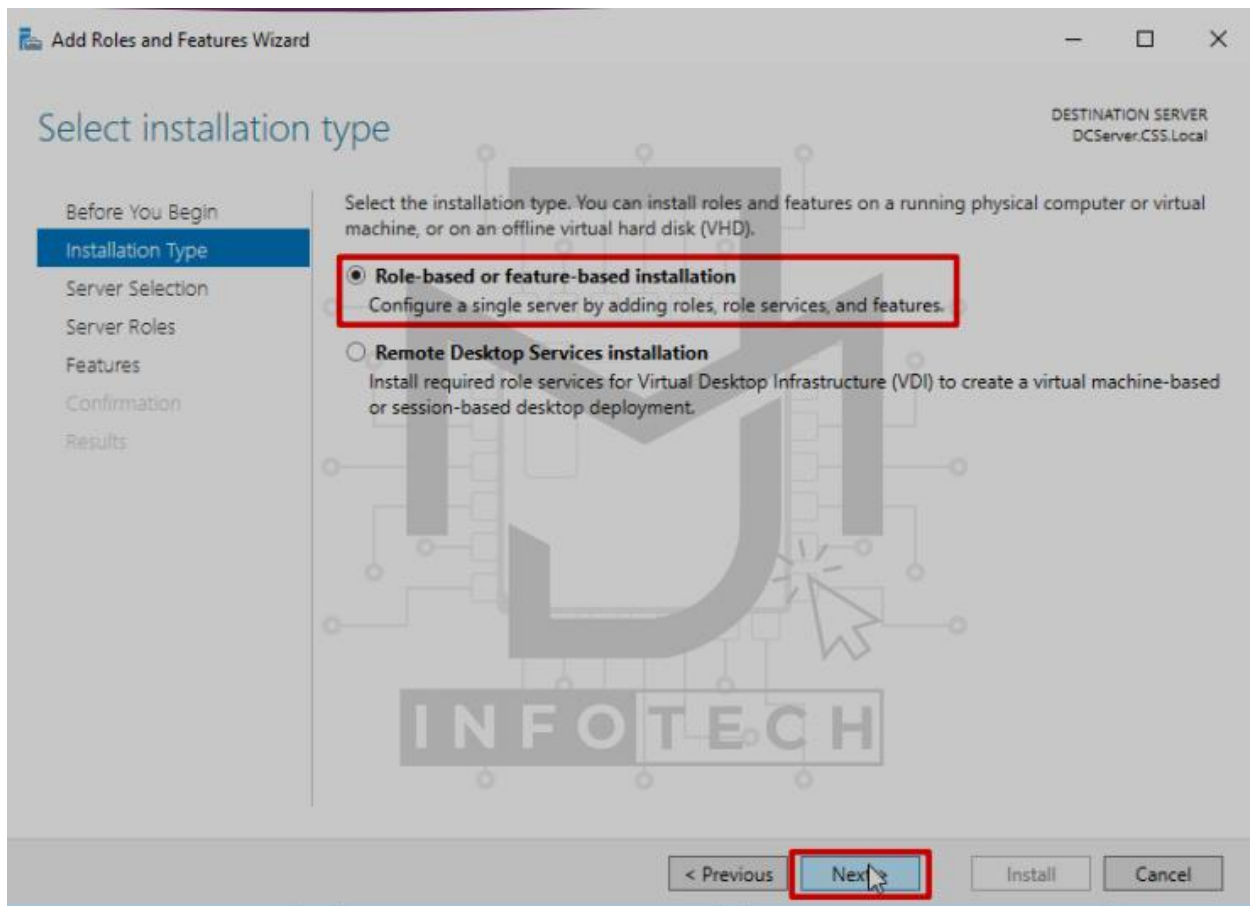
- A box around the initial `nslookup` command and its output, with an arrow pointing to a starburst labeled "1st Reply".
- A box around the `css.local` command and its output, with an arrow pointing to a starburst labeled "2nd Reply".
- A box around the `10.10.10.5` command and its output, with an arrow pointing to a starburst labeled "3rd Reply".

- Search Server Manager in the Start Menu and click it
- In the Server Manager – Dashboard Click Add roles and features

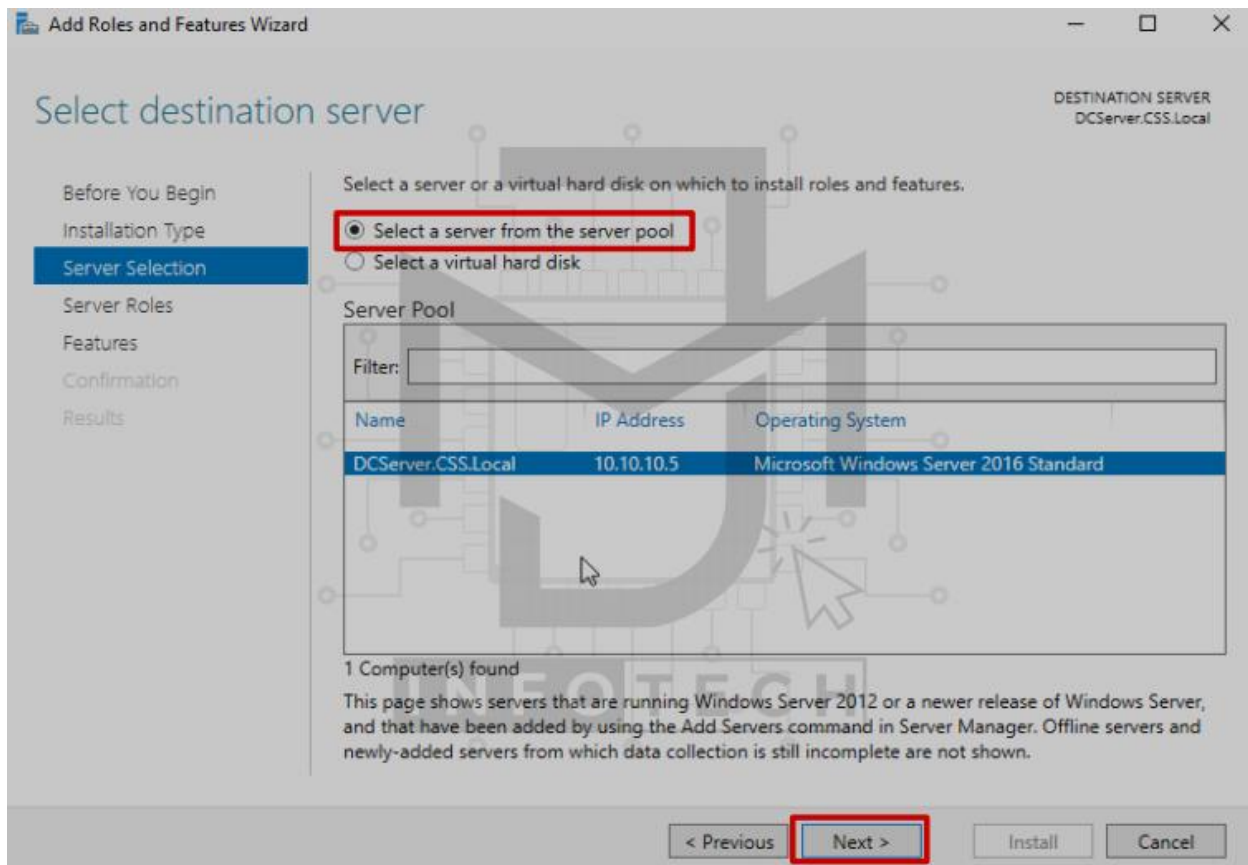


In the Before you begin, click Next to continue..

In installation Type select Role Base or feature-based installation, then click next to continue.

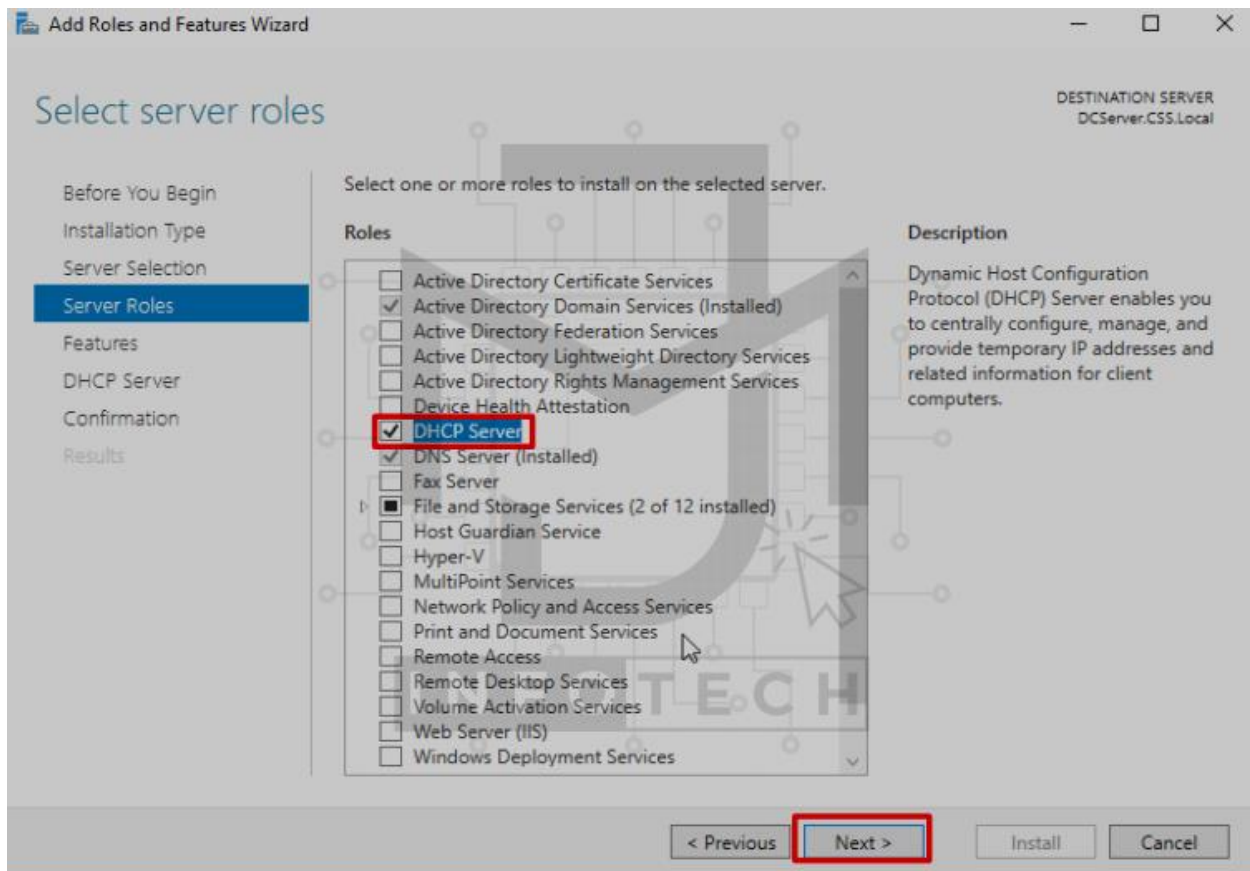


- On the Server Selection, let the Select a server from the server pool, selected
- Click Next to continue

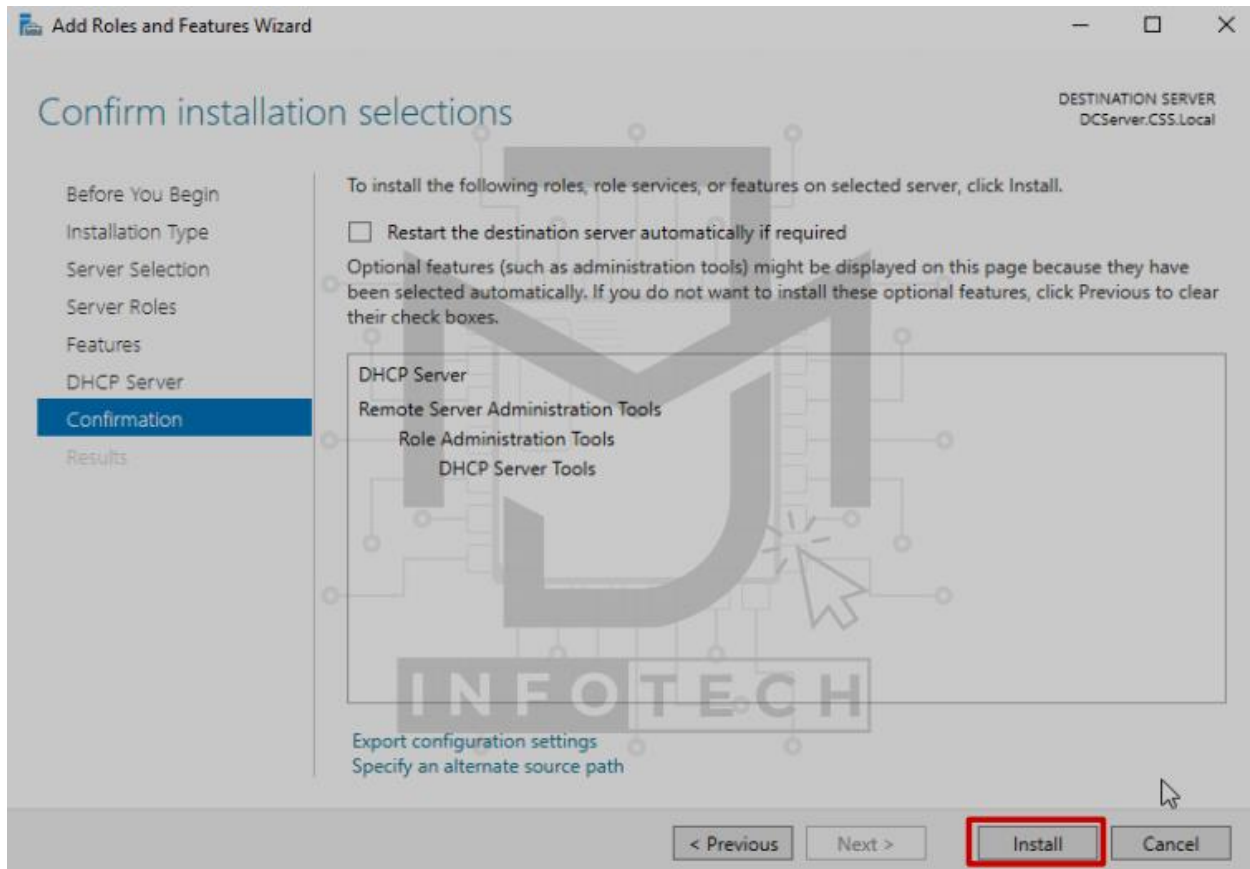


On the Server Roles,
Tick the following Roles - DHCP Server, then add features.

Click Next to continue

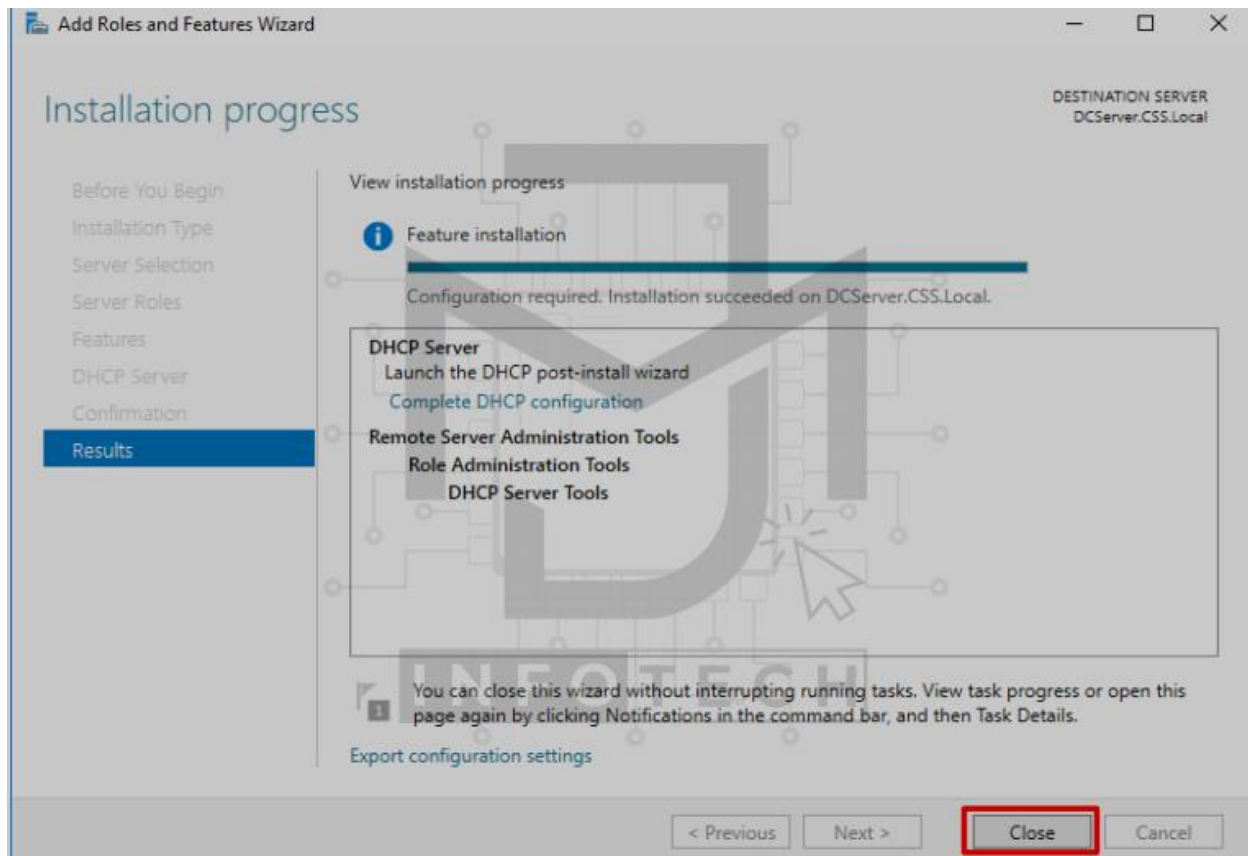


On the features, click Next to continue and do the same for DHCP Server Then at the confirmation click Install.



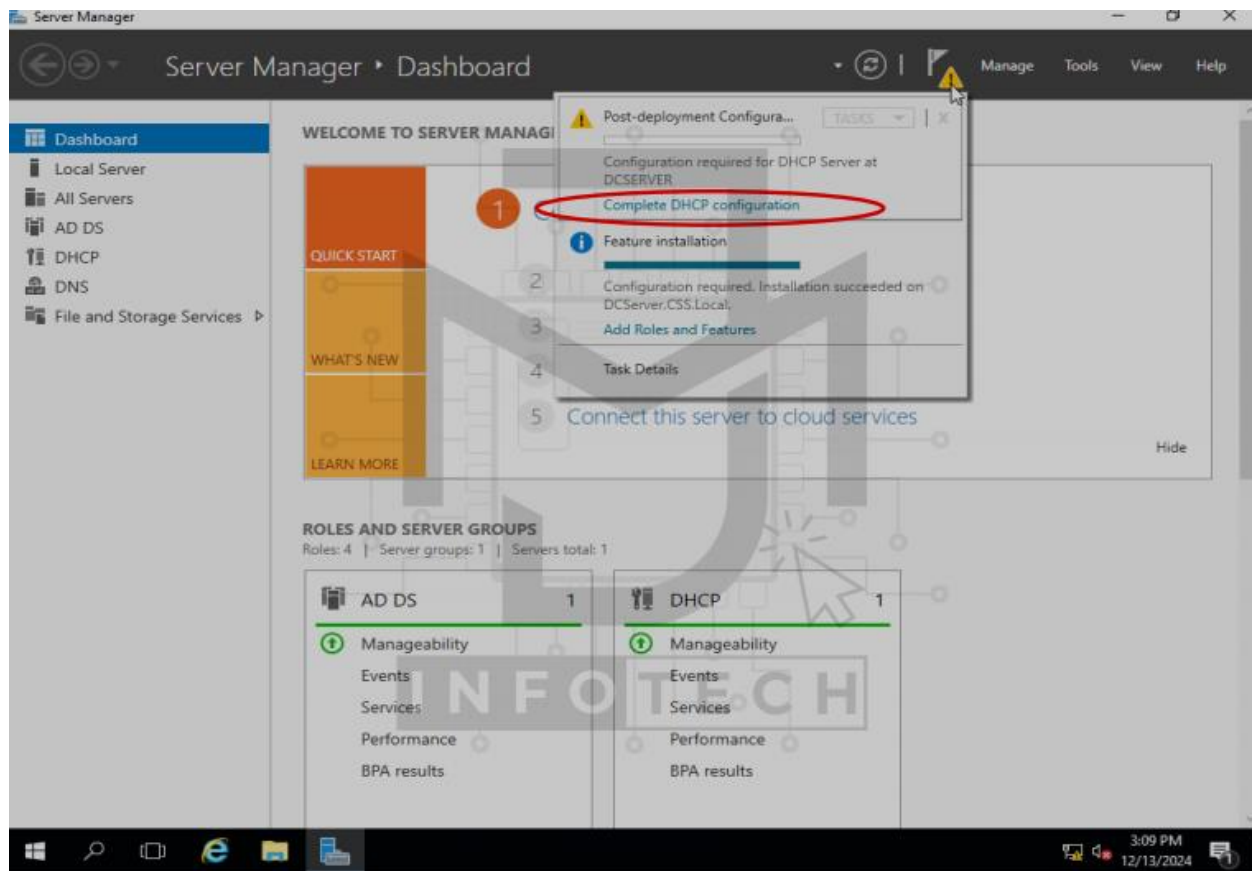
41

- On the Results, just wait until the installation progress is completed, depending on your system, it takes 3 to 5 minutes to complete the installation.
- Click Close to exit



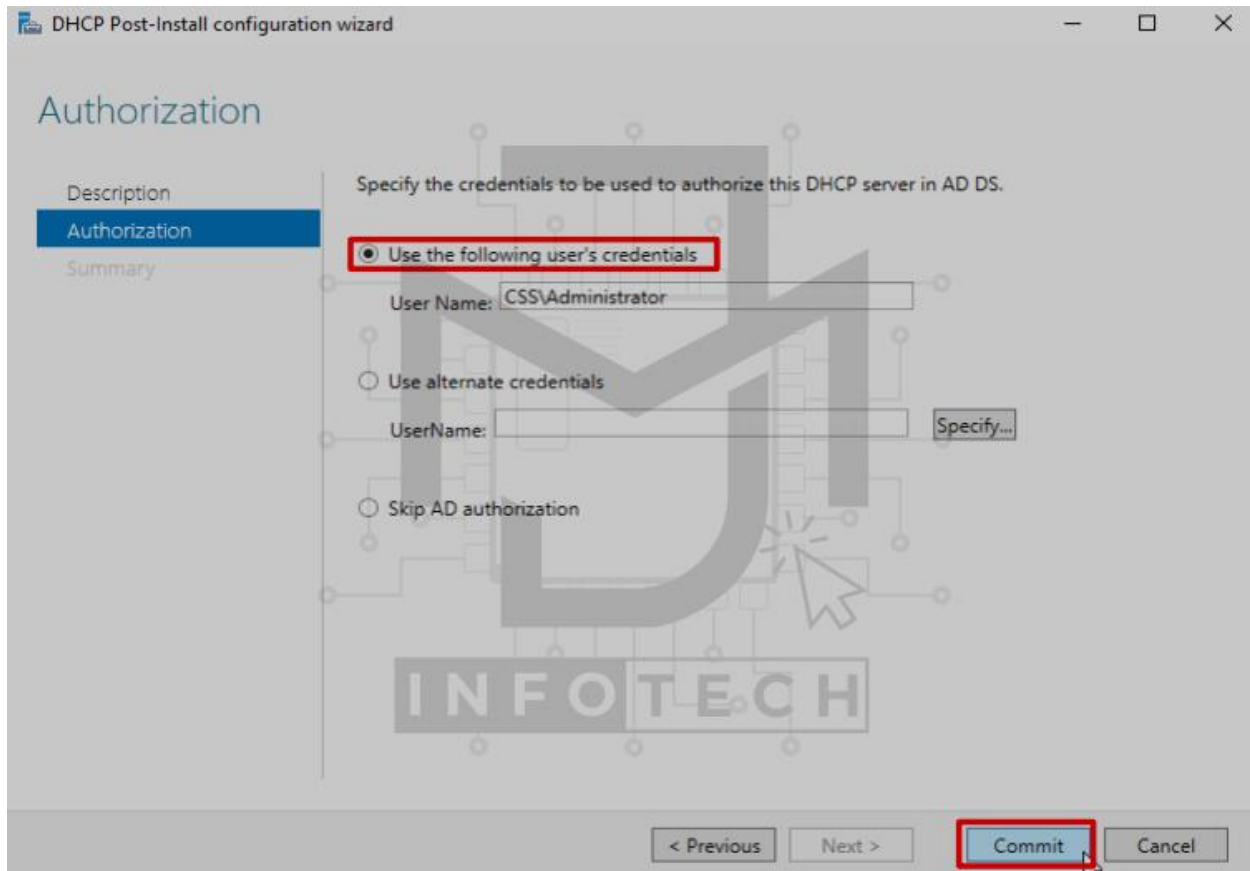
42(palatandaan) pls wag burahin - Andy

- In the server manager dashboard, you will see a yellow notification at the top right.
- Click on it.
- Now click on "Complete DHCP configuration"



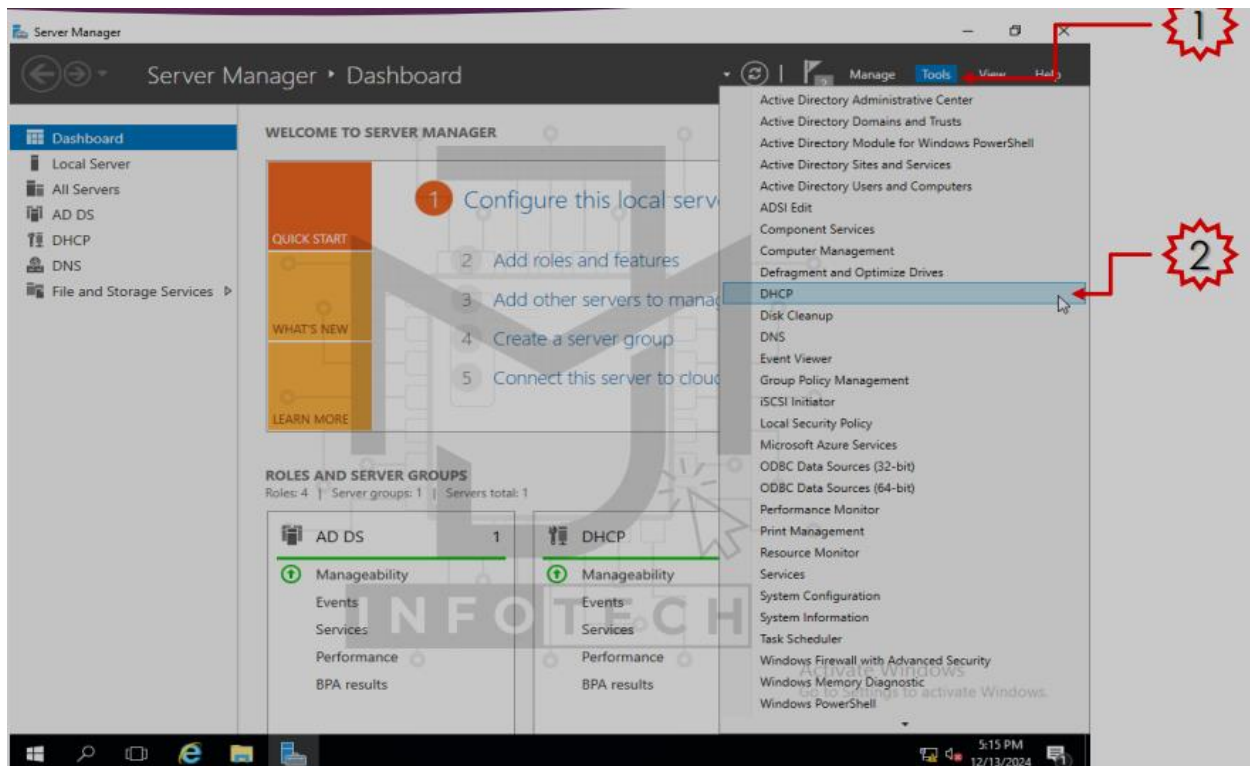
On the Description window click next to continue..

- On the authorization page, select "Use the following user's credentials"
- Click commit to continue

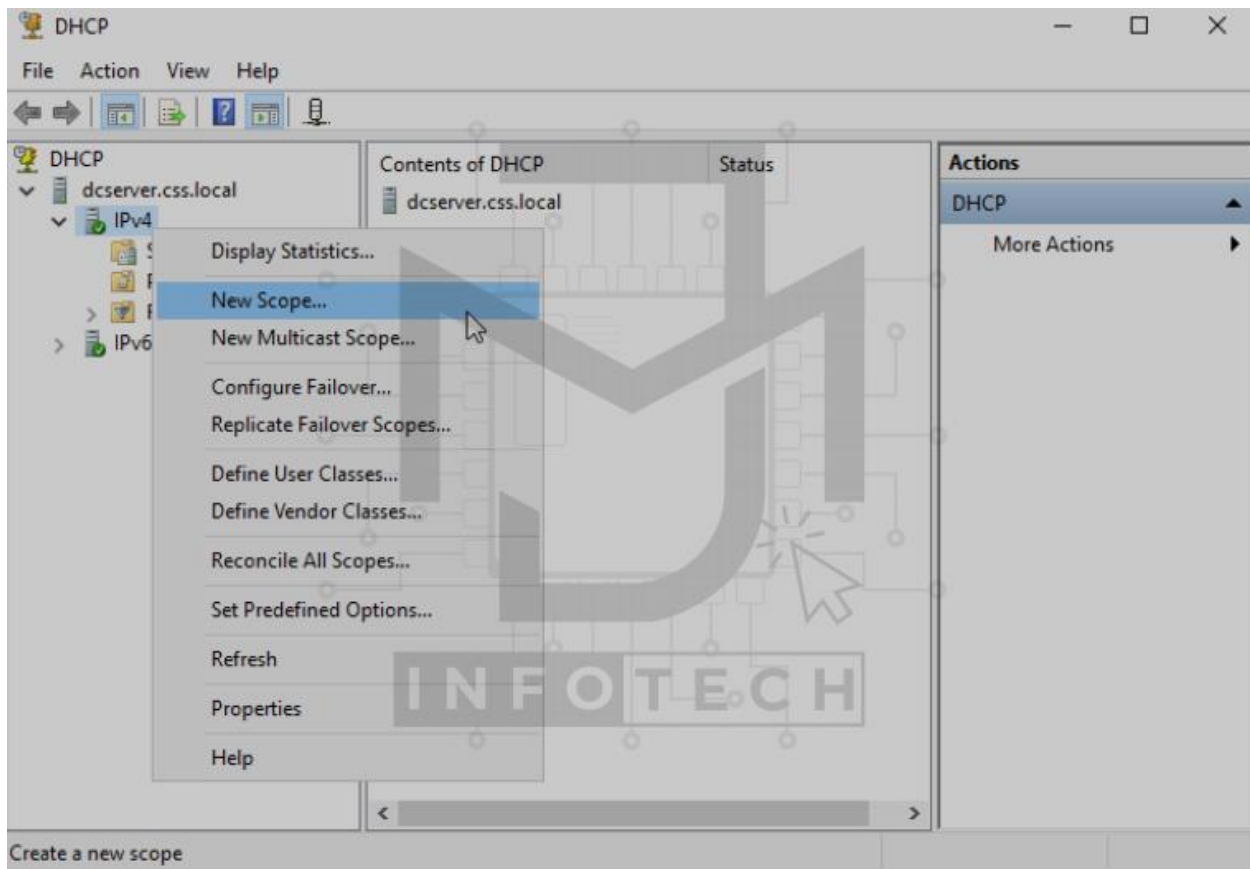


You will see a summary window just click close to continue.

In the server manager dashboard, click tools and select DHCP

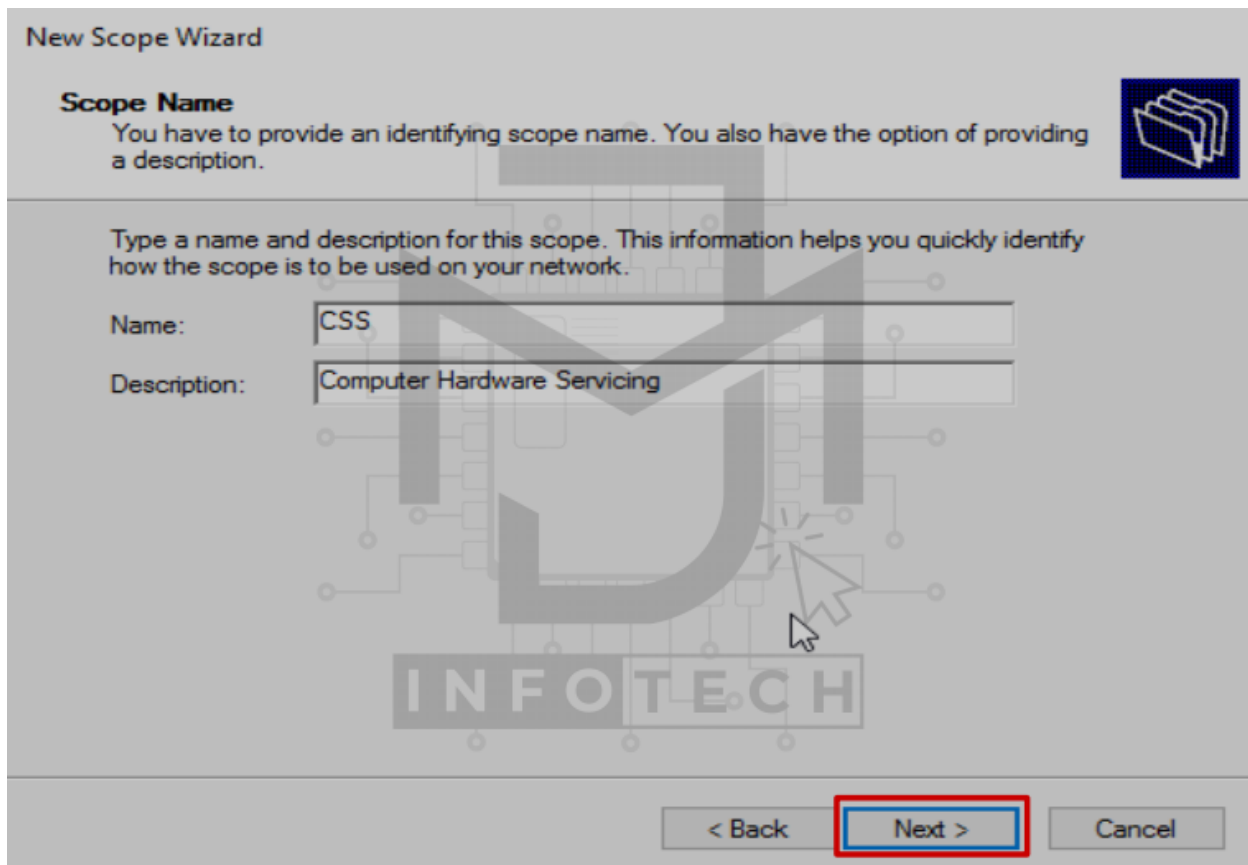


When the DHCP opened, expand both DHCP server name and IPv4, right-click on IPv4 then click on the **New Scope** option to create an IP scope for distribution.



In the welcome to the New Scope wizard click next to continue.

- In the Scope Name, type your desired name e.g. CSS
- In the description portion, you can put anything you want or leave it blank.
- Click Next to continue



New Scope Wizard

Scope Name
You have to provide an identifying scope name. You also have the option of providing a description.

Type a name and description for this scope. This information helps you quickly identify how the scope is to be used on your network.

Name:

Description:

< Back **Next >** Cancel

- Specify the IP Address Range by entering the start IP Address and End IP address. The length automatically assigns, based on you IP address class and the subnet mask assigns automatically.
- In this Demo we are going to Use:
Start IP address: 10.10.10.10
End IP address: 10.10.10.200

- Click Next to continue

New Scope Wizard

IP Address Range
You define the scope address range by identifying a set of consecutive IP addresses.

Configuration settings for DHCP Server
Enter the range of addresses that the scope distributes.

Start IP address: 10 . 10 . 10 . 10
End IP address: 10 . 10 . 10 . 200

Configuration settings that propagate to DHCP Client

Length: 8
Subnet mask: 255 . 0 . 0 . 0

< Back Next > Cancel

- You can exclude some of the IP addresses for the servers or especial services. The IPs excluded will not be assigned automatically. IPs before and after the exclusion range are assigned. IPs excluded are assigned on servers manually. Specify the IP range you want to exclude.
- In this Demo we are going to leave it blank
- Click Next to Continue

New Scope Wizard

Add Exclusions and Delay

Exclusions are addresses or a range of addresses that are not distributed by the server. A delay is the time duration by which the server will delay the transmission of a DHCP OFFER message.

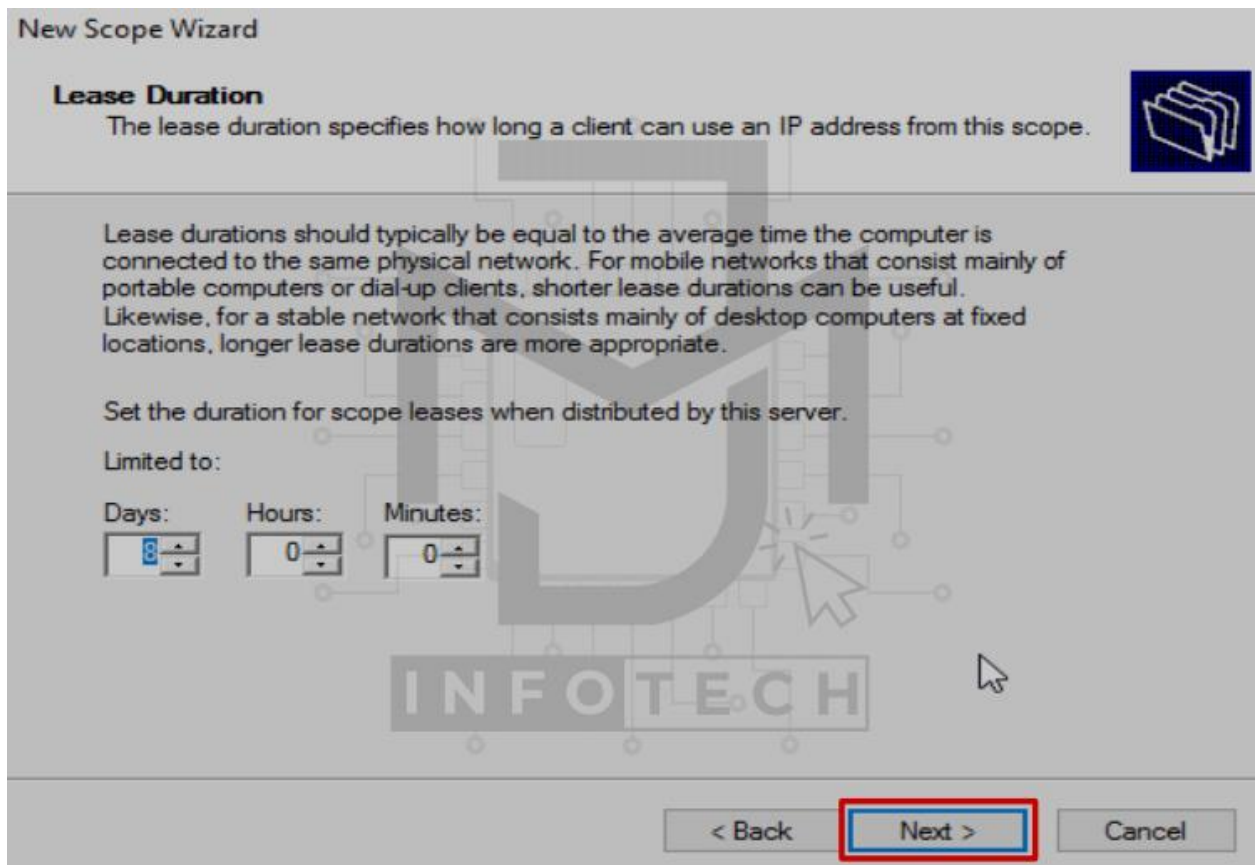
Type the IP address range that you want to exclude. If you want to exclude a single address, type an address in Start IP address only.

Start IP address: End IP address:

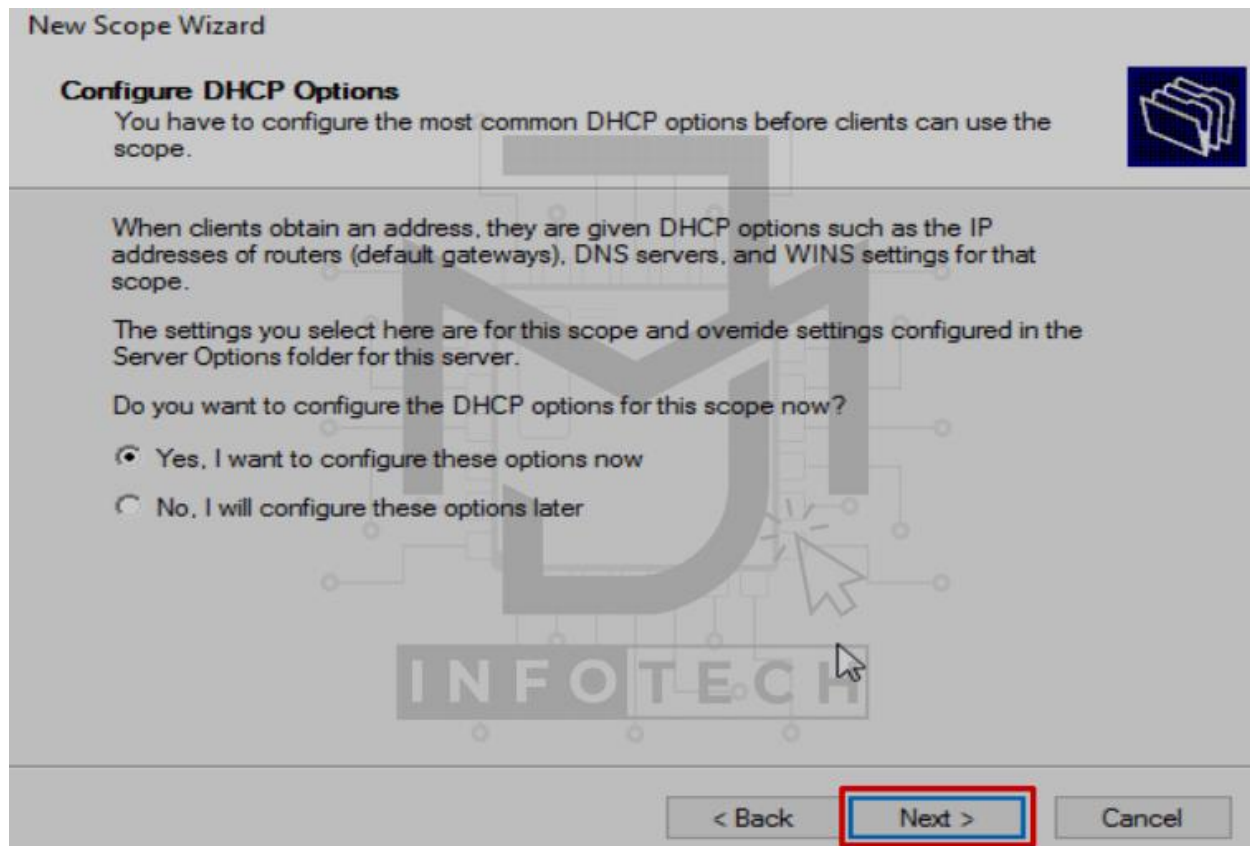
Excluded address range:

Subnet delay in milli second:

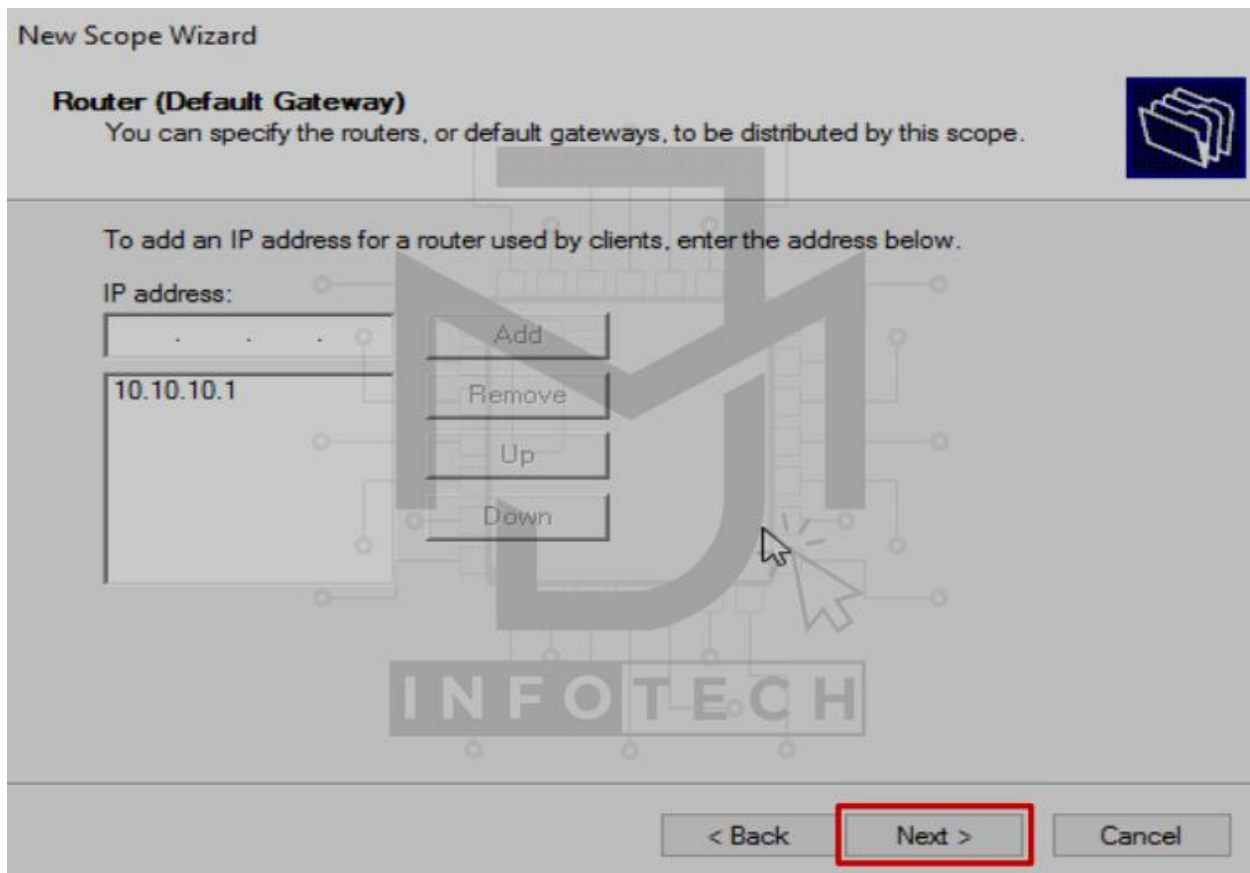
- On the Lease Duration window, we can specify how long we want the DHCP lease to last. The lease is how long the client can keep the TCP/IP settings before it needs to come back to the DHCP Server for a new lease or configuration.
- We are keeping the default settings of 8 Days. Click Next to continue.



- On the Configure DHCP option page, select this option: select Yes, I want to configure these options now.
- Click Next to continue



- Enter your Router's IP address in the box and click on add button to add the IP address.
- Click on Next button.



- DNS (Domain Name System) server translates the name to IP address and IP address to name. Type the DNS server's name in the box. Then type IP address in the box below then click on Add button to add to the IP address.
- Click Next to continue

New Scope Wizard

Domain Name and DNS Servers
The Domain Name System (DNS) maps and translates domain names used by clients on your network.

You can specify the parent domain you want the client computers on your network to use for DNS name resolution.

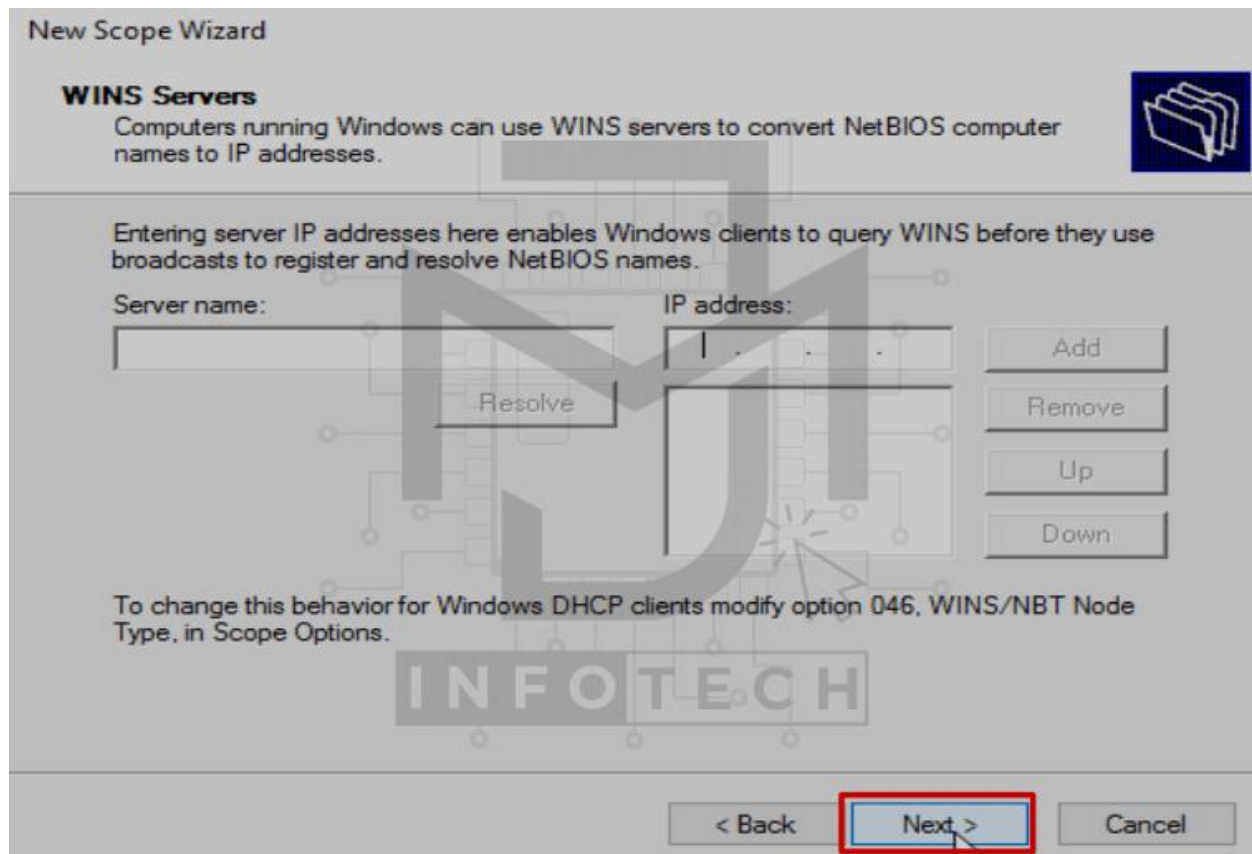
Parent domain:

To configure scope clients to use DNS servers on your network, enter the IP addresses for those servers.

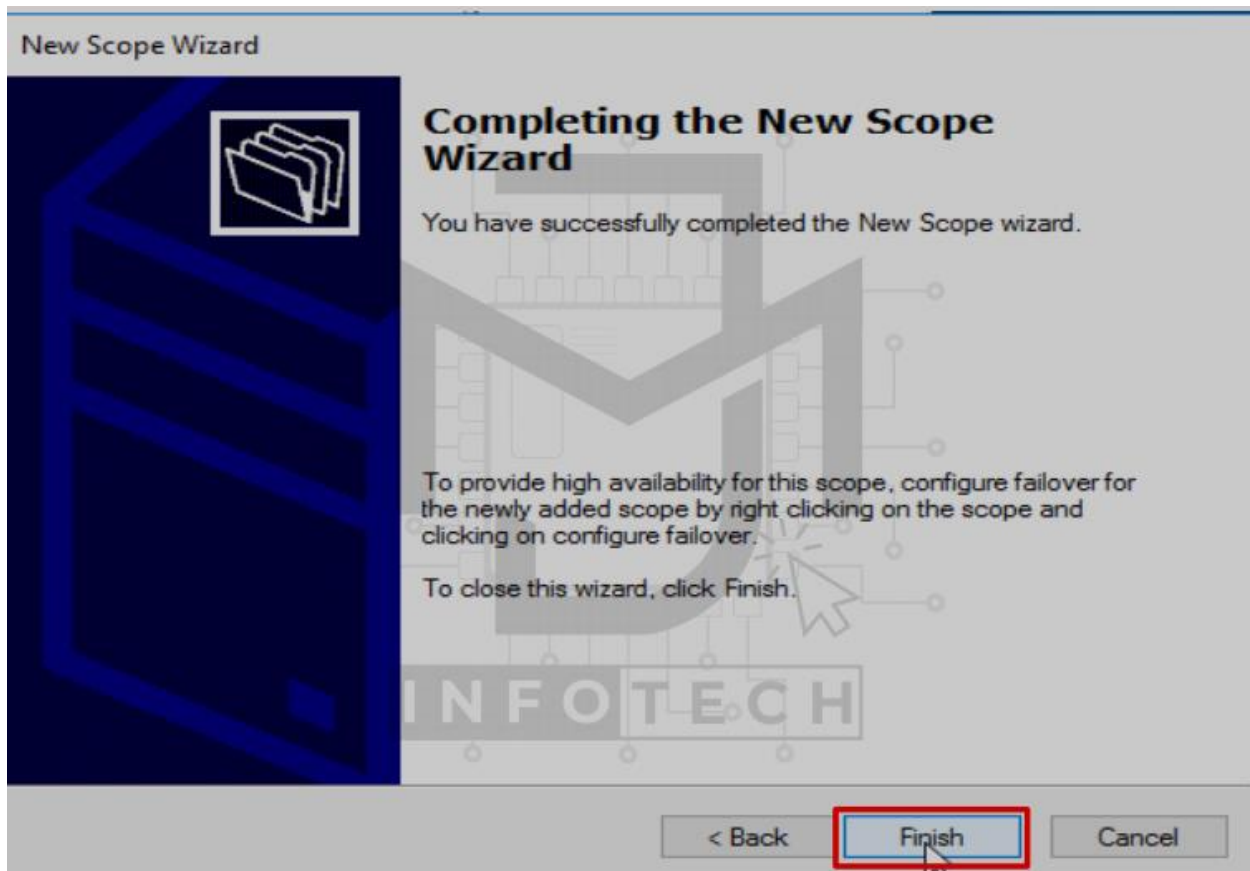
Server name:	IP address:	
dcserver	10.10.10.5	Add
Resolve		Remove
		Up
		Down

< Back **Next >** Cancel

- WINS Server translates NetBIOS computer name to IP address.
Type WINS server name in Server name box. Type the WINS server's IP address and click on Add button.
- In this Demo we going to leave it blank.
- Click Next to continue



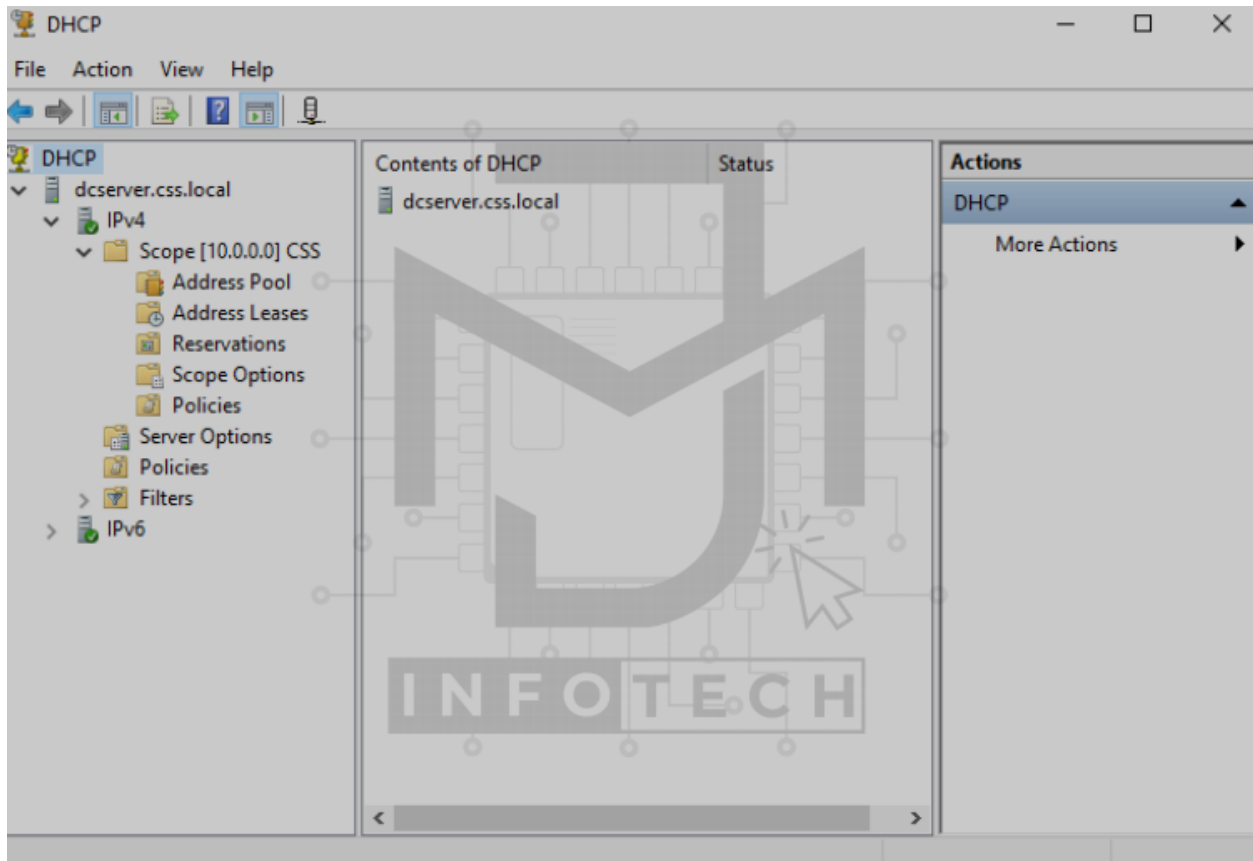
Finish the wizard by clicking on the finish button.



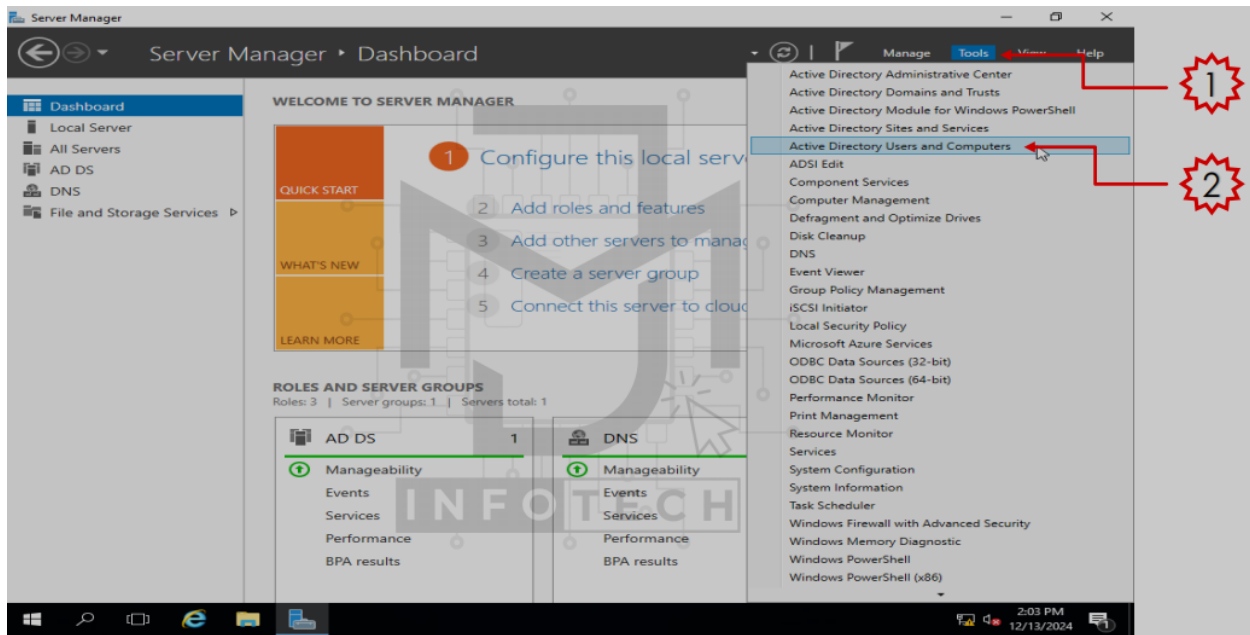
We are done with the creation of new scope. It is the time to know more about DHCP server console. Below the DHCP server name, you see IPv4 and IPv6 options. It means that Windows server 2016 can support both IPv4 and IPv6 versions of IP address. Expand IPv4 to see the scope you have created. Also, Server Options, Policies and Filters.

Expand the scope. Things you see are these:

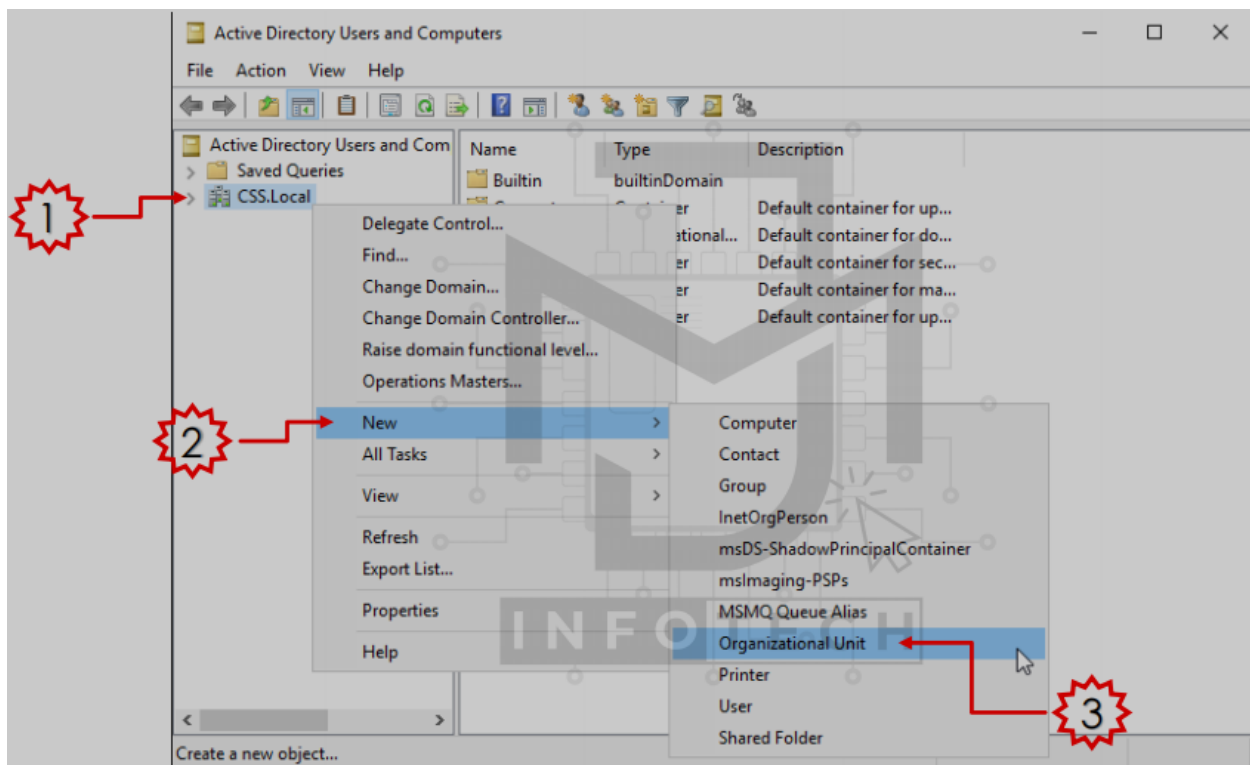
- **Address Pool:** IP address range can be assigned to clients and exclusion range is listed here.
- **Address Leases:** IP address which has been assigned to clients dynamically are listed here. IP address which has been assigned manually won't be listed here.
- **Reservations:** Reserved IP range is listed here.
- **Scope Options:** Some options like Router, PXE client and etc.
- **Policies:** Policies created for DHCP server are listed here.



On the Server Manager - Dashboard, click Tools, and Select **Active Directory Users and Computers**.

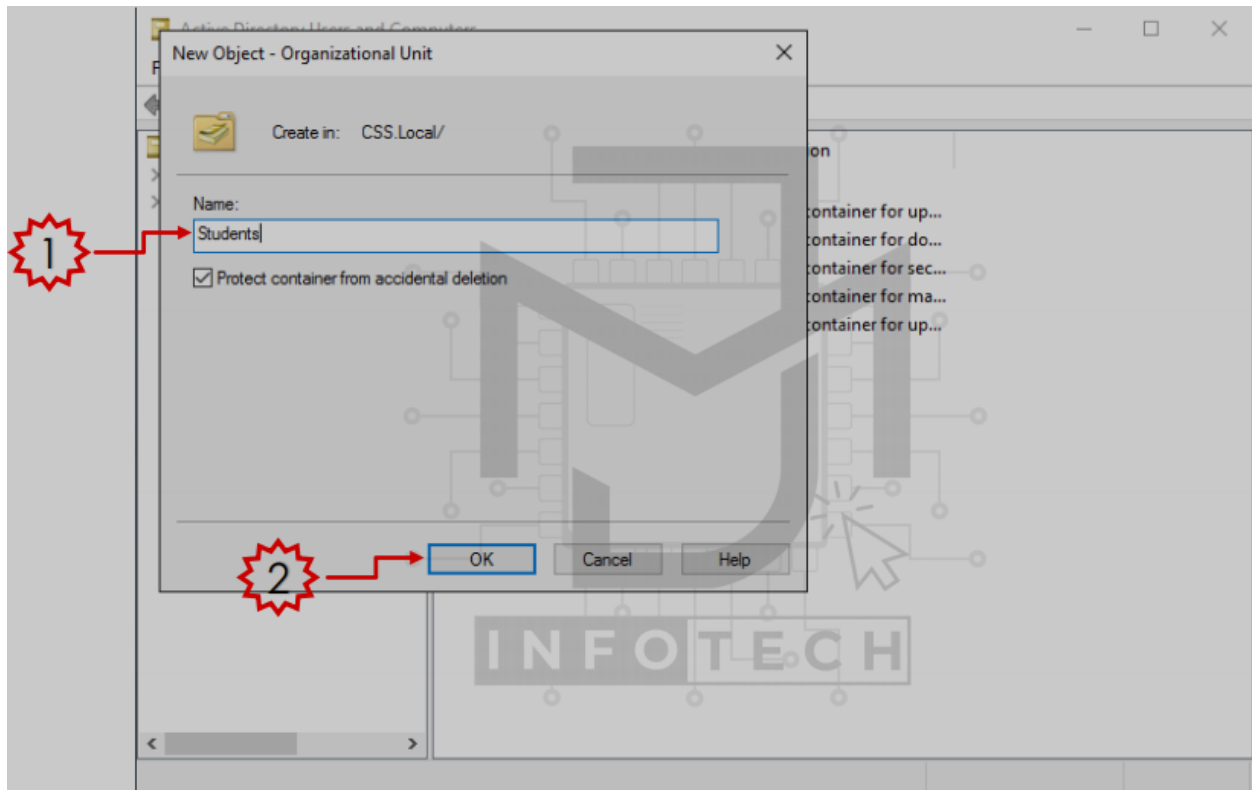


Right click on your **Domain**, select **New**, and choose **Organizational Unit**.

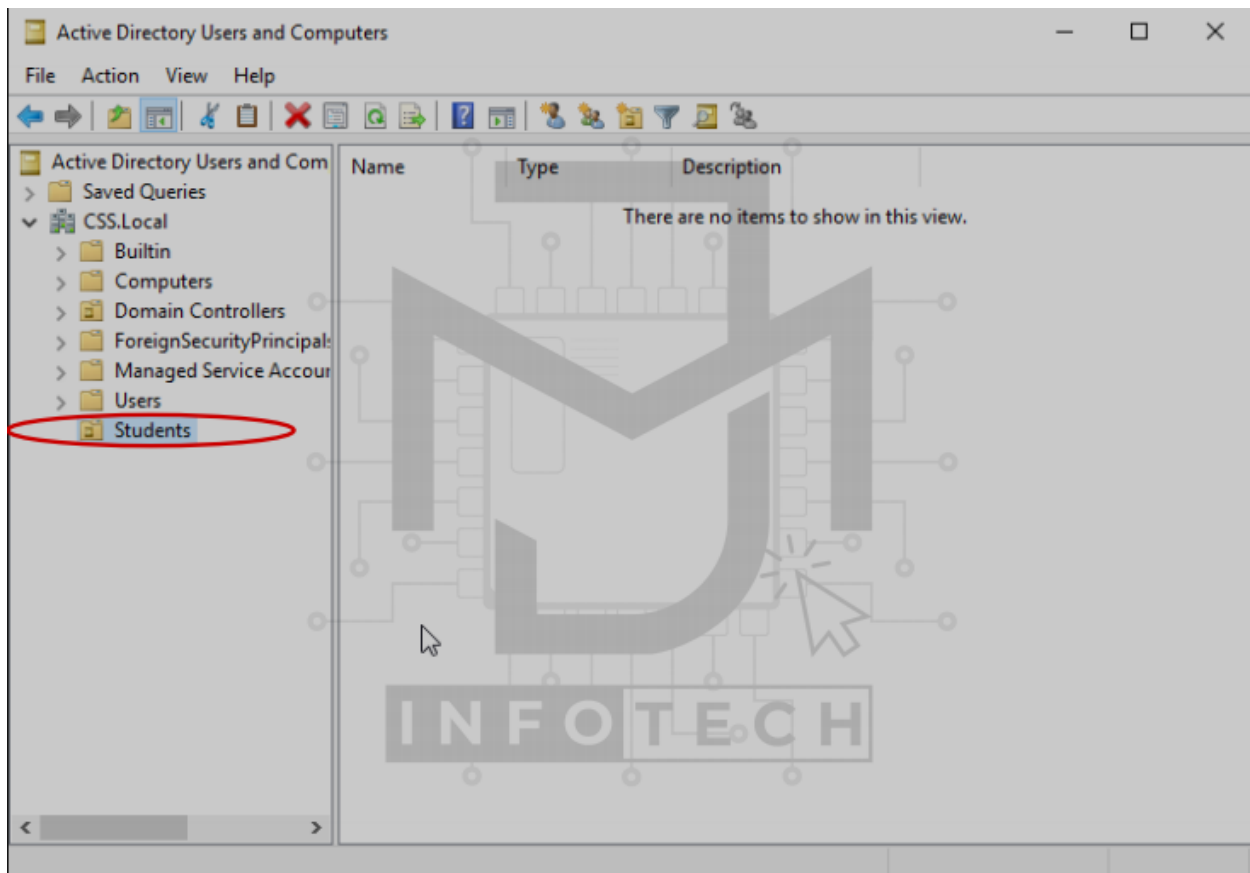


- Here type the name of the Organizational unit you want to create (for example Students).

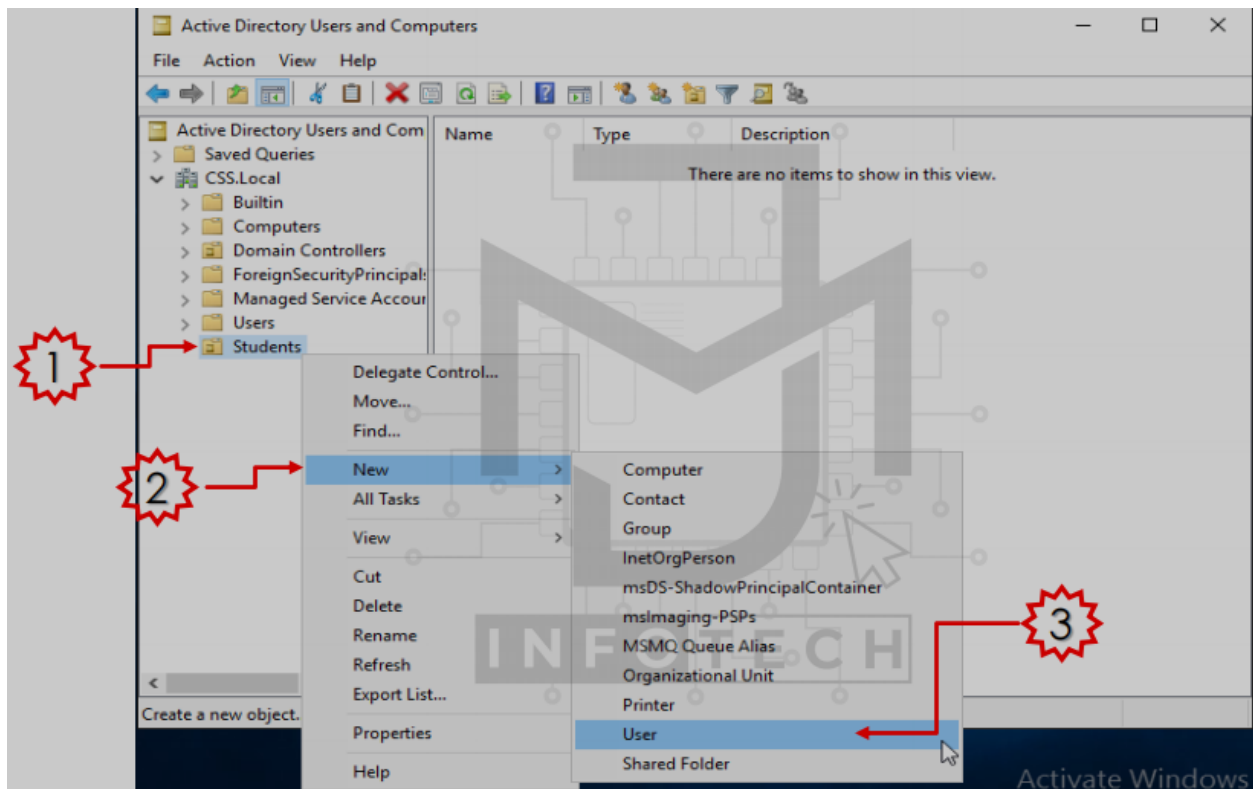
- Select the checkbox of "Protect container from the accidental deletion" option. This option will protect the object from accidental deletion.
- Click OK to finish.



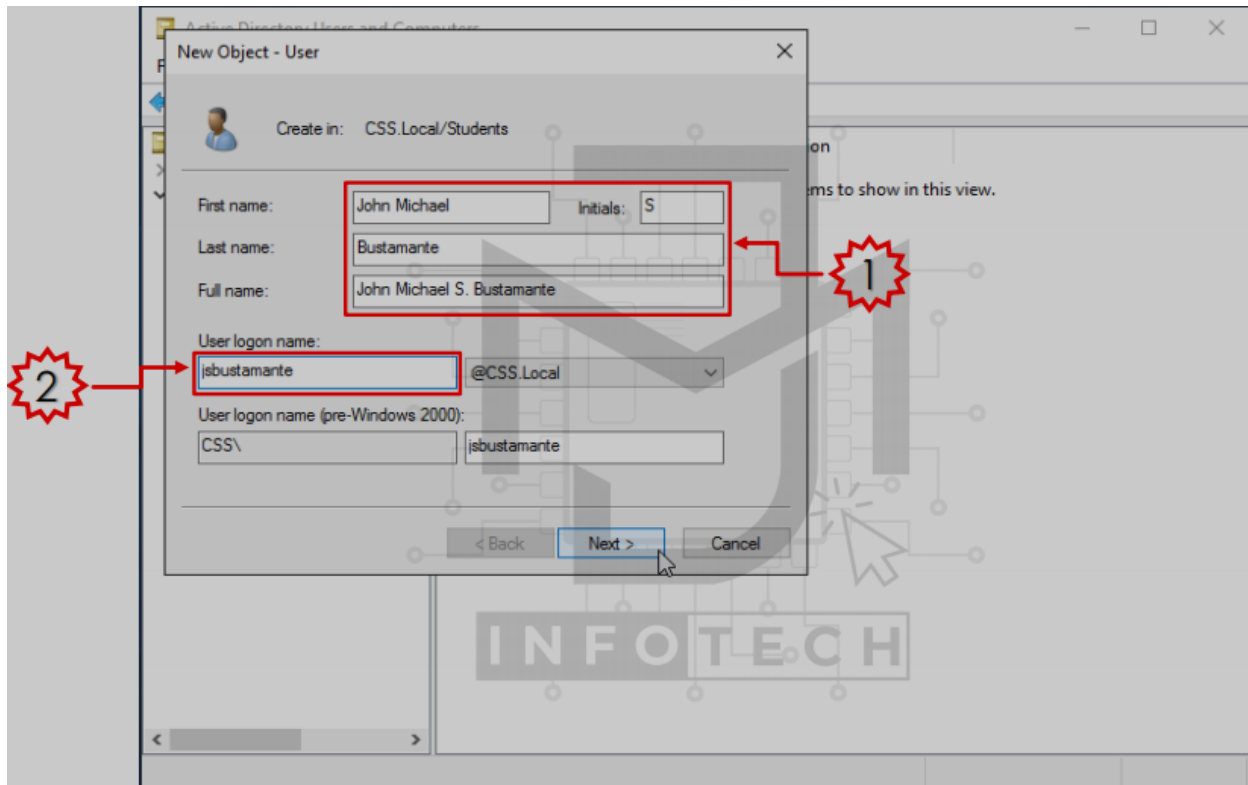
Now OU with the name "Students" has been successfully created.



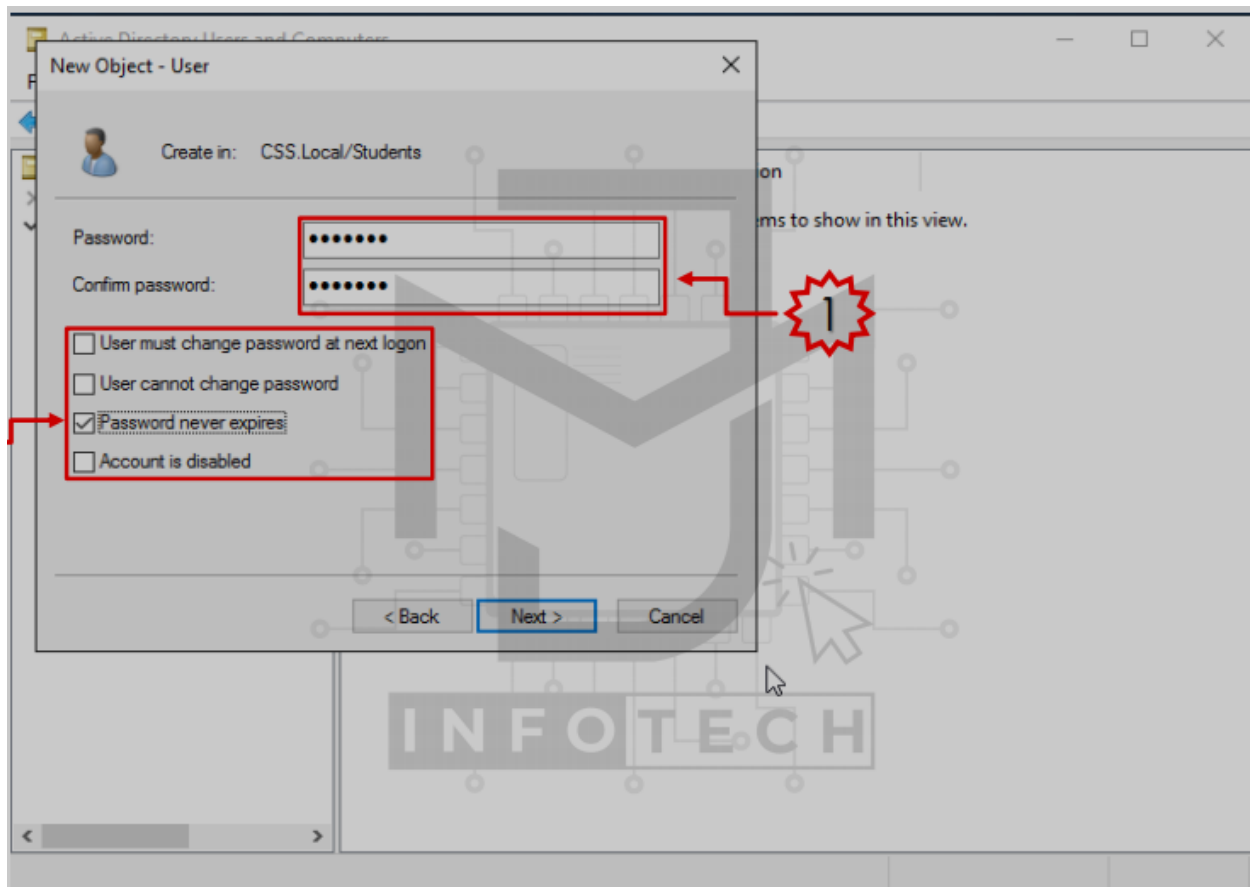
- Now we will add a user to this OU. To create a user, right Click on the "**Students**" OU, choose **New** and then click on "**User**".



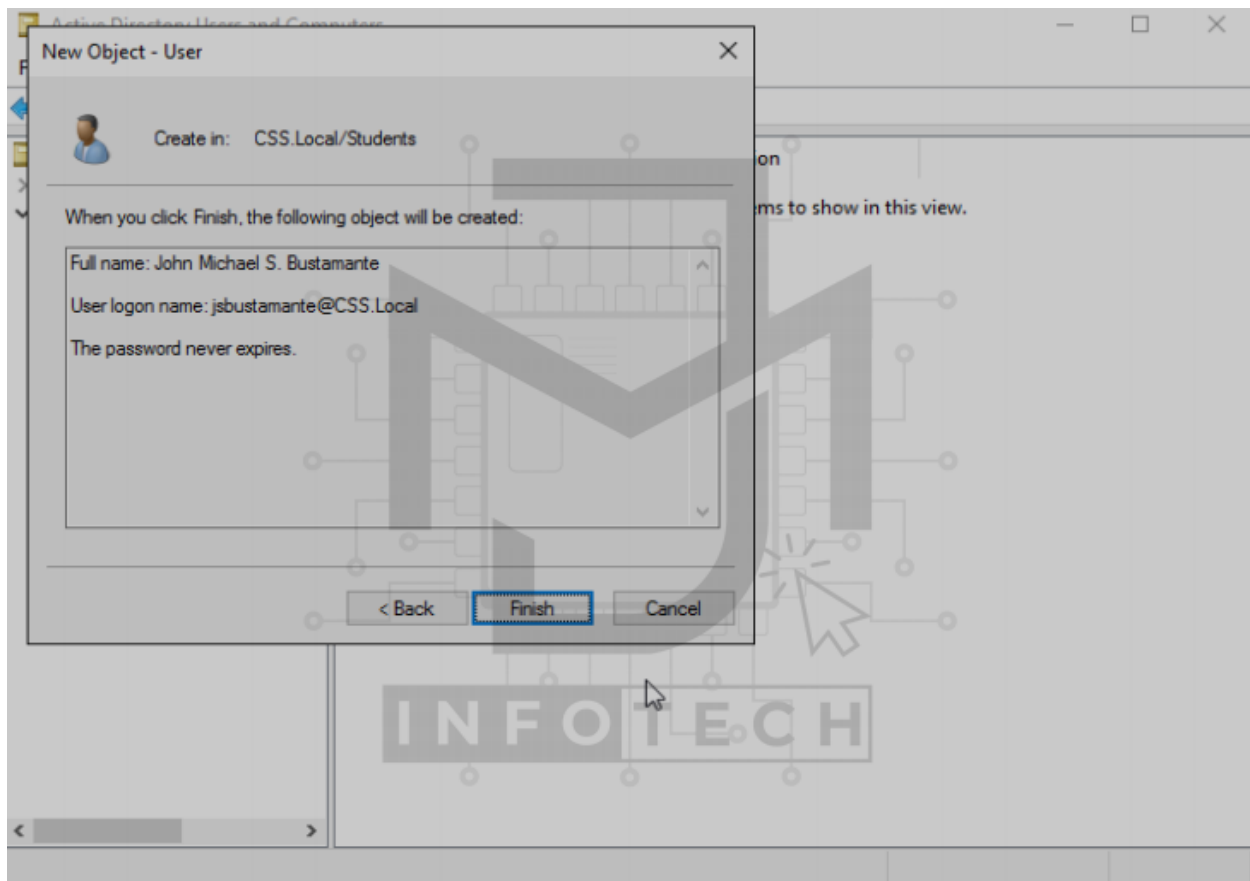
- A new window will appear, here we have to enter the user information.
- Type the user's First name, Middle Initial and Last name.
- On the User logon name type on the "Username" that the user will use to log in to the Computer in the Network.
- if the user wants to log in then he will type "abestipular@CSS. Local" or "CSS.Local /abestipular" in the username field.
- Click Next to continue



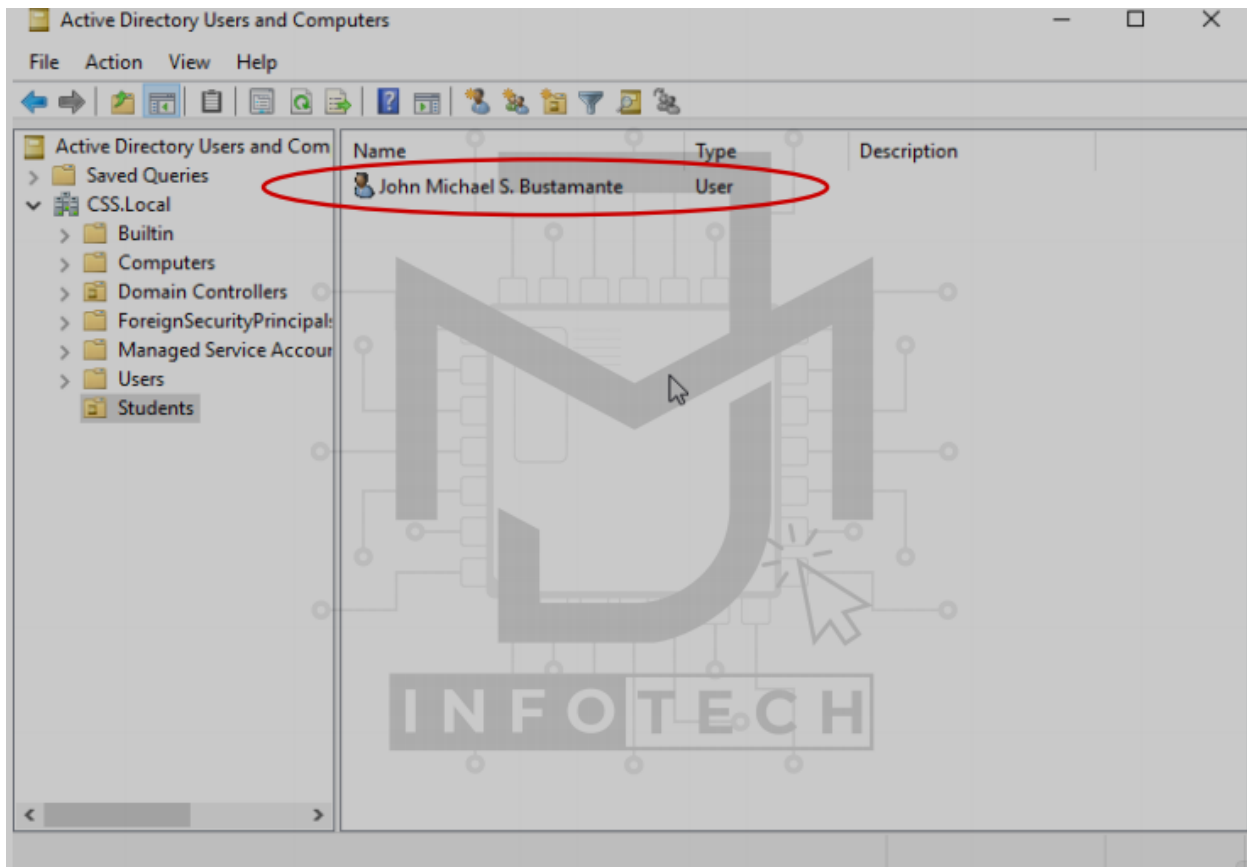
- You will now be prompted to enter a password for your new user as below, you will need to enter the password twice to ensure that the password has not been typed incorrectly.
- You can also set basic options for the user account here, (e.g. we choose "Password never expires". Once you have entered a password, and have chosen the required options.
- Click Next to continue



You will now be presented with a summary pane as shown, please review the details presented, once you have confirmed that they are correct, select Finish to exit.



Now you have successfully created a user under your Organizational Unit (OU).



What is Folder Redirection?

In this presentation shows you how simply you can redirect folder in Windows Server with group policy. The folder redirection is the way to keep a profile folder to a network location or other location in the local computer. Typically, user profiles and settings are stored in local profile. By redirecting folders, you can access to data regardless of which computers you are logs in.

In addition to the immediate benefit of having that data on a file server that is much easier to keep backed up, the user also gets the

benefit of being able to go to multiple computers in your organization and still have access to their data. Using the default Windows settings and the default share settings on your file server, these redirections will be even made available offline automatically for your users.



The Policy-Based QoS node

This quality of service (QoS) node, known as the Policy-Based QoS node, defines policies that manage network traffic. For example, you might want to ensure that users in the Finance department have priority to run a critical network application during the end-of-year financial reporting period. You can do that by using the Policy-Based QoS node. In the User Configuration node only, the windows Settings folder contains the additional Folder Redirection node. With folder redirection, you can redirect user data and settings folders such as **AppData**, **Desktop**, **Documents**, **Pictures**, **Music**, and **Favorites** from their default user profile location to an alternate location on the network, where you can manage them centrally.

Infrastructure Requirement:

1 DC SERVER (DCSERVER)

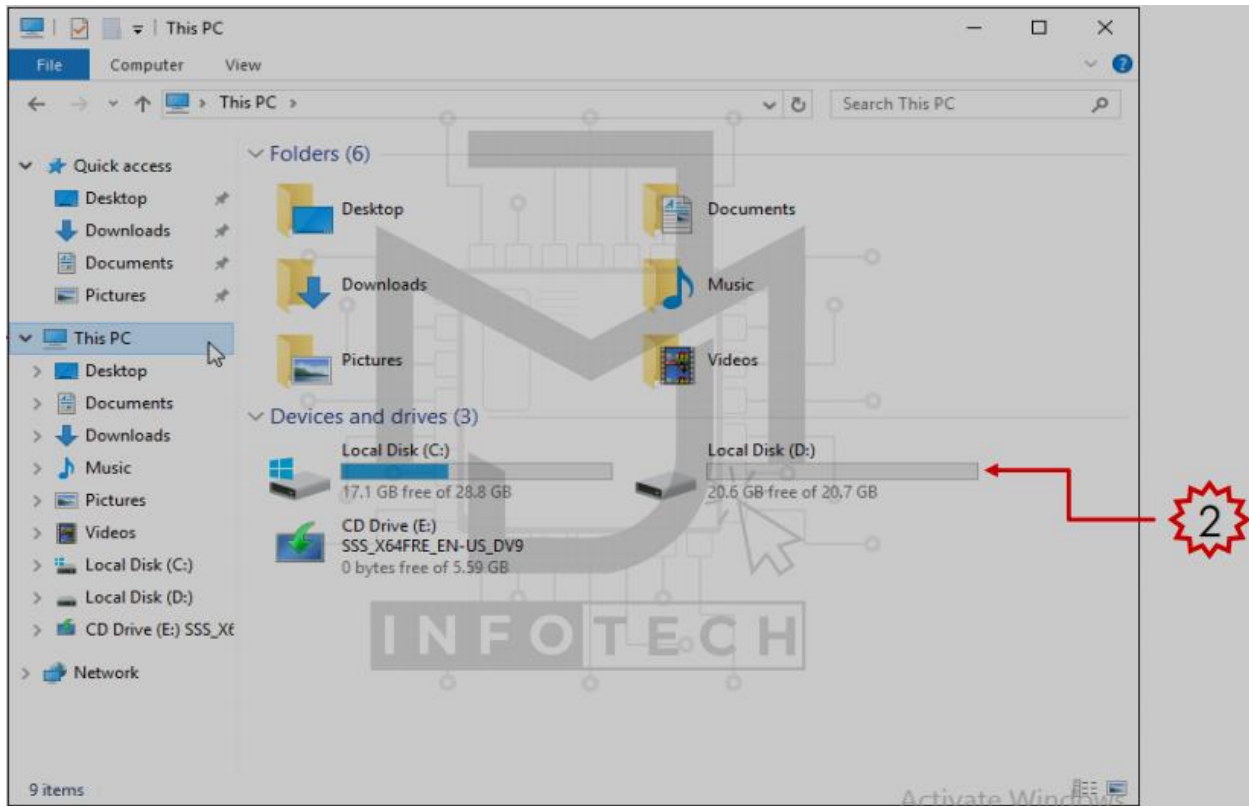
Client PC running on Windows 10 (CLIENT-01)

On your **DCSERVER**, at the **taskbar**, click the **File Explorer** icon.

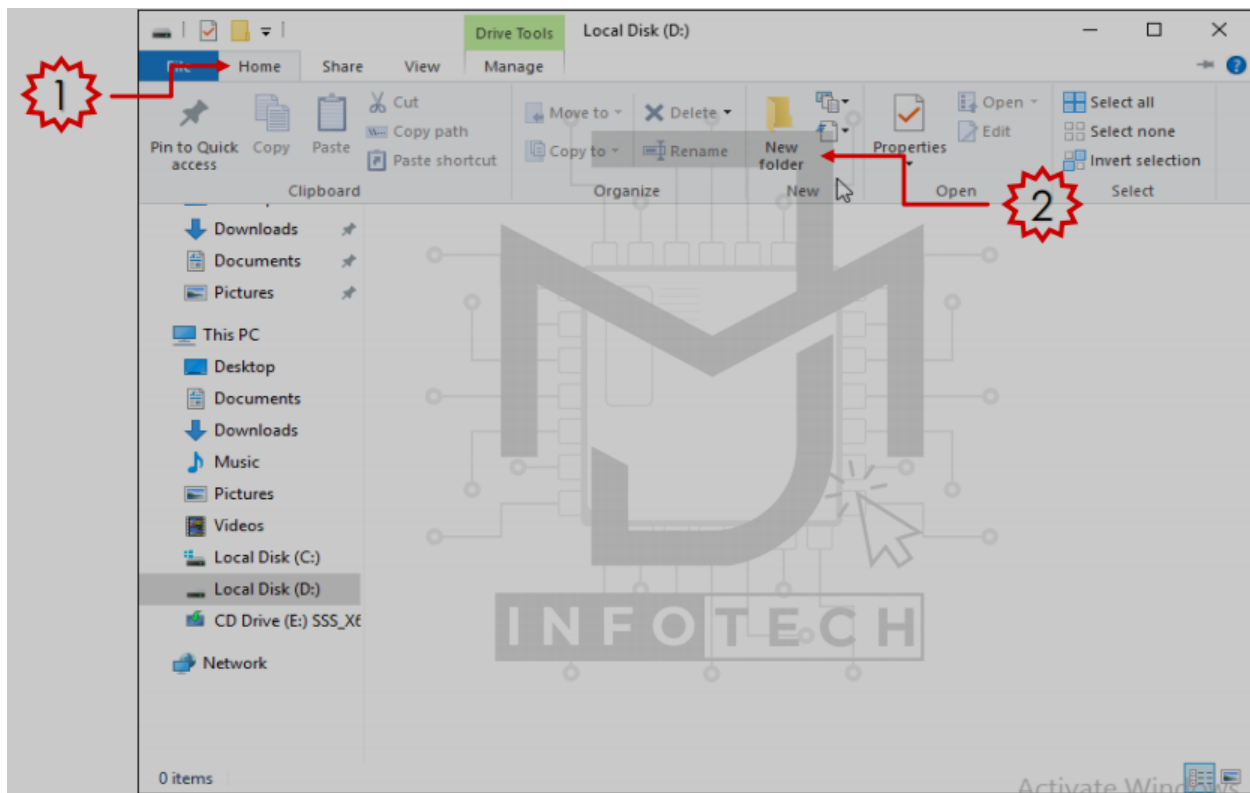


- In the navigation pane, click This PC.

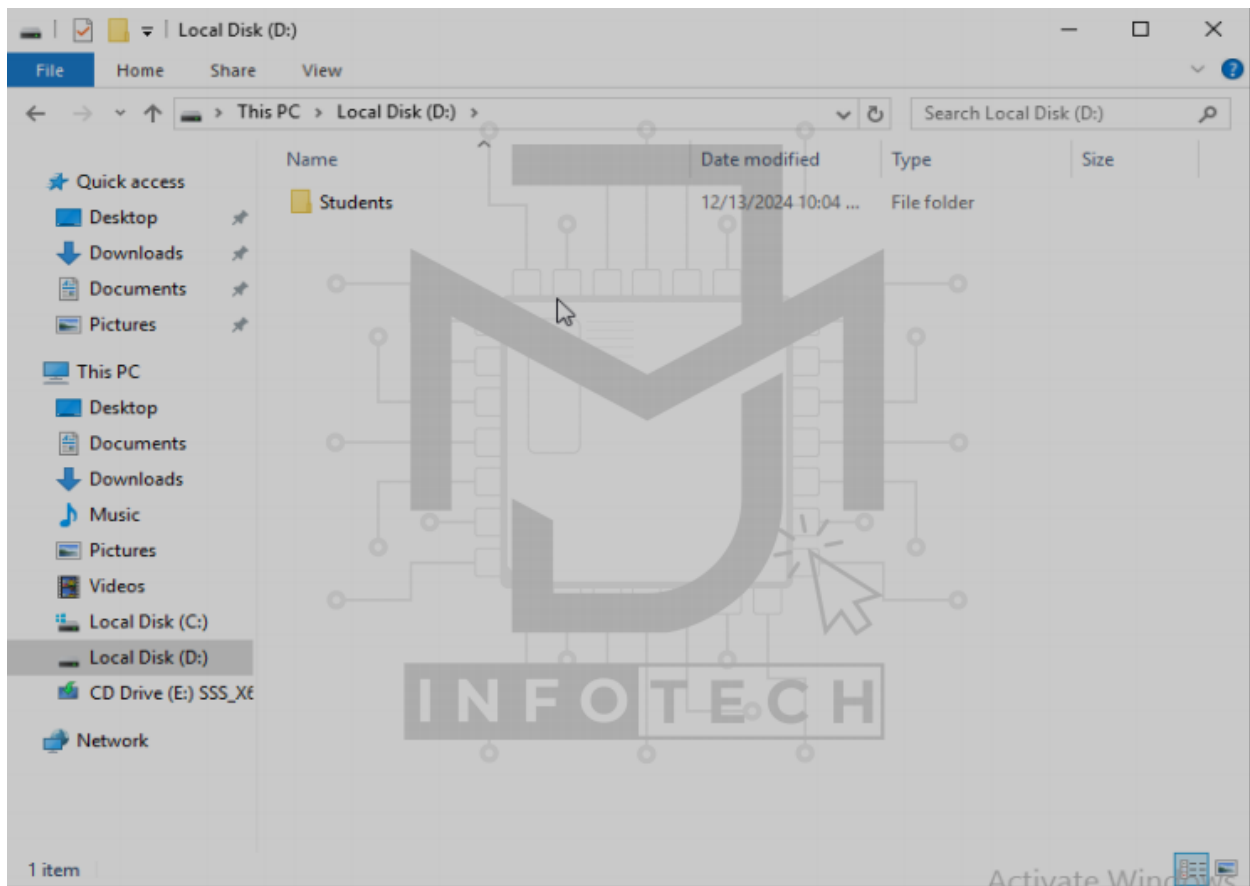
- In the details pane, double-click Local Disk (C:) or any Drive partition.
- In this demo we are going to select Local Disk (D:)



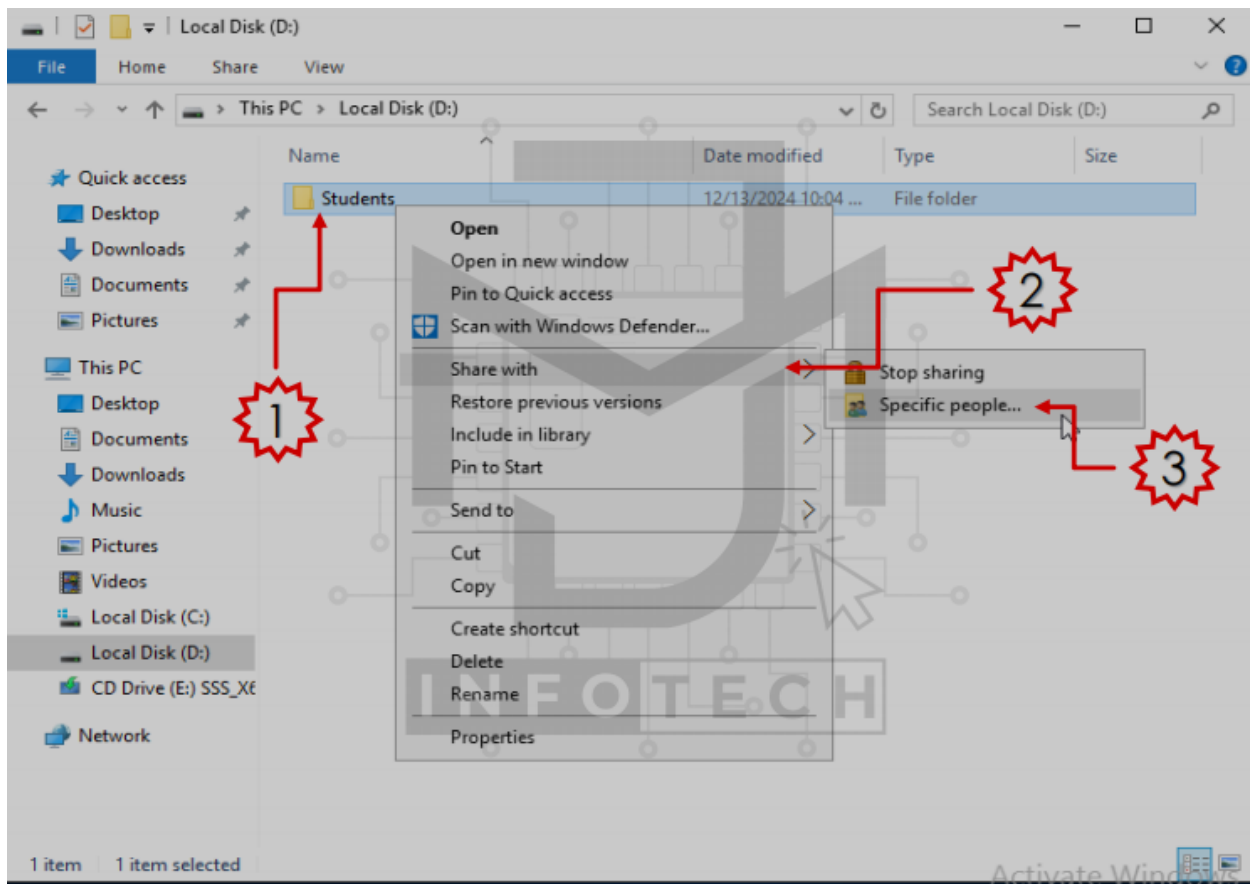
On the **Home** tab, click **New folder**.



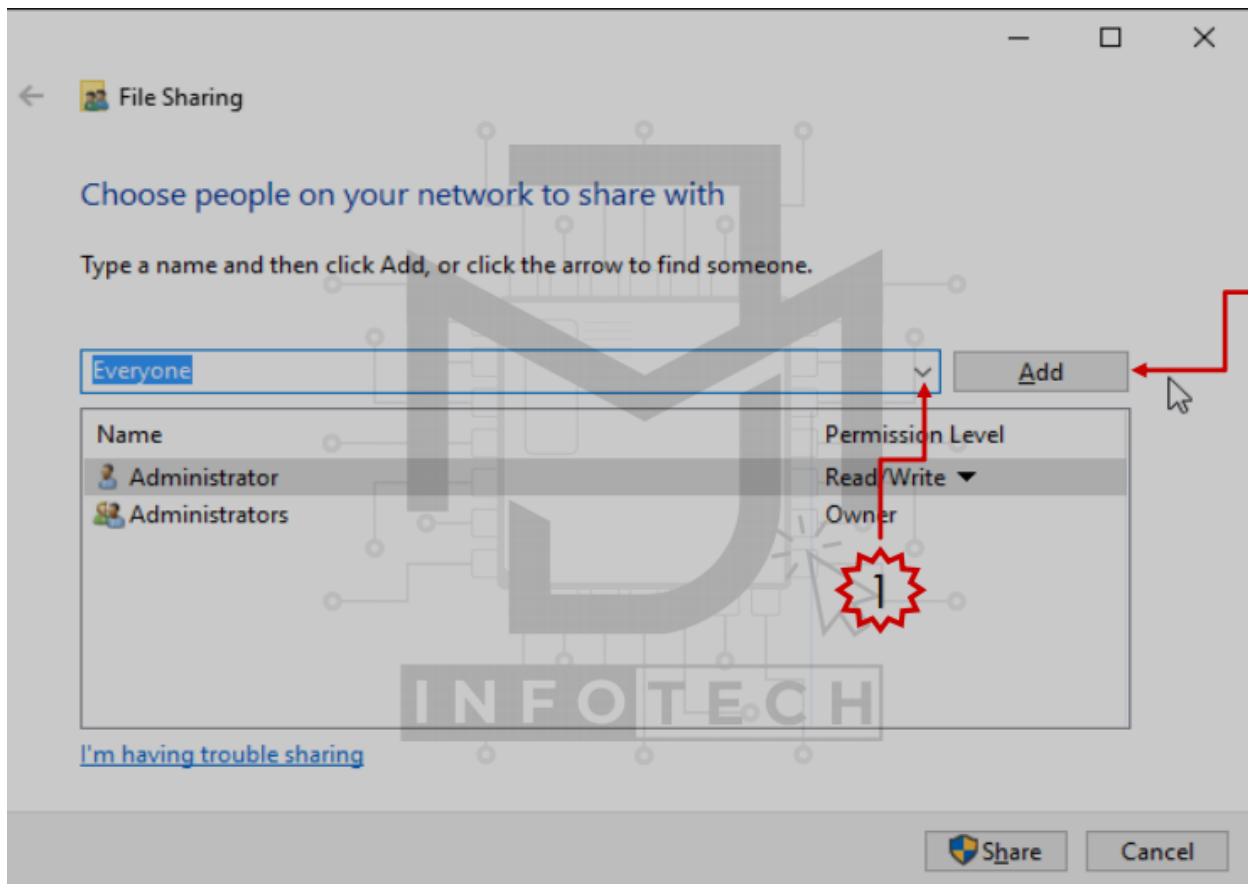
Type name of the new folder (e.g. Students), then press Enter.



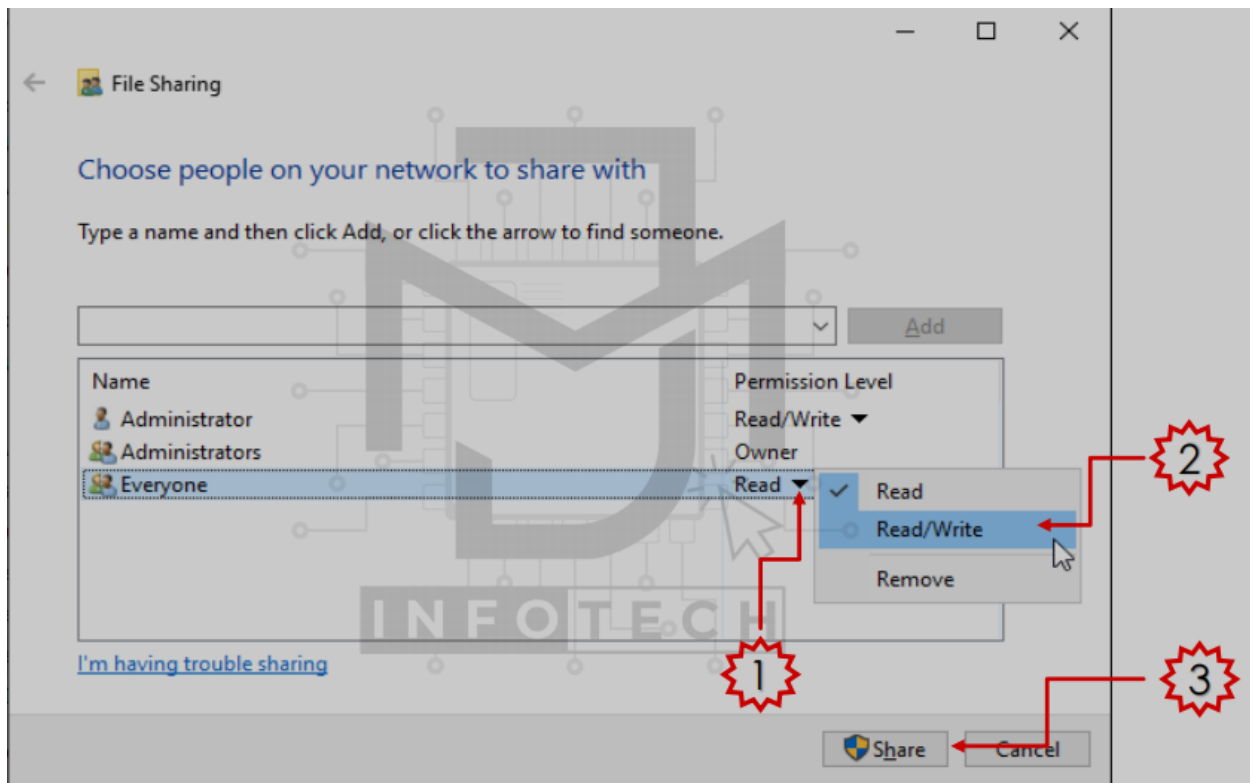
Right-click the Students folder, click Share with, and then click Specific people.



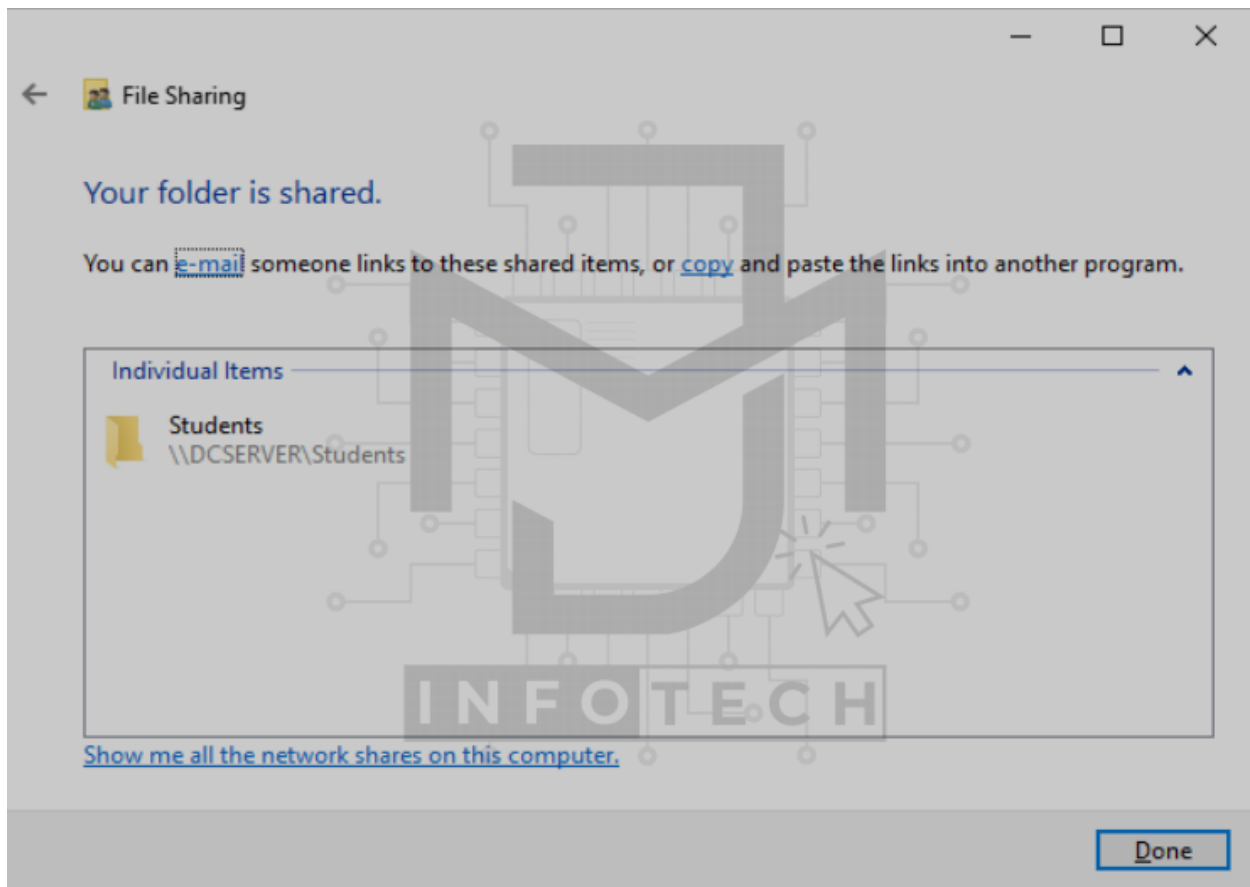
In the File Sharing dialog box, click the drop-down arrow, select Everyone, and then click Add.



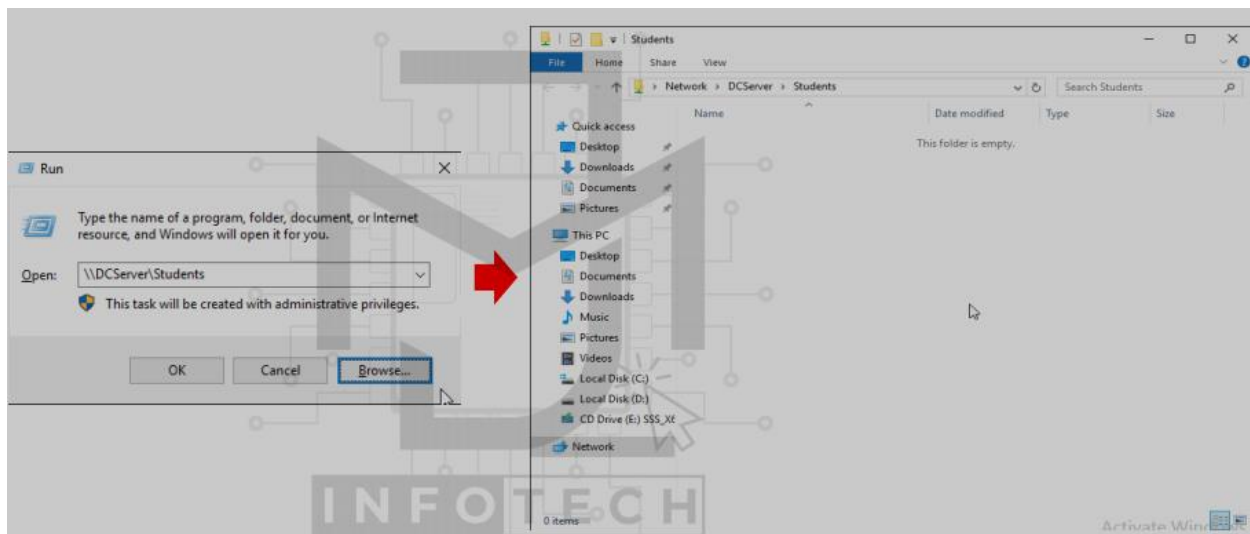
For the Everyone group, click the Permission Level drop-down arrow, and then click Read/Write.



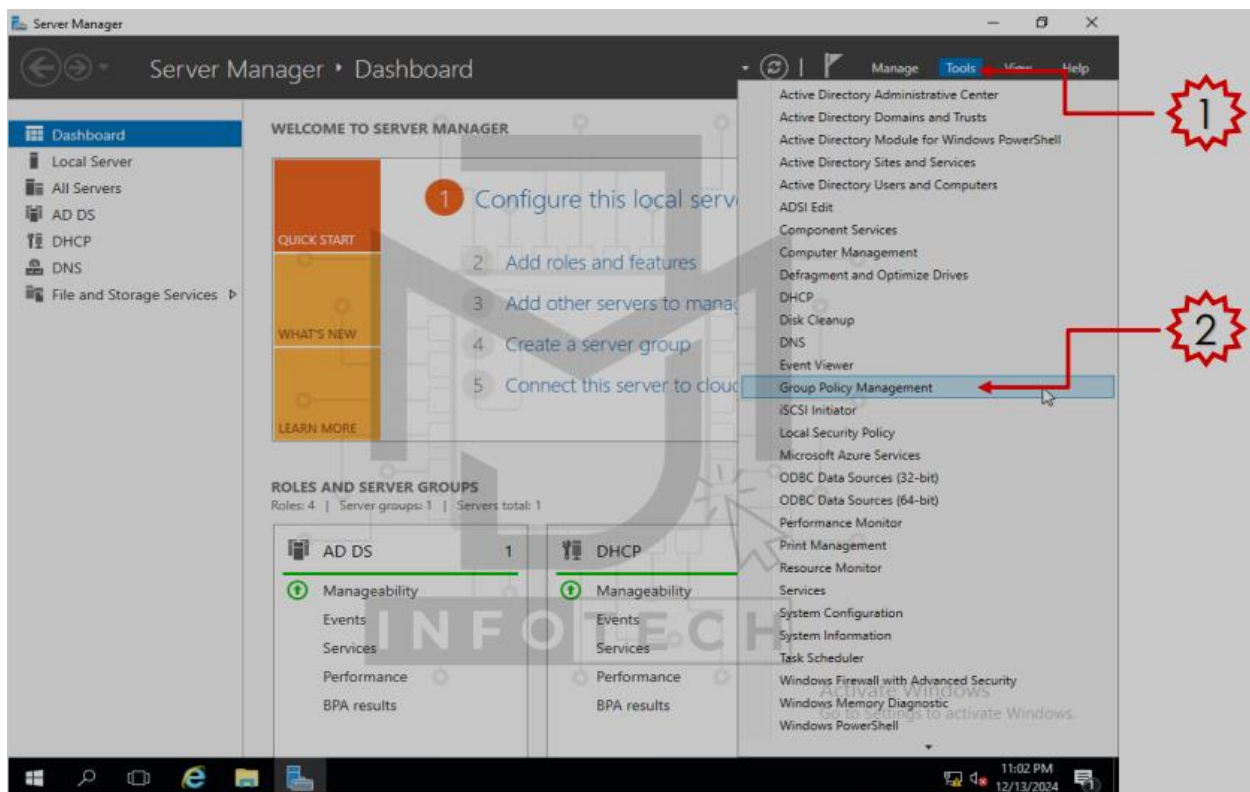
- Click Share, and then click Done.
- Close the Local **Disk (D:)** window



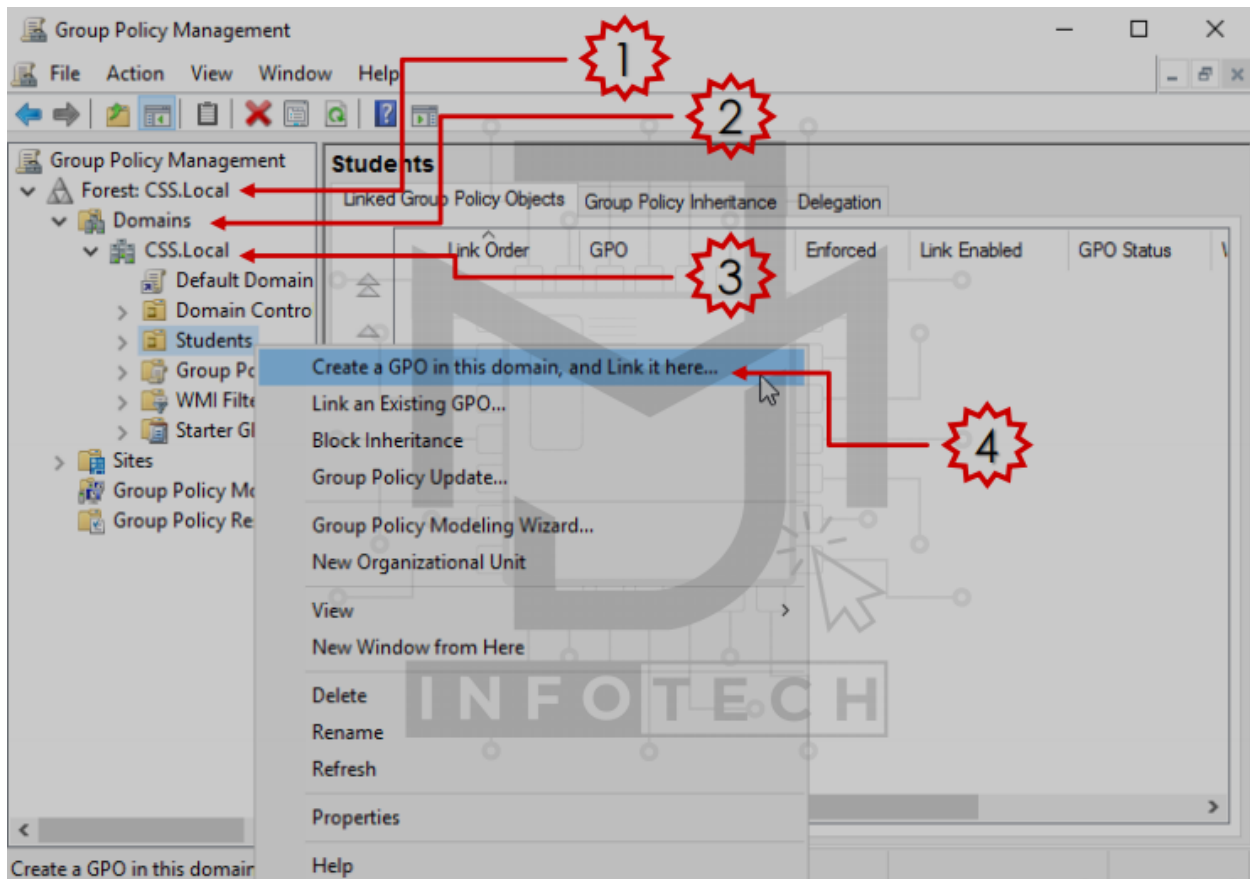
- On your keyboard press Win + R to open the Run dialog box.
- In the Run dialog box, type \\ComputerName\Shared Folder.
- For example, if the computer name is **DCServer** and the shared folder name is **Students**, you would type \\DCServer\Students in the Run dialog box. Press the Enter key to run it.



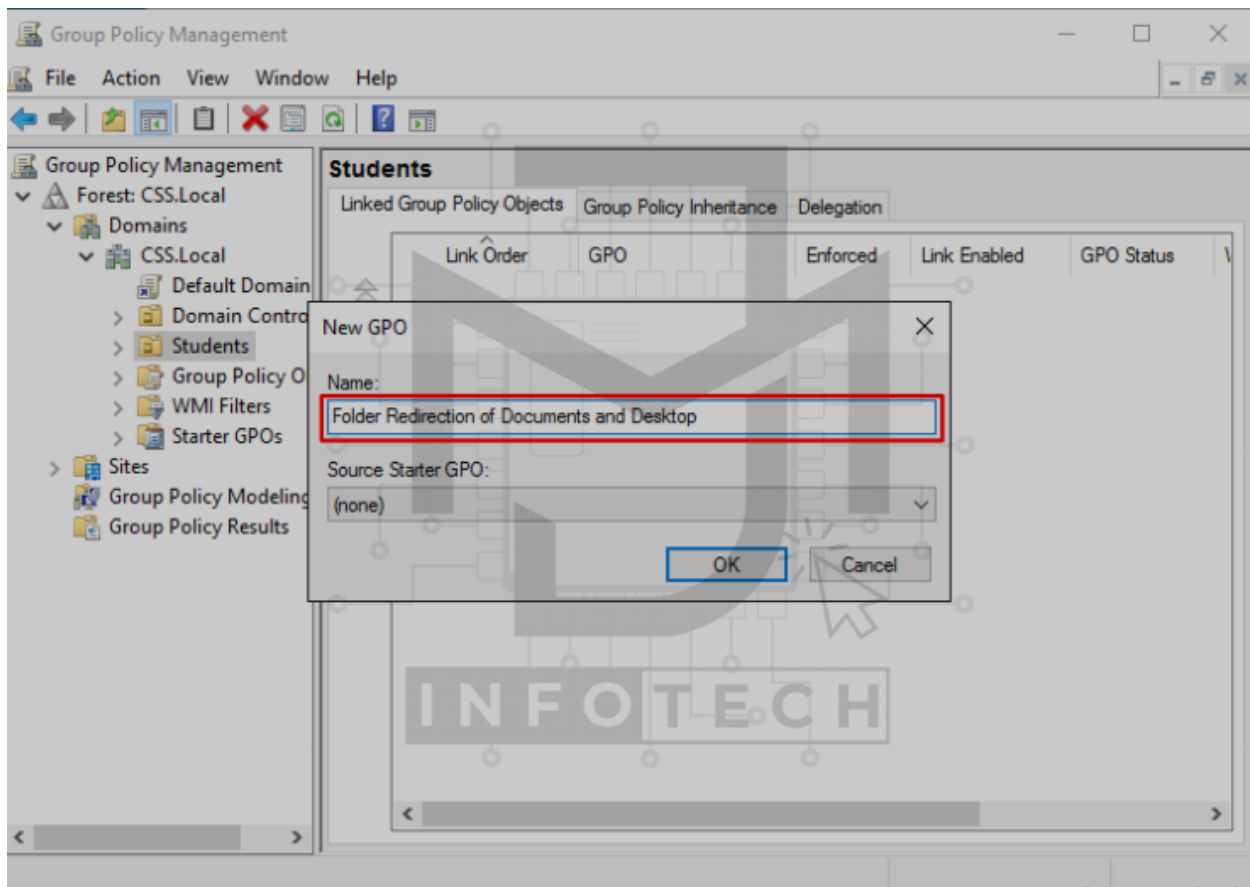
- In the Server Manager, click Tools and then click Group Policy Management.



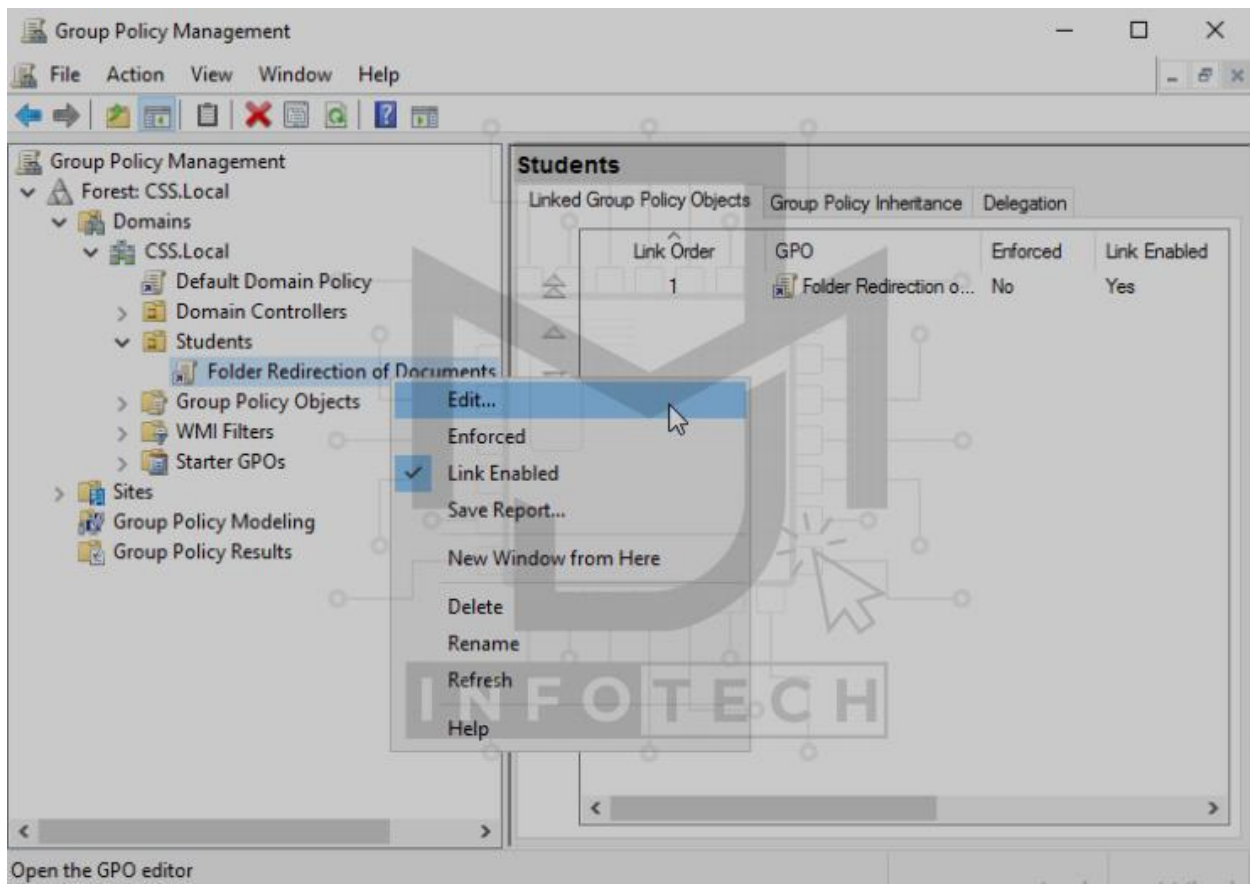
- In the navigation pane, expand "Forest: CSS.Local", then expand "Domains", then expand "CSS.Local"
- Right-click on the "Students" O.U, and then click **Create a GPO in this domain and Link it here...**



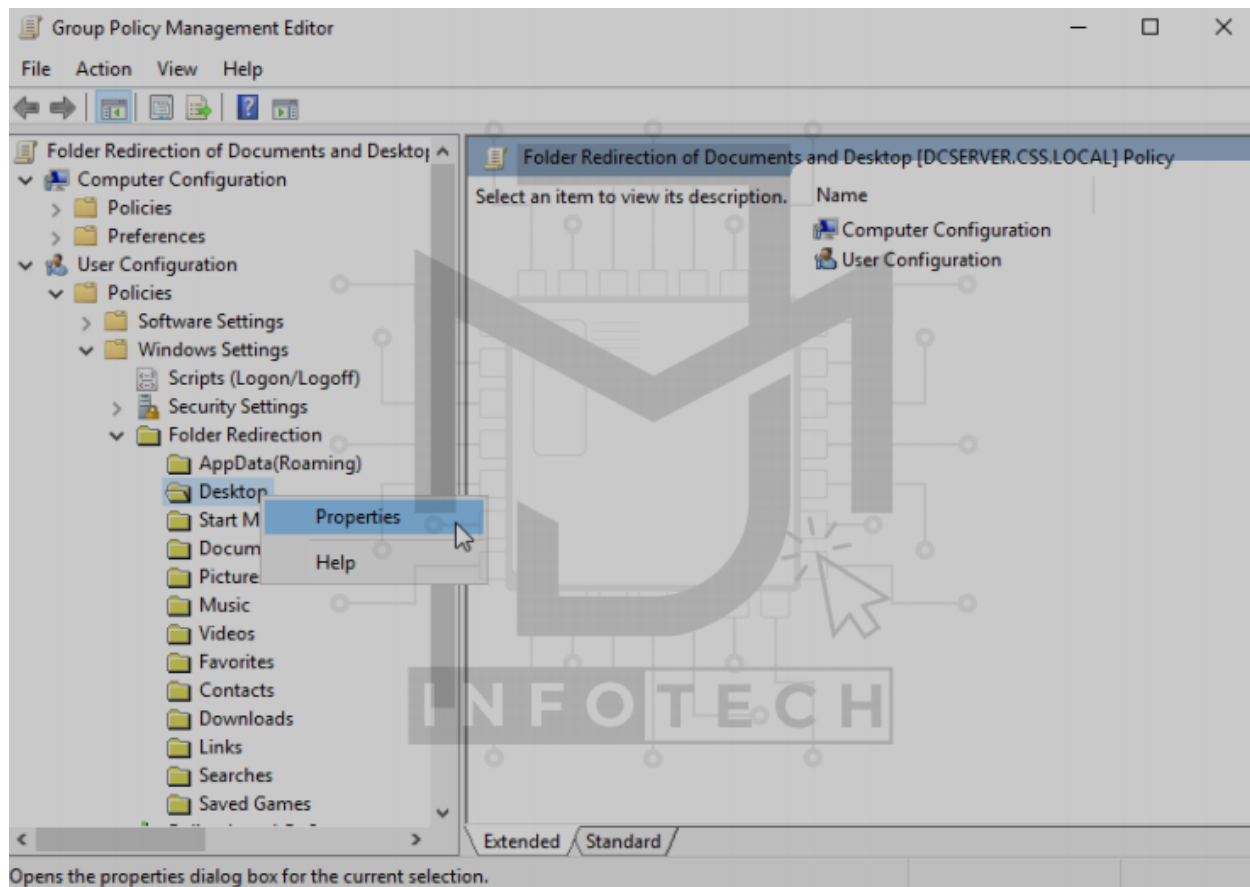
- In the New GPO dialog box, in the Name text box, type Folder Redirection of Documents and Desktop, and then click OK.



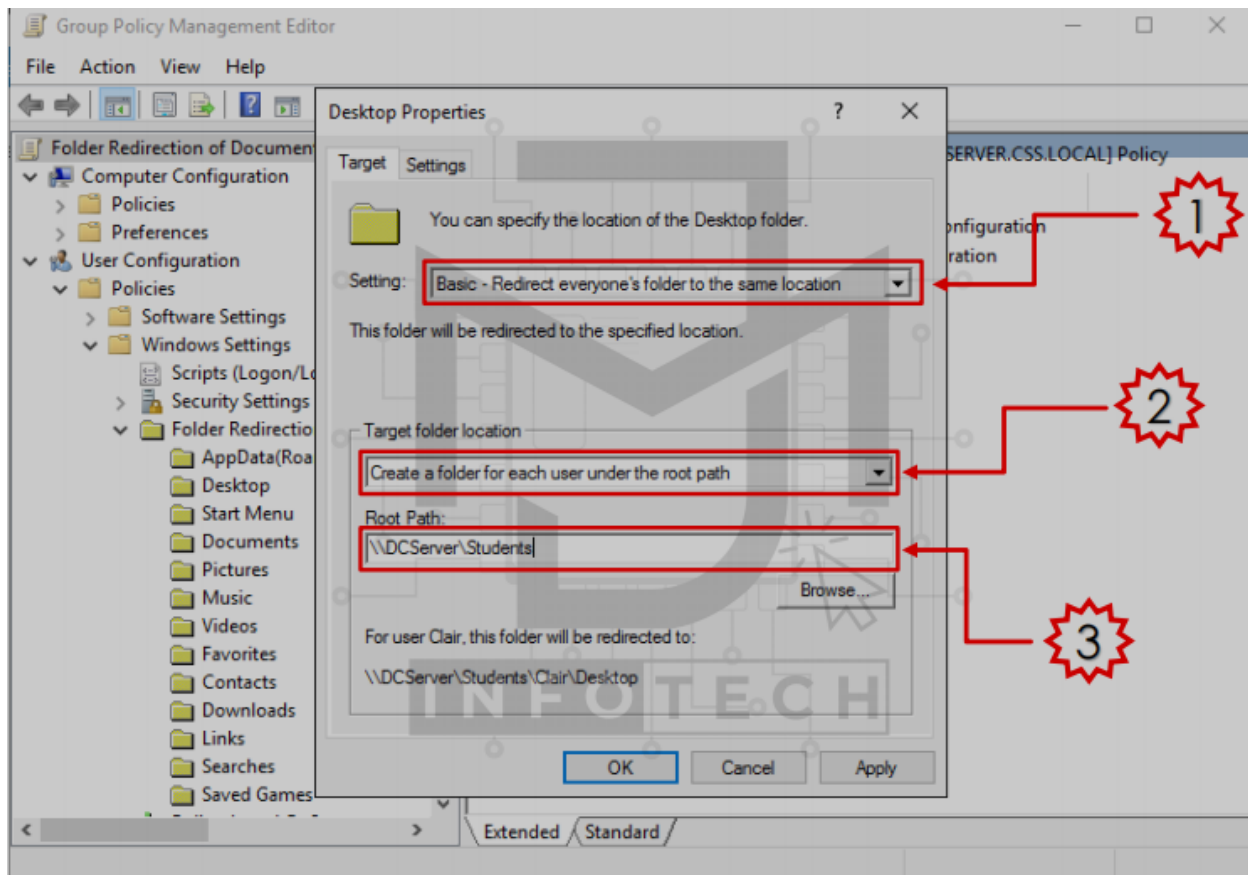
- In the navigation pane expand Students O.U
- In the navigation pane, right-click Folder Redirection of Documents &
- Desktop, and then click Edit.



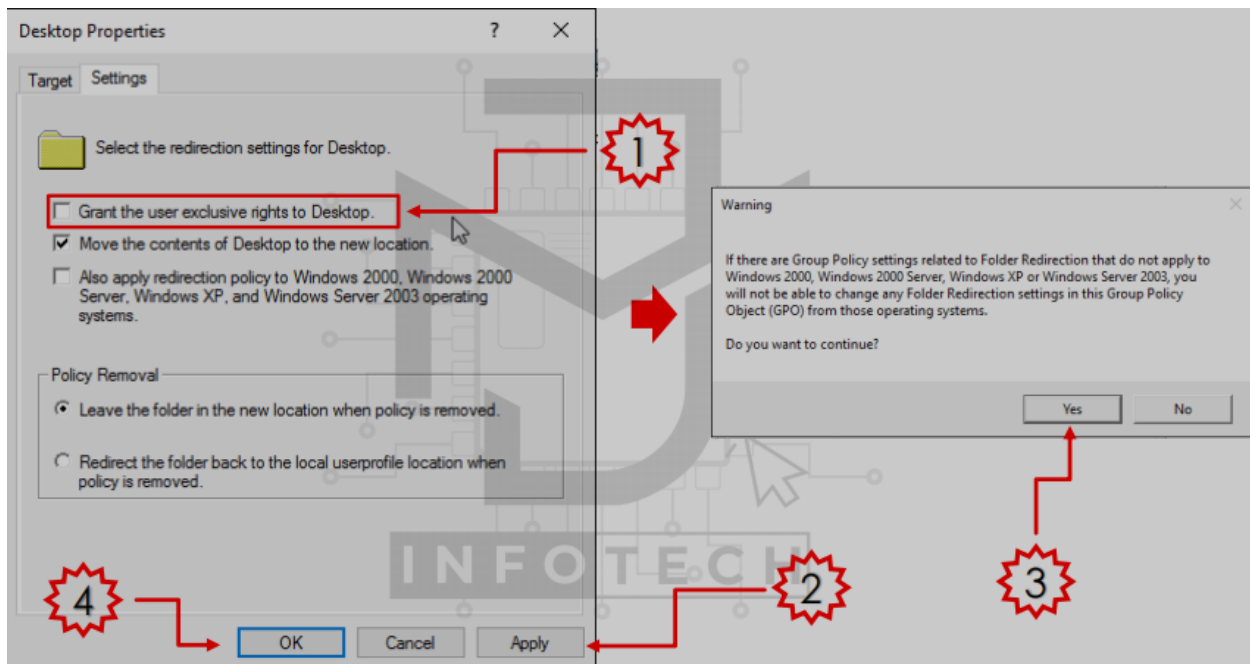
- In the Group Policy Management **Editor window**, under **User Configuration**, expand **Policies**, expand **Windows Settings**, and then expand **Folder Redirection**.
- Right click on Desktop and select Properties



- In the **Desktop Properties dialog box**, on the Target tab, click the Setting drop-down arrow, and then select **Basic-Redirect everyone's folder to the same location**.
- Ensure that the Target folder location box is set to **Create a folder for each user under the root path**.
- In the Root Path text box, type **\\DCServer\Students**



- In the Desktop Properties dialog box, on the Settings tab, uncheck the box Grant the user exclusive rights to Desktop.
- Click Apply to continue
- In the Warning dialog box, click Yes.
- Click OK to close.



~~REPEAT THE SAME PROCEDURES FOR DOCUMENTS~~



What is Domain Join?

Joining a domain when logging into a computer refers to the process of connecting that computer to a centralized network management system, typically found in enterprise environments.



Key Concepts

- **Domain:**

- A domain is a collection of computers, users, and devices that are managed under a common directory service, such as Microsoft Active Directory (AD). This allows for centralized management of resources, security policies, and user accounts.

- **Centralized Authentication:**

- When a computer is joined to a domain, user accounts are managed by the domain controller (DC). Users can log in using their domain credentials (username and password) rather than local accounts, allowing for easier access to network resources.

- **Group Policies:**

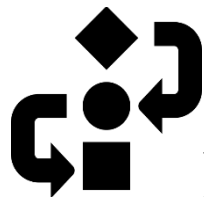
- Once a computer is part of a domain, administrators can apply group policies to enforce security settings, software installation, and other configurations across all computers in the domain.

- **Resource Access:**

- Being part of a domain allows users to access shared resources, such as files, printers, and applications, that are available on the network, based on their permissions.

- **Single Sign-On (SSO):**

- Users can often access multiple services and applications with a single set of credentials, simplifying the login process and improving security.



Process of Joining a Domain

- **Configuration:**

- The computer must be configured with the correct network settings to communicate with the domain controller (e.g., DNS settings).

- Only a Windows 10 Pro, Enterprise, or Education edition PC can join a domain.

- **Domain Credentials:**

- The user must have appropriate permissions to join the computer to the domain, usually requiring administrative credentials for the domain.

- **Domain Join Action:**

- The computer is added to the domain using the system settings (e.g., in Windows, this can be done through the "System Properties" window).

- You must be signed in to an Administrator account on your Windows 10 PC to join a domain Reboot:

- After joining the domain, the computer typically needs to be restarted for the changes to take effect.



Benefits of Joining a Domain

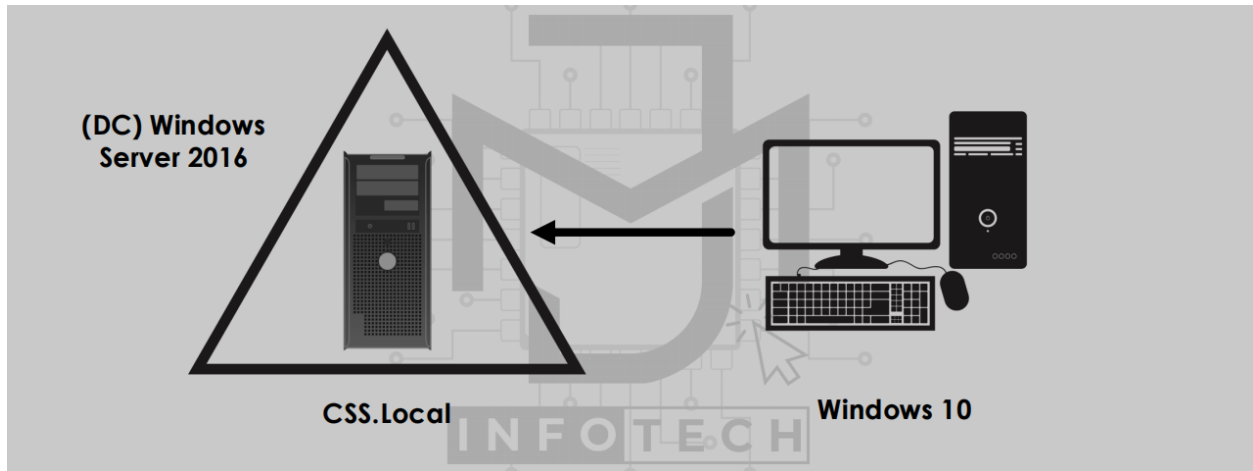
► **Improved Security:** Centralized management allows for better security practices and monitoring.

► **Easier Management:** Administrators can manage multiple computers and user accounts from a single location.

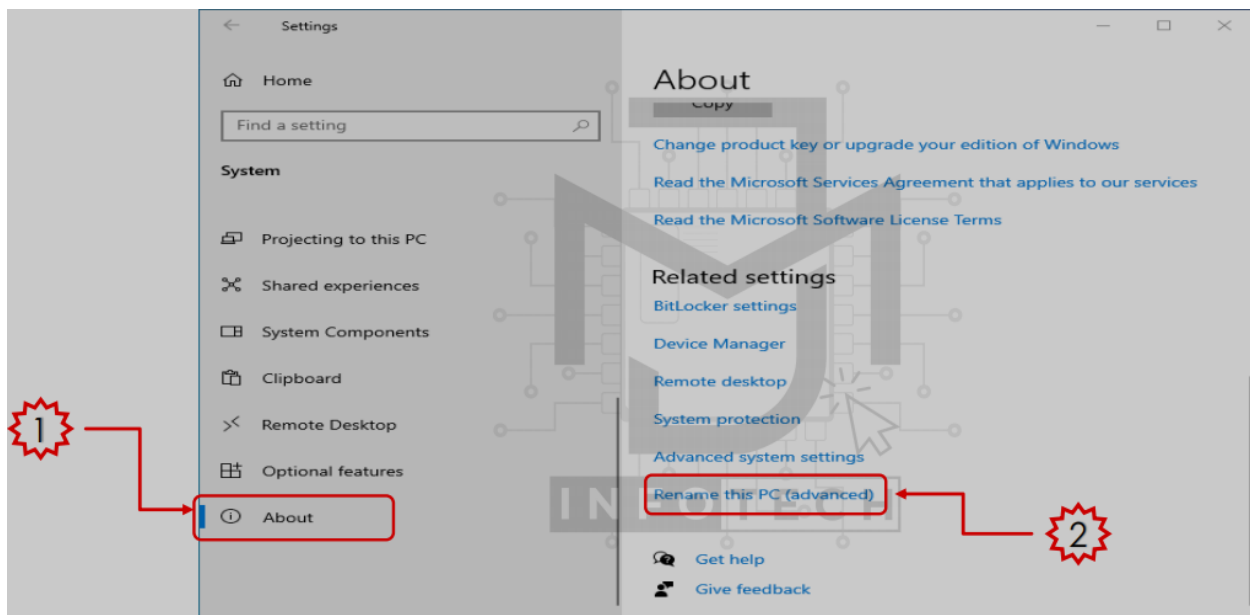
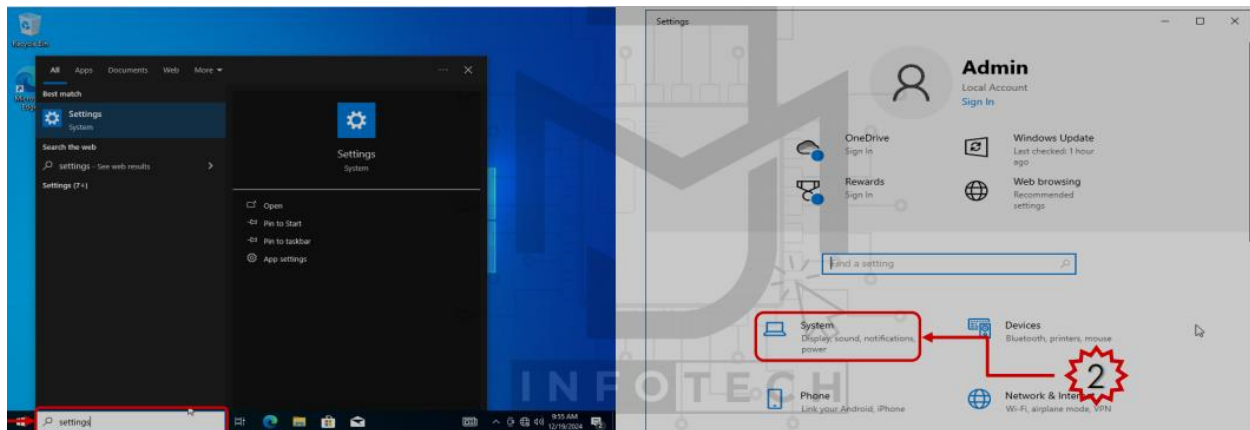
► **Scalability:** As an organization grows, adding new users and computers to the domain is relatively straightforward.

In summary, joining a domain enhances the management, security, and accessibility of resources within an organizational network.

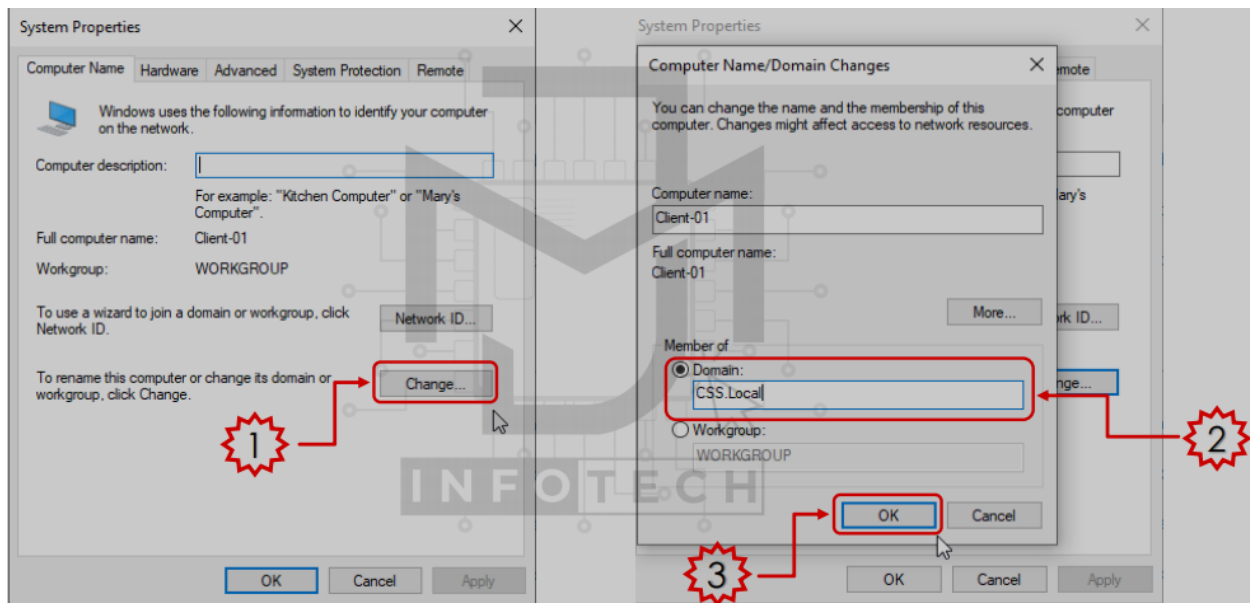
Set up diagram:



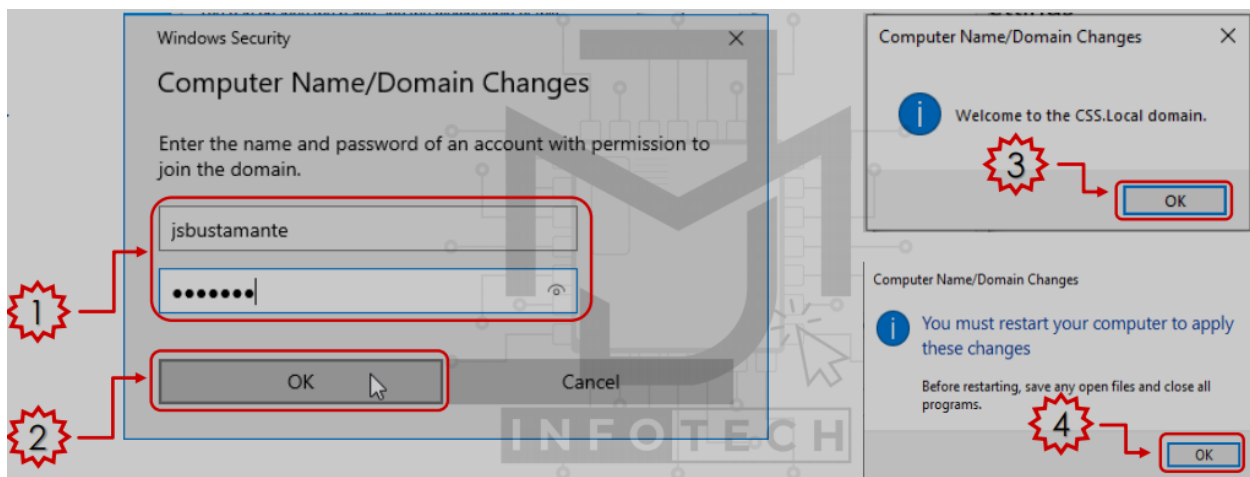
- On the Windows 10 PC sign in as **Administrator**.
- On the search bar, type **settings**, then press enter or use the keyboard shortcut **Win+I**
- Click **System**
- Click **About**
- Then click **Join a Domain**



- Click/tap on the Change button in the Computer Name tab.
- Select (dot) Domain type the domain name (e.g. "CSS.Local") of the domain you want to join
- Click OK to continue.

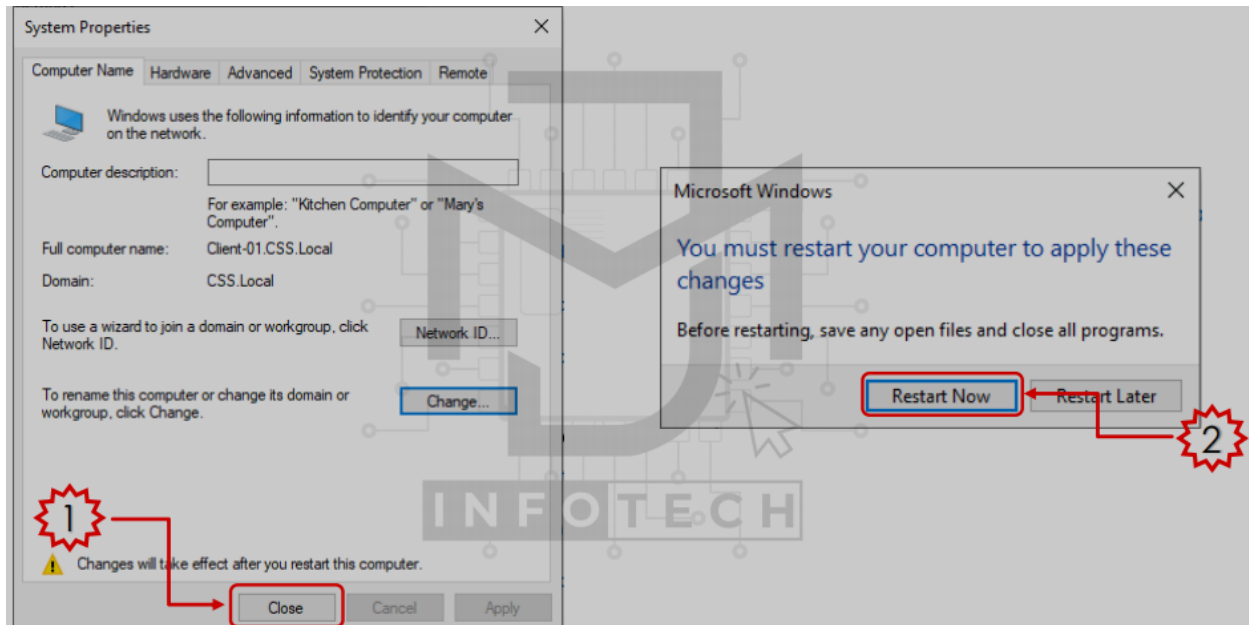


- Type the domain user name (e.g. "**abestipular**") and **password** for your domain account.
- Click **OK** to proceed.

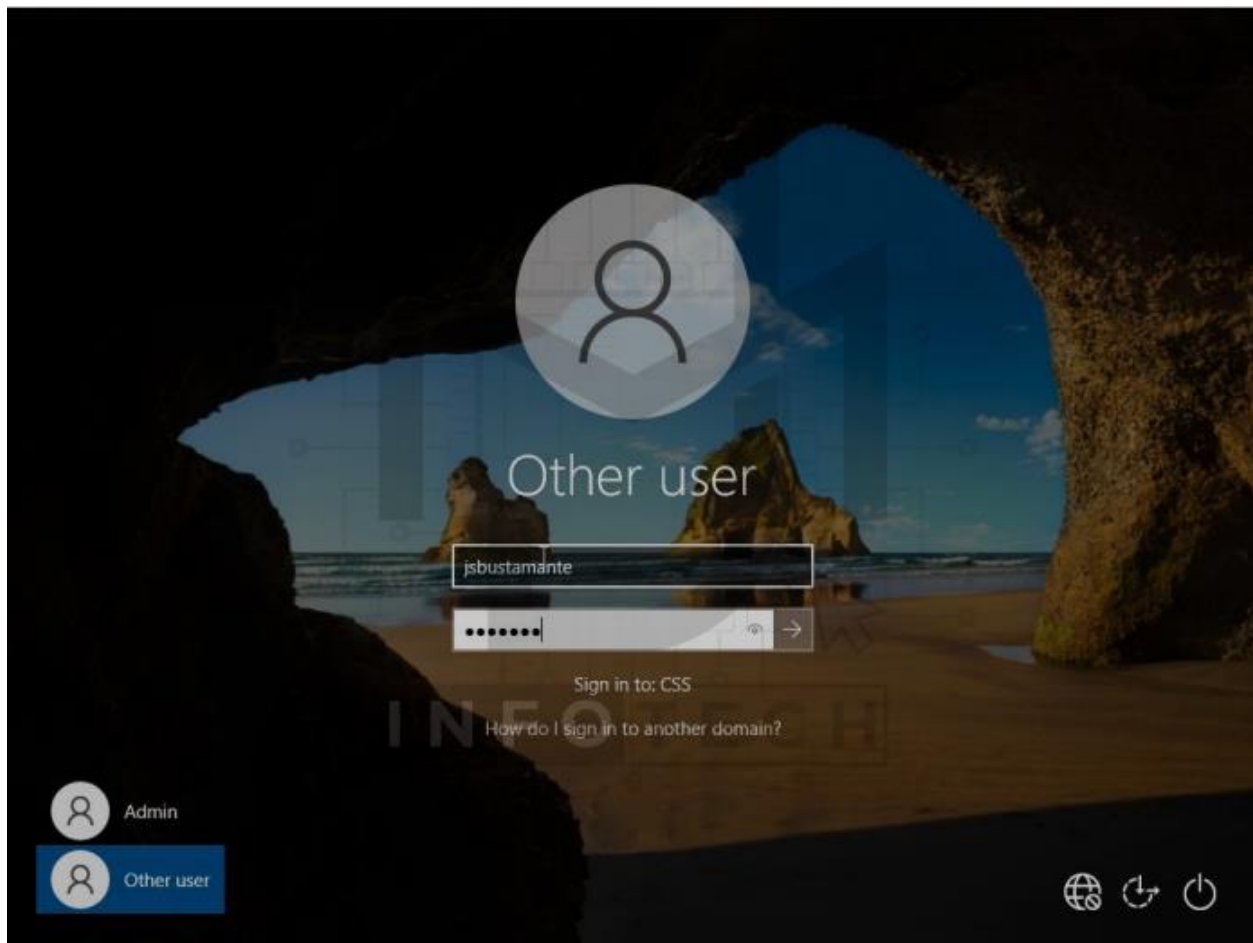


- Click Close to exit.
- Click on Restart Now.

- After you restart, your PC will be joined to the domain.



- When the sign-in screen appears, you will notice the DOMAIN\User account is displayed.
- Enter your password, and you will now be logged onto your domain.



Congratulations!

After all those instructions and guides, were you able to succeed this practice? If so, congratulations on you and if not look again and take your time.



Apply and perform!

Performance Task: Network Services Setup

Objective:

Students will configure a DHCP (Dynamic Host Configuration Protocol) server, DNS (Domain Name System) server, and FTP (File Transfer Protocol) server for a company network. Each student must follow the provided instructions while using the designated IP address and company name.

Task Description:

You are responsible for setting up network services for **CP**. Follow these steps for proper configuration:

Task: DHCP, DNS, and FTP Configuration

Steps:

1. Configure the DHCP Server

- Define the IP address range based on the assigned network (Example: 192.168.50.100 - 192.168.50.150).
- Set the appropriate subnet mask and default gateway.

- Ensure the DHCP server provides a DNS server address to clients.
- Test the configuration by connecting a client device and verifying automatic IP assignment.

2. Set Up the DNS Server

- Configure a custom domain (Example: cp.local) to resolve to a specific IP address (Example: 192.168.50.10).
- Set up forward and reverse lookup zones.
- Ensure that client devices use the DNS server and verify domain resolution with nslookup or ping.

3. Install and Configure an FTP Server

- Choose an appropriate FTP server software (e.g., FileZilla Server, vsftpd).
- Create at least two user accounts with different access levels (Read-Only and Read-Write).
- Assign shared directories and configure file permissions.
- Test the setup by logging in with each user and verifying upload/download access.

Expected Output:

- A step-by-step configuration guide.
- Screenshots of each server setup and settings.
- Screenshots of successful connectivity tests and file transfers.

Summary:

This task provides hands-on experience in configuring essential network services for CP. Students must follow the given IP address and company name while completing the setup, ensuring practical application of networking concepts.

The Criteria for the Performance Task

Criteria	Excellent (10)	Good (8)	Satisfactory (6)	Needs Improvement (4)	Incomplete (0)
DHCP Configuration	Fully functional with correct IP range, subnet mask, gateway, and DNS settings.	Minor misconfigurations but functional.	Some settings incorrect, partial DHCP function.	Major errors, DHCP fails to assign IPs.	No attempt or incorrect configuration.
DNS Configuration	Domain correctly resolves with forward and reverse lookup zones.	Minor misconfigurations, but functional.	Some settings incorrect, domain resolution works inconsistently.	Major errors, DNS resolution fails.	No attempt or incorrect configuration.
FTP Configuration	FTP server set up correctly with both user access levels and working file transfers.	Minor issues with user access or file transfers.	Some settings incorrect, partial functionality.	Major errors, FTP server does not work.	No attempt or incorrect configuration.
Testing and Evidence	Screenshots provided for all steps, showing successful configuration and testing.	Screenshots provided but missing some steps.	Some missing screenshots or unclear evidence.	Minimal evidence of testing.	No screenshots or testing evidence.
Documentation	Detailed and well-structured step-by-step guide.	Mostly detailed, minor missing information.	Basic documentation with some missing details.	Poorly structured and missing critical steps.	No documentation or extremely incomplete.



What to Check

Now that you've explored the contents, practice and activities of this

lesson, it's time to reflect on what you've learned. Answer this short quiz to assess your understanding on **Configuring Network Services**. Apply what you've learned and carefully consider each question. Good luck!

 How to Access the Short Quiz:

 Click this link: <https://forms.gle/RUnHvXGjU19kMfiQ7>

OR

 Just kindly scan this QR Code to access the Short Quiz via Google Forms.

Scan this QR Code



What I have learned

Essay

Reflect on your experiences or observations with network services, such as DHCP, DNS, and FTP. Why do you think proper configuration is the crucial part in ensuring a smooth and secure technological experience? How might your attitude or approach toward managing these services impact others (users) who rely on the network?

References: